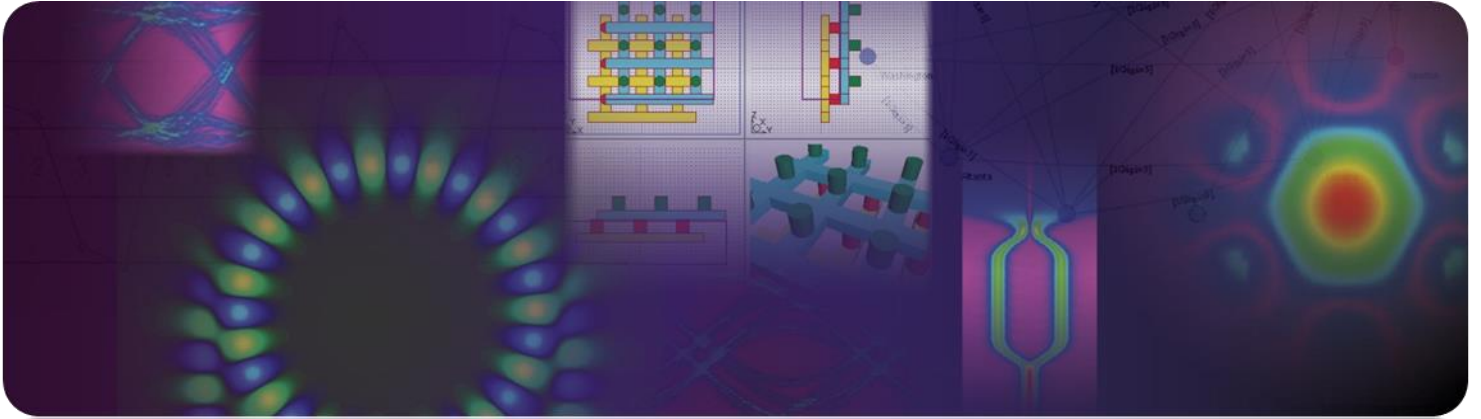




Photonic Solutions

FemSIM FEM

v2024.09

User Guide

Copyright Notice and Proprietary Information

Copyright © 2025 Synopsys, Inc. All rights reserved. This software and documentation contain confidential and proprietary information that is the property of Synopsys, Inc. The software and documentation are furnished under a license agreement and may be used or copied only in accordance with the terms of the license agreement. No part of the software and documentation may be reproduced, transmitted, or translated, in any form or by any means, electronic, mechanical, manual, optical, or otherwise, without prior written permission of Synopsys, Inc., or as expressly provided by the license agreement.

Right to Copy Documentation

The license agreement with Synopsys permits licensee to make copies of the documentation for its internal use only. Each copy shall include all copyrights, trademarks, service marks, and proprietary rights notices, if any. Licensee must assign sequential numbers to all copies. These copies shall contain the following legend on the cover page:

"This document is duplicated with the permission of Synopsys, Inc., for the exclusive use of _____ and its employees. This is copy number _____."

Destination Control Statement

All technical data contained in this publication is subject to the export control laws of the United States of America. Disclosure to nationals of other countries contrary to United States law is prohibited. It is the reader's responsibility to determine the applicable regulations and to comply with them.

Disclaimer

SYNOPSYS, INC., AND ITS LICENSORS MAKE NO WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, WITH REGARD TO THIS MATERIAL, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE.

Trademarks

Synopsys' company and certain product names are trademarks of Synopsys, as set forth at:

<http://www.synopsys.com/Company/Pages/Trademarks.aspx>. All other product or company names may be trademarks of their respective owners.

FOSS Notices

Free and Open Source Software (FOSS) notices are located in the \readme subfolder of the RSoft installation folder.

Contents

Preface	1
Foreword	1
How to Read This Manual	1
1 Introduction	4
1.A. Program Installation & System Requirements	4
1.B. Physical Conventions	4
1.C. Program Executables	5
1.D. Program Operation	6
1.D.1. Graphical Program Operation	6
1.D.2. CLI Operation	9
1.D.3. Log Files	9
1.E. Example Files	9
1.F. README Files	10
1.G. Product Support, Upgrades, and Resources	10
2 Background	11
2.A. Overview	11
2.B. The Electromagnetic Boundary Value Problem	12
2.C. The Solution via Finite Elements	12
2.D. References	14
3 Using FemSIM	15
3.A. The Mesh & Boundary Conditions	15
3.A.1. Computational Domain & Grid	16
3.A.2. Mesh Options	17
3.A.3. Boundary Conditions	19

3.A.4. Viewing the Mesh	21
3.B. Simulation Options	21
3.B.1. Mode Options	21
3.B.2. Radial Calculations	23
3.B.3. Polarization	23
3.B.4. Mode Dispersion Calculation	24
3.C. Display Options	26
3.D. Using Monitors to Make Measurements	27
3.D.1. Using Monitors in FemSIM	28
3.D.2. Computing Overlap Integrals	29
3.D.3. Computing Confinement Factors	30
3.D.4. Computing Mode Spot-Size	30
3.D.5. Computing Mode Area	30
3.D.6. Built-In Values for Editable Monitor Type Option	31
3.E. Output Options	31
3.F. Advanced Options	34
3.F.1. Advanced Algorithm Options	34
3.F.2. Advanced Simulated Bend Options	35
3.G. Using Multiple Cores/CPU's (Multi-Threading)	38
3.H. Supported Material Types	39
3.I. RSoft Utilities	39

4 Basic Tutorials 41

Basic Tutorial 1: Simple Waveguide Mode Calculation	41
CAD Window Basics	42
Creating a New Circuit	43
Creating Variables via the Symbol Table	44
Adding the Waveguide	46
Viewing the Structure in 3D	49
Creating the Mesh and Computing the Mode	49
The Simulation Window	50
Accessing Saved Data	52
Areas for Further Exploration:	54
Basic Tutorial 2: Simple Fiber Mode Calculation	55
Creating the Structure	55
Creating Variables via the Symbol Table	57
Adding the Fiber Segment:	57
Viewing the Structure along the Z axis	57
Creating the Mesh and Computing the Mode	58
The Simulation Results	60
Areas for Further Exploration:	61

5 Advanced Tutorials 63

Tutorial 1: Leaky Modes in a Multilayer/Rib Waveguide.....	63
Structure Overview	64
Creating the Structure in the CAD Environment.....	64
Setting the Mesh Parameters.....	67
Calculating the Fundamental TE & TM Modes	68
Calculating Higher Order Modes	68
Areas for Further Exploration	70
Tutorial 2: Modes of a Slot Waveguide Structure	72
Structure Overview	72
Creating the Structure in the CAD Environment.....	72
Calculating the Fundamental TE Mode with Coarse Grid	74
Performing a convergence study.....	75
Convergence Study Using Advanced Grid Features	77
Areas for Further Exploration	79
Tutorial 3: VCSEL Cavity Mode	80
Structure Overview	80
Device Setup	81
Computing the Supported Cavity Modes.....	85
Scanning Aperture Radius to Study Device Characteristics	88
Areas for Further Exploration	91
Tutorial 4: Bending Modes.....	92
Background	92
Structure Overview	92
Calculating the Straight Waveguide Mode	93
Calculating the Bending Mode	94
Scanning over the Bend Radius.....	96

Appendix A Tips and Traps **98**

A.A. Common FemSIM mistakes.....	98
A.B. Some Good FemSIM habits to learn	98

Appendix B FemSIM File Formats **100**

B.A. Standard Data File Format.....	100
B.B. Mode Files	100
B.C. Neff File Format.....	101
B.D. Monitor File Format	101

Appendix C Symbol Table Variables **102**

C.A. FemSIM Simulation Parameters	102
C.B. Mesh & Boundary Options.....	103
C.C. Mode Options.....	103
C.D. Display Options	104

Appendix D Release Notes	105
Index	108

Preface

Foreword

FemSIM FEM uses the Finite Element Method (FEM) to calculate modes, both waveguide and cavity, of any 1D or 2D cross-section. A non-uniform hybrid triangular/rectangular mesh is used to solve for all field components of supported modes, both guided and leaky. FemSIM can easily find modes for highly hybrid structures, polarization-sensitive structures, highly leaky structures, and structures with small feature sizes.

How to Read This Manual

While this manual can (and should!) be used as a reference manual, it is recommended that you read the first few chapters in their entirety to get a firm foundation from which to interpret the rest of the documentation.

If you are a new user, please reject the notion that you can simply turn to a section of interest in this manual. It can be hard to resist this urge, but doing so will reward you with better usage habits, a fuller understanding of how the software works, and will help you from suffering from common usage problems. Furthermore, we recommend that you look through the RSoft CAD manual before proceeding too far into this manual as it provides a foundation for creating design files for use with FemSIM.

How is This Manual Organized?

The manual can be logically split into several main parts:

This manual should be used alongside the RSoft CAD manual as described in the next section.

Introductory Information

Chapters 1 and 2 provide an overview of the installation of the program, an introduction to the program and its components, information about technical support and product upgrades, and background of the FEM algorithm used. All new users should read these chapters in their entirety.

FemSIM Usage

Chapter 3 explains the usage of FemSIM including setting mesh and boundary conditions, simulation options, as well as output and display options, measurement and analysis. All new users should read these chapters in their entirety.

Tutorials

Chapters 4-5 contain tutorial examples that span both basic and advanced FemSIM topics. The basic tutorials (Chapter 4) are recommended for all new users.

Appendices

The appendices provide detailed discussion of file formats and other relevant information. The appendices can be treated as a reference manual and do not need to be read by all new users.

Where Is The Documentation For...

The documentation for the RSoft Photonic Device Tools is divided into several manuals. The manuals are structured using a simple rule:

Anything defining geometry and/or material parameters is in the CAD manual. Anything else is in an appropriate simulation manual.

Using this rule, almost any topic can be found. As with any rule, there are a few exceptions. The major exceptions are:

- *Installation*

The installation procedure is covered in the Photonic Solutions Installation Guide.

- *Parameter Scanning/Scripting/Batch Operation*

These topics are very similar, and are shared by all the simulation modules. They are discussed in Chapter 10 of the CAD manual.

- *Computing the Index Profile*

Computing/Displaying the index profile is discussed in Section 3.G of the CAD manual.

- *Pathways*

Pathways define the location of pathway monitors (used by BeamPROP and FemSIM) and the location and geometry of launch (initial) fields in BeamPROP, FullWAVE, and ModePROP. They are documented in Section 6.G of the CAD manual.

- *Command Line Utilities*

The RSoft Photonics Device Tools ships with several command line utilities that perform a variety of tasks. These utilities are documented in Appendix E of the CAD manual.

- *RSoft Expressions*

Virtually any numeric field can accept an analytical expression involving pre-defined and user-defined variables. The form of these expressions, including valid arithmetic operators and functions can be found in Appendix C of the CAD manual.

Anytime this rule is violated, a note will direct the reader to the proper section in the proper manual.

Where are these manuals located?

All documentation is placed on your computer during the program installation. Online versions can be accessed through the RSoft CAD via the Help menu item, or the two help buttons on the right of the top toolbar. The actual files can be found in the subdirectory `help` in the installation directory. Additionally, PDF versions can be found in the subdirectory `docs`. These files require the Adobe Acrobat Reader, which can be obtained from Adobe (www.adobe.com) at no charge.

Typographical Conventions

A number of typeface and layout conventions are followed in this manual.

- The *names* of fields and controls in the GUI dialogs are written in **boldface**
- The *values* of pull-down menus and radio button controls are written in *italics*.
- File names and paths, symbol table variables and values, expressions typed in GUI edit fields, and code snippets are written in `monospace`.
- In referring to example CAD files, the installation directory for the CAD tool is specified as `<rsoft_dir>`, and should be replaced with the correct value for your installation.

1

Introduction

This chapter explains the installation procedure for FemSIM, a discussion of physics conventions used in this manual, information on running the program, and notes about getting product updates and technical support.

1.A. Program Installation & System Requirements

The installation process is outlined in the Photonic Solutions Installation Guide. Please check the 'System Requirements' section of the relevant product on Synopsys' Photonic Solutions website (<https://www.synopsys.com/photonic-solutions/product-system-requirements.html>) for a list of the specific OS versions we currently support.

Note that once a product has been successfully installed, it may not function correctly after changes such as OS updates or other software/hardware modifications are made to the computer system. In such cases, we will attempt to resolve any issues for customers that have a current annual maintenance contract, but we do not guarantee success.

1.B. Physical Conventions

As with any branch of science, there are a number of concepts in the study of photonic devices for which there exist several different definitions exist in the literature. These are our conventions:

Units

- The standard unit of length is measured in microns [μm].
- The angular unit used is degrees.
- The units of imaginary refractive index are defined as:

$$n_{imag} = \frac{\gamma\lambda}{4\pi}$$

where λ is the wavelength and γ is the usual exponential loss coefficient defined such that the power decays as $e^{-\gamma z}$ and is given in units of μm^{-1} .

- FemSIM displays computed mode profiles in arbitrary units that can be scaled to match the power in any experimental or theoretical result. We normally work without worrying about the units of the field in FemSIM for this reason.

Polarization

Polarization is defined relative to the **Dimensions** setting in the Global Settings dialog:

Dimensions	TE	TM
2D	Ey	Ex
3D	Ex	Ey

Please consult Appendix G in the CAD manual for a more detailed discussion.

1.C. Program Executables

The directory `<rsoft_dir>\bin\` contains many executable files. The following is a list of the executable for the RSoft CAD, FemSIM, the WinPLOT graphing tool, and MOST, our scanning and optimization tool.

Product	Windows Name	Linux Name
The RSoft CAD tool	<code>bcadw32.exe</code>	<code>xbcad</code>
FemSIM simulation tool	<code>femsim.exe</code>	<code>xfemsim</code>
WinPLOT graphing tool	<code>winplot.exe</code>	<code>xplot</code>
MOST tool	<code>rsmost.exe</code>	<code>xrsmost</code>

These executable are usually referred to via their Windows names in most documentation. These programs can also be run from the command line; more information on this can be found in the next section.

1.D. Program Operation

A FemSIM simulation can be started via either the Graphical User Interface (GUI) or a Command Line Interface (CLI). Both of these methods are equally powerful, and are useful in different circumstances. Beginning users are encouraged to use FemSIM via the GUI interface in order to understand the usage of the program. Advanced users who wish to incorporate FemSIM into scripts/batch operations are encouraged to use the scripting capabilities of FemSIM.

1.D.1. Graphical Program Operation

The GUI operation of FemSIM is a very straightforward process and can be broken down into 5 basic steps:

Step 1: Creating the structure

We begin by assuming that the main RSoft CAD program has been started, and that a layout window has been opened which contains the design to be simulated; this could be a newly created file, or a previously saved file that was loaded via the File/Open menu item. The general operation of the RSoft CAD program is described in the RSoft CAD manual.

FemSIM will solve for the mode of any 1D or 2D profile, so it might not be necessary to create an entire structure in the CAD. FemSIM only supports 2D profiles defined in the XY plane. For this case, a 3D CAD files must be created.

Step 2: Choosing the FemSIM simulation engine

Set the **Simulation Tool** to *FemSIM* in the Global Settings dialog box.

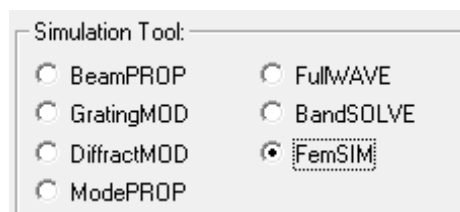


Figure 1-1: The Simulation Tool option as shown in the Global Settings dialog box.

Step 3: Setting the Simulation Parameters

Click the **Perform Simulation**, or green light icon, in the left toolbar to open the FemSIM Simulation Parameters dialog shown in Fig. 1-2. This dialog sets basic numerical parameters such as the domain, grid, and boundary, and other settings. A detailed description of these settings can be found in [Chapter 3](#).

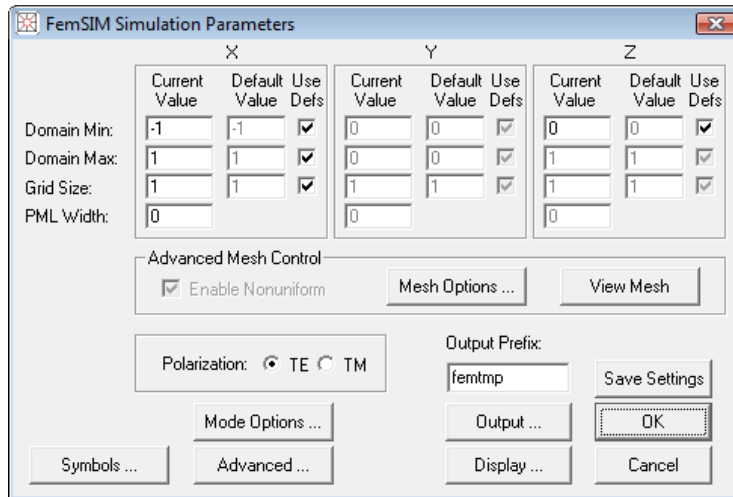


Figure 1-2: The FemSIM Simulation Parameters window where basic numerical simulation parameters are entered.

The default values can be accepted or different values for any parameter can be entered by simply editing the current value. To save simulation results, enter a prefix in the **Output Prefix** field and select the desired data files via the **Output...** button. FemSIM's output options are discussed in more detail in [Section 3.E](#). It is also advisable to check the simulation mesh to be used via the **View Mesh** button.

To check that the refractive index values entered are correct, use the **Display Material Profile** button in the left toolbar of the CAD. For FemSIM, which uses a generally non-uniform and irregular mesh, the index profile shown may not accurately represent the mesh to be used, but will show the correct index values.

Step 4: Running the Simulation

Click **OK** in the FemSIM Simulation Parameters dialog to begin the simulation. Note that **Cancel** rejects any changes and returns to the CAD interface, and **Save Settings** saves any changes made. Once a simulation is started, the computation begins in a new window separate from the CAD as shown in Fig. 1-3.

Within the simulation window shown in Fig. 1-3, the convergence process is shown in the lower part, and the simulation results are shown in the upper part as they are computed. The **Output** menu item and the simulation toolbar allow the simulation mesh and the different components of any mode profiles found to be viewed. During a simulation, only the first mode found will be displayed; to view other modes, use the **Output** menu options or the arrows on the simulation toolbar. Modes will be available to view as they are computed.

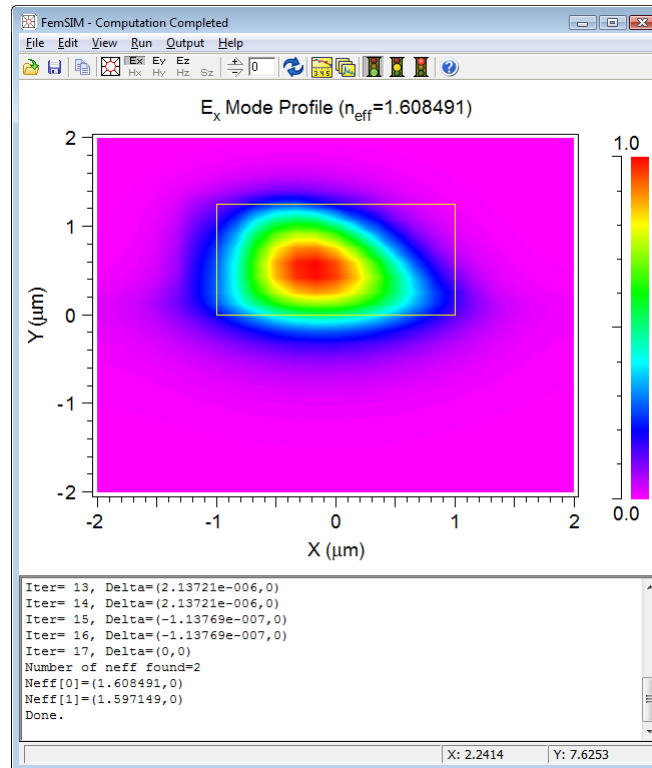


Figure 1-3: The FemSIM Simulation dialog where simulation results are displayed.

The simulation program runs as a background task so that you can switch away to another program, or back to the main program to edit another circuit or begin another simulation. When the simulation is finished, the title bar and status bar of the simulation program indicates “Computation Completed”, and displays the coordinates of the cursor for reference.

When viewing a mode profile, right-clicking on an axis will display the cross-section of the mode at that point. Right clicking again will return the view to the contour plot. The mouse can also be used to zoom in by dragging a box around the area of interest. Right clicking will zoom back out.

Step 5: Accessing Results

FemSIM saves simulation results to the same directory that the design file (.ind) was opened from. To view or print saved simulation results, click the WinPLOT icon on the top toolbar of the CAD window (the yellow graph icon), and select the desired graph from the dialog that is presented. The graph types are listed in the **Files of type** field. RSoft’s DataBROWSER, which can also be opened by an icon in the top toolbar of the CAD, can also be used to quickly view several graphs.

Each output is contained in two ASCII files: one file contains the raw data, and the other contains WinPLOT plotting commands. The file chosen via the above method is the WinPLOT command file. To access the raw data, open the data file with any text editor.

1.D.2. CLI Operation

As an alternative to using the main RSoft CAD program to initiate a FemSIM simulation it can sometimes be useful to run the simulation program from a CLI. This is particularly true when a series of simulations needs to be run in which some parameter is to be varied, or when incorporating FemSIM into a larger scripting environment.

The syntax to run a design file named `mydesign.ind` and to set the **Output Prefix** to `run1` is simple:

```
femsim mydesign.ind prefix=run1
```

When operating FemSIM via the command line, the user can access the exact same feature set as if they were running FemSIM through the GUI. All simulation options within the GUI are controlled by variables. A list of these variables can be found in [Appendix C](#).

For a complete discussion of the command line syntax, please see Chapter 10 in the CAD manual.

1.D.3. Log Files

A log file which contains information about each simulation is saved, useful for debugging or other purposes. By default, this file is named `log.txt` and appended to each time a simulation is run in a particular directory. There are several options to control this:

- To create a new file each time the simulation runs (i.e., delete the old file), use the Menu option in the RSoft CAD View/Clear Log/Always.
- The variable `rsoft_logbyprefix` can be used to name the log file using the simulation prefix. A value of 0 uses the default `log.txt` name, a value of 1 uses the simulation prefix (i.e. `<prefix>_log.txt`) and appends data for subsequent simulations, and a value of 2 uses the simulation prefix and creates a new file for each simulation

1.E. Example Files

The subdirectory `<rsoft_dir>\examples\FemSIM` contains example files for FemSIM. These files have an extension `*.ind` (named for the refractive index distribution which describes the circuit). In addition to these examples, there is a `Tutorials` subdirectory which contains the index files described in the Tutorial chapters of this manual.

1.F. README Files

The latest product information can be found in the README files located in the `/readme` subdirectory of the installation directory. These files contain important last minute information about RSoft software that is not contained in other documentation, including new or improved features and options.

1.G. Product Support, Upgrades, and Resources

The software includes technical support, product updates, and other resources. Software sales and technical support worldwide is handled by Synopsys, except in those countries where we have local representatives whom you can contact directly.

Visit <https://www.synopsys.com/optical-solutions/support.html> for information about how to access training resources, knowledge articles, product downloads/updates, documentation, how to open support cases, and other resources. Many of these resources are in Synopsys' SolvNetPlus portal (<https://solvnetplus.synopsys.com/>), eligible users can find information about how to create an account at the main support link above.

2

Background

This chapter provides technical information and references on the simulation methods used in FemSIM. This material may be skipped initially; however it introduces terminology and notation that is important for understanding other discussions in the manual.

2.A. Overview

The Finite Element Method (FEM) method is generally advantageous in complex geometries and/or high index contrast materials. Presented here is the basic theoretical background for the FEM for Electromagnetics as implemented in FemSIM. It is a full-vector implementation for both propagating and leaky waveguide modes, and cavity modes for 1D and 2D Cartesian cross-sections and 3D BORs (bodies of revolution) in cylindrical coordinates. The method uses state-of-the-art Hybrid Edge/Node Elements that allow for efficient separation of the spurious modes from the physical ones. First and second order interpolant basis are provided for both rectangles and triangles. PEC (perfect electrical conductor) or PML (perfectly matched layer) boundary conditions may be selected independently for each direction. It may be used to find all the modes or a small group of modes about a given wavelength.

2.B. The Electromagnetic Boundary Value Problem

We begin with the source-free time-harmonic form of the vector wave equation in an arbitrary, anisotropic, lossy media [1,2],

$$\nabla \times \left(\frac{1}{\bar{s}} \cdot (\nabla \times \mathbf{E}) \right) - k_0^2 \bar{\epsilon}_r \mathbf{E} = 0 \quad (1)$$

subject to vanishing field boundary conditions at the domain edges.

$$\hat{n} \times \mathbf{E} = 0 \quad (2)$$

The complex diagonal tensors, $\bar{s}$ and $\bar{\epsilon}$, represent coordinate-stretching and the dielectric material, respectively. Throughout the domain $\bar{s}$ is the identity tensor, but in the boundary layer it has the following form

$$\begin{aligned} \bar{s} &= \left(\frac{s_y s_z}{s_x} \right) \hat{x}\hat{x} + \left(\frac{s_x s_z}{s_y} \right) \hat{y}\hat{y} + \left(\frac{s_x s_y}{s_z} \right) \hat{z}\hat{z} \\ s_{\alpha=x,y,z} &= 1 - j \left(\frac{\alpha - L}{L} \right) \delta_{\max} \end{aligned} \quad (3)$$

where $\delta_{\max}$ is the loss tangent, α is the distance from the edge, and L is the thickness of this layer, known as the perfectly matched layer (PML). The tensor elements in the PML are matched to those in the rest of the domain according the prescription

$$\bar{\epsilon}_2 = \epsilon_1 \bar{s} \quad (4)$$

to produce arbitrarily small reflections at the PML interface for all frequencies and angles of incidence. The PML is terminated at the domain edge with a perfect electrical conductor (PEC) boundary condition [3].

2.C. The Solution via Finite Elements

The Finite Element Method (FEM) does not solve the boundary value problem (1-3) directly, but rather a related one based on a variational expression, or functional, constructed from the

operator of the differential equation (1). This functional in two dimensions, over domain A, is given by [4]

$$F(\mathbf{E}) = \iint_A \left[(\nabla \times \mathbf{E})^* \cdot \frac{1}{S} \cdot (\nabla \times \mathbf{E}) - k_0^2 \mathbf{E}^* \cdot \bar{\epsilon} \cdot \mathbf{E} \right] dA \quad (5)$$

For propagating and leaky modes [5], a separable field ansatz, $E(x,y,z) = E(x,y)\exp(-j\beta z)$, where β is the modal propagation constant along z .

Instead of finding an expansion basis over the entire domain, which can be difficult in general, the Finite Element Method subdivides the domain into a collection of elements, for which a simple basis can be defined. This basis vanishes outside the element, so that the final solution is just a summation over the solutions of all the elements. For hybrid Node/Edge FEM, the transverse components are expanded in a vector (edge element) basis,

$$\mathbf{E}_T(x, y)e^{-j\beta z} = \sum_{i=1}^n \mathbf{N}_i E_{Ti} = \sum_{i=1}^n \{U\hat{x} + V\hat{y}\} E_{Ti} \quad (6)$$

where the E_{Ti} are the values of the field along each edge. The longitudinal component (perpendicular to the plane of the element) is represented by a scalar (node element) basis,

$$E_z(x, y)e^{-j\beta z} = \sum_{i=1}^n N_i E_{zi} \quad (7)$$

where the E_{zi} are the values of the field at each node. The basis dimension, n , depends on the geometry of the element and the order of the interpolation. Numerous interpolant basis have been described in the literature [6-9], and will not be repeated here, though most are based on the edge element proposed by Nedelec [9].

Lastly, the functional is minimized according to (8), yielding a matrix eigenvalue equation with β^2 as the eigenvalue, and the field components at the nodes and edges as the eigenvector.

$$\frac{\partial F}{\partial E_i} = 0 \quad (8)$$

A similar formulation can be made for cavity modes [10,11].

2.D. References

- 1 J. Jin, *The Finite Element Method In Electromagnetics*, 2nd Ed., John Wiley & Sons, New York, 2002.
- 2 K. Saitoh and M. Koshiba, "Full-Vectorial Imaginary-Distance Beam Propagation Method Based on a Finite Element Scheme: Application to Photonic Crystal Fibers," *IEEE JQE*, **38-7**, p.927 (2002).
- 3 D. Schulz, C. Glingener, M. Bludszweit, and E. Voges, "Mixed Finite Element Beam Propagation Method," *IEEE JLT*, **16-7**, p.1336 (1998).
- 4 M. Koshiba, S. Maruyama, and K. Hirayama, "A Vector Finite Element Method With the High-Order Mixed-Interpolation-Type Triangular Elements for Optical Waveguiding Problems," *IEEE JLT*, **12-3**, p.495 (1994).
- 5 Y. Tsuji and M. Koshiba, "Guided-Mode and Leaky-Mode Analysis by Imaginary Distance Beam Propagation Method Based on Finite Element Scheme," *IEEE JLT*, **18-4**, p.618 (2000).
- 6 Y. Tsuji, M. Koshiba, and T. Shiraishi, "Finite Element Beam Propagation Method for Three-Dimensional Optical Waveguide Structures", *IEEE JLT*, **15-9**, p. 1728 (1997).
- 7 M. Koshiba and Y. Tsuji, "Curvilinear Hybrid Edge/Nodal Elements with Triangular Shape for Guided-Wave Problems," *IEEE JLT*, **18-5**, p.737 (2000).
- 8 A.F. Peterson, "Vector Finite Element Formulation for Scattering from Two-Dimensional Heterogeneous Bodies", *IEEE Trans. On Antennas and Propagation*, **43-3**, p. 357 (1994).
- 9 J. C. Nédélec, "Mixed finite elements in R^3 ," *Numerische Mathematik*, **35**, p. 315 (1980).
- 10 A. Greenwood and J. Jin, "A Novel Efficient Algorithm for Scattering from a Complex BOR Using Mixed Finite Elements and Cylindrical PML," *IEEE Trans. on Antennas and Propagation*, **47-4**, p.620 (1999).
- 11 J-F. Lee, M. Wilkins, and R. Mittra, "Finite-Element Analysis of Axisymmetric Cavity Resonator Using a Hybrid Edge Element Technique," *IEEE Trans. on Microwave Theory and Techniques*, **41-11**, p. 1981 (1993).

3

Using FemSIM

This chapter describes FemSIM's mesh definition, simulation options, display options, and output options.

3.A. The Mesh & Boundary Conditions

FemSIM solves for the modes of a 1D or 2D profile on a non-uniform hybrid mesh which is automatically calculated based on the structure defined in the RSoft CAD as well as a variety of simulation options. It is composed of a mix of triangular and rectangular shaped elements with constant refractive index which cover the computational domain, including the PML.

The parameters which control the mesh are the computational domain, grid sizes, boundary conditions, and other mesh properties.

FemSIM can be time and memory intensive. For this reason the default grid size (1 μm) is coarse and the default PML width is zero. It is recommended to start with these settings and change them slowly as the time and memory requirements grow nonlinearly with the number of mesh elements. In some cases, it can be beneficial to ignore one or more components from the calculation of the simulation parameter defaults. This can be done with the `skip_est` component tag discussed in Chapter 6 of the RSoft CAD manual.

3.A.1. Computational Domain & Grid

The primary controls for the mesh are the computational domain and the grid sizes. The upper part of the FemSIM Simulation Parameters dialog allows the user to specify this information for each spatial direction.

	X			Y			Z		
	Current Value	Default Value	Use Defs	Current Value	Default Value	Use Defs	Current Value	Default Value	Use Defs
Domain Min:	$-2 \cdot H \cdot \text{gric}$	-2	☐	$-2 \cdot H \cdot \text{gric}$	-1	☐	0	0	☒
Domain Max:	$2 \cdot H \cdot \text{gric}$	2	☐	$2 \cdot H \cdot \text{gric}$	2.3333	☐	1000	1000	☒
Grid Size:	$1/\text{NG}$	0.5	☐	$1/\text{NG}$	0.5	☐	0.5	0.5	☒
PML Width:	0			0			0		

Advanced Mesh Control
☒ Enable Nonuniform Mesh Options ... View Mesh

Figure 3-1: The section of the FemSIM Simulation Parameters dialog where computation domain, grid, and PML information is set. The PML specifications are discussed in the next section.

The coordinate(s) used depend on type of modes solved for: 1D modes are solved for on the X axis at a fixed Z position and 2D modes are solved for in the XY plane at a fixed Z position. For a 1D mode, a 2D CAD file should be used; for a 2D mode, a 3D CAD is required. To change the dimensionality of the CAD file, set the **Model Dimension** option in the Global Settings dialog.

Each spatial direction can be defined separately, and are (all units are in μm):

Domain Min and Domain Max

These options set the domain limits to be used during the simulation. For accurate results, the simulation domain must include all parts of the structure that you want to simulate, including any evanescent fields outside of the structure. It shouldn't be excessively large as nothing can be gained by simulating parts of the domain where no field is present. The Z position at which the mode is calculated can be set by the Z **Domain Min**.

Grid Size

These options set the grid size for each dimension and are the primary option to control the mesh generation. Since FemSIM uses a non-uniform and irregular mesh the values specified here represent the 'average' or 'nominal' grid size for the simulation.

There are several options which control the irregularity and non-uniformity of the mesh including interface alignment and the use of triangular mesh elements. These options are discussed in the next section.

3.A.2. Mesh Options

The mesh options are set via the Advanced Mesh Controls dialog which can be opened by clicking the **Mesh Options...** button in the FemSIM Simulation Parameters dialog.

The majority of the options described in the Advanced Mesh Controls dialog are described in Chapter 9 of the RSoft CAD manual. This section describes several notes about using the options with FemSIM, as well as some options which only FemSIM supports. For a complete understanding of FemSIM's grid/mesh options, both this section and Chapter 9 in the RSoft CAD manual must be read.

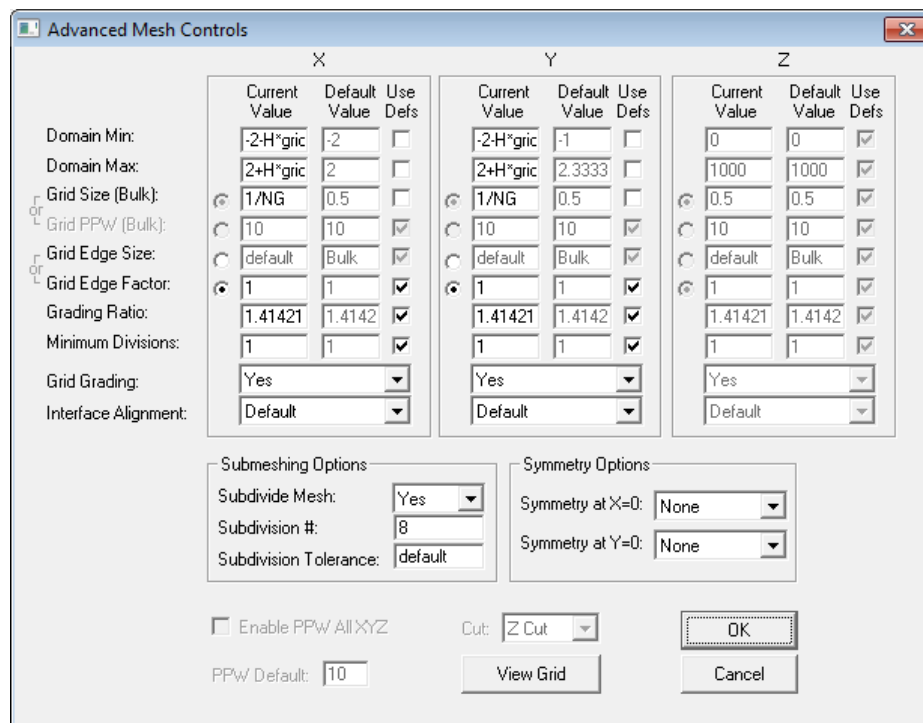


Figure 3-2: The FemSIM Advanced Mesh Control dialog. The options in the top portion of the dialog are described in Chapter 9 of the CAD manual. The options in the bottom portion are described in this section. The points-per-wavelength (PPW) options are currently not enabled for FemSIM.

The grid/mesh options are:

FemSIM Submeshing Options

FemSIM supports the subdivision of each grid cell into triangular elements to represent the structure. These options are:

Subdivide Mesh

In order to better conform to the structure geometry, FemSIM can use an irregular mesh composed of triangular segments. This option enables the use of this type of mesh, and is the second most important option after the grid size for non-rectangular structures. As a general rule this option should always be enabled for these structures. Its choices are:

- *Default*

This choice automatically enables this option when the CAD global setting **3D Structure Type** is set to *Fiber*. For all other structure types, it is automatically disabled. This is the default setting for this parameter.

- *No*

This choice disables the use of an irregular mesh.

- *Yes*

This choice enables the use of an irregular mesh.

Irregular meshing with FemSIM is created within the base grid defined by both the **Grid Size** and **Interface Alignment** settings. Within each element of this base grid, FemSIM will detect curves and angled interfaces between material regions, i.e. regions with the same refractive index. This is done by subdividing the base grid by the factor given by the value of the **Subdivision #** field. Then, once the interfaces are found, triangular mesh elements are used to create the mesh. Furthermore, FemSIM will force an element edge to the base grid if the distance from the element edge to the base grid is less than the value of the **Subdivision Tolerance** field.

Note that index averaging at material interfaces occurs when an irregular mesh is not used.

Subdivision #

This number indicates the number of subdivisions to make within the base grid in order to detect material interfaces. The default value is 8.

Subdivision Tolerance

This number represents the tolerance to force an element edge to the base grid. The default is the keyword `default`, and is automatically calculated by the software. This parameter usually does not have to be changed.

FemSIM Symmetry Options

FemSIM supports convenient ways to define symmetry:

Symmetry at X=0 and Symmetry at Y=0

These options are convenient ways to apply symmetric boundary conditions and reduce the required simulation time and memory considerably if the structure exhibits symmetry at X=0

and/or $Y=0$ plane. They automatically crop the simulation domain and PML at the symmetry axis so that only the positive coordinate region is used, and then apply the appropriate boundary conditions to yield the requested symmetry of the fields. The choices are:

- *None*

This choice disables any use of symmetry, and is the default value.

- *Symmetric*

This choice enables symmetric calculations about the indicated axis.

- *Antisymmetric*

This choice enables antisymmetric calculations about the indicated axis.

Additional Notes about FemSIM Interface Alignment

A discussion of the Interface Alignment Option is described in Chapter 9 of the RSoft CAD manual. However, note that when **Interface Alignment** is set to *Default*, interface alignment will always occur. For curved or angled geometries, elements near the interface between material regions may straddle the interface, and an average index of refraction is used unless the **Subdivide Mesh** option is used.

3.A.3. Boundary Conditions

FemSIM needs to define how the field is treated at the boundary of the simulation domain. The supported boundary conditions are:

PML

A PML (Perfectly Matched Layer) boundary consists of several mesh points which are added to the edge of the domain. Its width is defined by the **PML Width** field in the FemSIM Simulation Parameters dialog. The use of a PML is enabled by setting a non-zero **PML Width**. A good general rule of thumb is to use 10 grid points for the PML.

When using PML you should follow these guidelines:

- *Guidelines for Solving for Guided Modes*

PML should be used whenever the field does not go to 0 at the edge of the simulation domain. Note that even if PML is used, it is desirable for the simulation domain to be sufficiently large for the field to decay to near 0 at the edge of the domain whenever the modes calculated by FemSIM are to be used as a launch field in another tool.

- *Guidelines for Solving for Leaky Modes*

When solving for Leaky Modes, PML should always be used, and the **Allow Leaky Modes** option in the Advanced Options dialog ([Section 3.F.1](#)) must be enabled. Note that when this option is used, and the actual mode field is evanescent instead of leaky on one or more

boundaries, it is very important to either position the domain so that any evanescent waves are almost zero at the boundary, or use a sufficiently large PML, or a combination of both.

When solving for leaky modes, the **PML Width** must be large enough so that the PML functions correctly. A thicker PML is always better, but at the cost of computer memory and time. A simple way to determine if your PML is sufficiently thick is to look for ripples in the leaky part of the mode amplitude. The default real/imaginary output always shows sinusoidal oscillation for leaky waves (Fig. 3-3a) but the amplitude should not. Any reflection from a boundary will interfere with the leaky wave and form ripples that can be seen in the amplitude (Fig 3-3b).

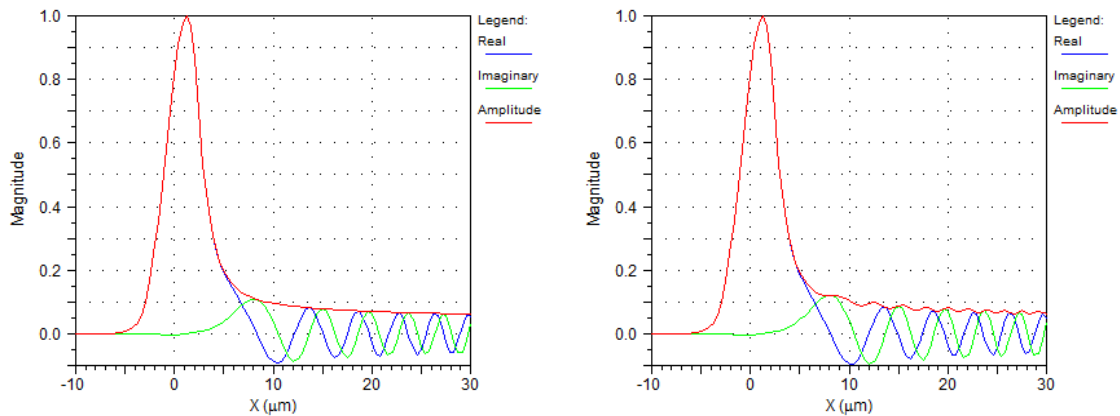


Figure 3-3: An example leaky mode profile a) with no reflection from PML, and b) with reflection from PML. The reflection appears as ripples in the amplitude (red) curve.

Symmetric (Neumann)

This condition sets the normal derivative of the field to zero at the boundary and implies a symmetric field pattern. To use this boundary condition, use the symmetry options in the FEM Mesh Parameters dialog described above. This option can be set for X and Y independently.

When solving for transverse modes, it is important to consider the effect of the symmetry on the polarization. The above symmetric/antisymmetric conditions apply to only one of the field components. This field component is defined via the variable `polarization` in the symbol table. A value of 0 (default) means the field symmetry refers to E_x , and E_y has the opposite symmetry. A value of 1 means the field symmetry refers to E_y , and E_x has the opposite symmetry.

AntiSymmetric (Dirichlet)

This condition sets the field to zero at the boundary and implies an antisymmetric field pattern. To use this boundary condition, use the symmetry options in the FEM Mesh Parameters dialog

described above. This option can be set for X and Y independently. See note above on polarization.

3.A.4. Viewing the Mesh

Before performing a simulation, it is important to ensure that the mesh accurately reflects the structure. The simulation mesh can be viewed using the **View Mesh** button in the FemSIM Simulation Parameters dialog. The mesh will be displayed, along with any PML. If the mesh is not acceptable, then the mesh parameters should be modified until a satisfactory mesh is achieved.

3.B. Simulation Options

FemSIM will find any guided, leaky, and radiation modes (or cladding modes if the PML width is zero) as may be present. It will find modes of all polarizations simultaneously.

If symmetry options are used, only modes which have the specified symmetry will be found. Multiple simulations would be required to find modes with other symmetry.

3.B.1. Mode Options

The options which control the types of modes to be found are located in the FEM Mode Options dialog and can be opened by clicking the Modes... button in the FemSIM Simulation Parameters dialog.

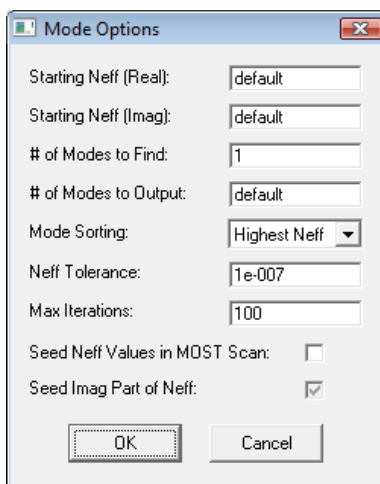


Figure 3-4: The FEM Mode Options dialog.

The options in this dialog are:

Starting Neff (Real) and Starting Neff (imag)

These parameters set the real and imaginary value of the starting effective index. The program will find the requested number of modes closest to the value given. The default real value corresponds to the highest index in the simulation domain, so therefore the program searches for modes with the highest effective indices. The default imaginary value is 0 which is useful for finding guided modes. For a cavity mode calculation where the effective index represents the frequency, the default values are 0 so that the program tries to find the lowest frequency modes.

It is recommended and usually sufficient to only set the real part of this parameter. Without good knowledge about the imaginary part you may unnecessarily limit the search for modes.

of Modes to Find

This field sets the number of modes to find and has a default value of 1. The eigenvalue solver can generally find any number of modes without assistance. The time and memory increase only slowly with the number of requested modes, thus, for example, finding two modes is nearly as easy as finding one.

For guided modes, this parameter rarely needs to be changed unless higher order modes are desired. For leaky modes, however, there may be many modes with an effective index higher than the one of interest, and so it may be better to select a starting point using the **Starting Neff** option rather than to select a large number of modes to solve for.

of Modes to Output

This field sets the number of modes to output. This is useful when used in conjunction with the **Mode Sorting** option as only modes that meet the desired criteria will be saved. The default value is the keyword `default`, which indicates that all modes found should be output.

Mode Sorting

This field allows the modes found to be sorted by different criteria. Possible choices are:

- *None*
This choice indicates that the modes should not be sorted; they will be saved in the order of their proximity to the **Starting Neff** value.
- *Highest Neff*
This choice indicates that the modes should be sorted by the real part of their effective indices from highest to lowest values and is useful when solving for guided modes. When the default value of **Starting Neff** is used, this choice reproduces the behavior of the RSoft BPM-based mode solvers: The fundamental mode will be first ($m=0$), and higher order modes will follow ($m=1$, $m=2$, etc.).
- *Lowest Loss*

This choice indicates that the modes should be sorted by the imaginary part of their effective indices from highest to lowest values and is useful when solving for leaky modes.

Neff Tolerance and Max Iterations

These options control how the FemSIM calculation converges: **Neff Tolerance** controls the tolerance for effective index convergence, and **Max Iterations** sets the maximum number of iterations to use.

Note that the convergence rate is also related to the number of modes being solved for. Specifically, the convergence rate can be increased by solving for more modes than required: increase the **# of Modes to Find** option, and use the **# of Modes to Output** option to ignore these extra modes. Alternatively, if you want to maintain a set convergence rate for multiple simulations, solve for the same number of modes so that the results are easier to compare.

Seed Neff Values in MOST Scan and Seed Imag Part of Neff

These options enable the seeding of the neff values during a MOST scan. This is useful as the neff values can shift during a scan. When this option is set, the user-defined starting neff values are used for the first point in the scan, while subsequent points use the neff of the first mode found for the previous point. Note that this option cannot be used with MOST clustering as the modes need to be found in a serial fashion.

3.B.2. Radial Calculations

FemSIM can exploit the azimuthal symmetry of cylindrical structures by performing a radial calculation in which the coordinates (x,y) map to (r,z). This is primarily useful for VCSEL cavity modes. The **Radial Calculation** and **Azimuthal Mode #** options in the Advanced options dialog enable and set the radial mode number respectively. The mode number corresponds to the m in the HE_{mn} type designation, which differs by 1 from the LP_{mn} designation. For example, the true fundamental mode of a fiber is the HE_{11} mode, which is approximated by the LP_{01} mode.

This setting corresponds to the azimuthal behavior of the longitudinal field components, which is typical for full-vector treatments of structures with circular symmetry. The azimuthal behavior of the transverse components differs by 1. Thus, $m=1$ corresponds to the fundamental, and is the default value and produces a Gaussian-like intensity pattern for the power flow. The $m=0$ case corresponds to the pure TE and TM waves (which do not always exist) and produces a doughnut-shaped intensity pattern.

3.B.3. Polarization

It is important to understand how FemSIM treats polarization. The FemSIM Simulation options dialog has a **Polarization** control which is active for 2D simulations (1D mode profiles) and inactive for 3D simulations (2D mode profiles). Please see [Section 1.B](#) for the polarization convention used by FemSIM.

- In a 3D simulation (2D mode profiles) Maxwell's equations are not completely separable into TE or TM versions – all modes have at least some TE and/or TM component. As such, FemSIM solves for both TE and TM modes together. The mode results can be filtered by the normalized peak values, or through the `fem_sort_pol` and `fem_show_pol` options described below.

The symmetry options described in [Section 3.A.3](#) can be applied to the boundaries (if applicable) to essentially solve for a specific polarization. It is important to note that as described in [Section 3.A.3](#), the value of the `polarization` variable (which can be visually seen by the **Polarization** control in the FemSIM Simulation Properties dialog) is used to determine which field component the symmetry is applied to and therefore indirectly controls the polarization of the modes found.

- In a 2D simulation (1D mode profiles), Maxwell's equations are separable and so TE and TM modes can be found separately. By default, FemSIM will solve for TE modes. Control over the polarization can be done via the **Polarization** option in the FemSIM Simulation Properties dialog.
- The `fem_sort_pol` and `fem_show_pol` symbols can be used to indicate which modes are 'TE' or 'TM'. These symbols affect the information shown in the info windows during a simulation, and how the modes are ordered or sorted:
 - *Mode Polarization:* The `fem_show_pol` symbol enables the display of polarization information for each mode in the info window. Set to 0 to disable showing polarization info (default), to 1 to show TE or TM, or 2 to show TE and TM and the polarization measure. The polarization measure is a number between 0 and 1, a mode in the range (0,0.5) is TE, the range (0.5,1) is TM. A value of 0 is pure TE, a value of 1 is pure TM.
 - *Mode Sorting:* The `fem_sort_pol` symbol sets how modes are sorted, in conjunction with the **Mode Sorting** option. A value of 1 interleaves TE and TM modes (TE first), a value of 4 interleaves TM and TE modes (TM first), a value of 2 puts all TE modes first in the list, and a value of 3 puts all TM modes first in the list.
- The `/vectc` WinPLOT option can be used to analyze polarization using two field components, say E_x and E_y . See [Section 3.E](#).

3.B.4. Mode Dispersion Calculation

The **Dispersion** option in the bottom of the Mode Calculation dialog enables the automatic output of dispersive properties, including group index, dispersion, and real/imaginary effective indices, all as a function of wavelength. Click the **Opts...** button to open the

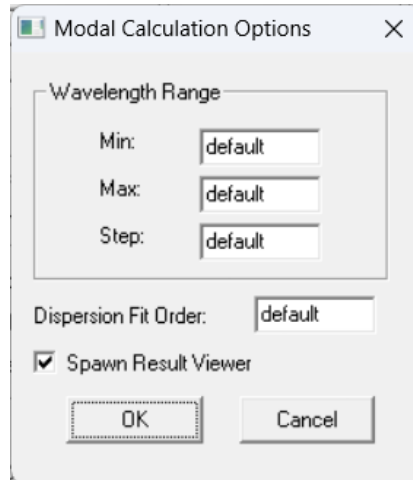


Figure 3-5: The Mode Dispersion Options dialog.

The **Wavelength Range** can be set, the default corresponds to 33 points $\pm 33\%$ of the **Free Space Wavelength**. The **Dispersion Fit Order** controls the order of the polynomial that is used to fit the dispersion data. Like fields elsewhere in the RSoft CAD, the current (or default) value of the parameter, can be viewed by holding the **CTRL** key and double-clicking on the option. The **Spawn Result Viewer** will open the S-Matrix Viewer to display the dispersion properties. More complicated scans can of course be done manually with MOST.

Clicking **OK** will run the wavelength scan via MOST. The plotted quantity can be controlled through the drop-down named **Quantity**, with the options being **Neff (real)**, **Neff (imag)**, **Ngrouop**, and **Dispersion**. Sample results are shown here:

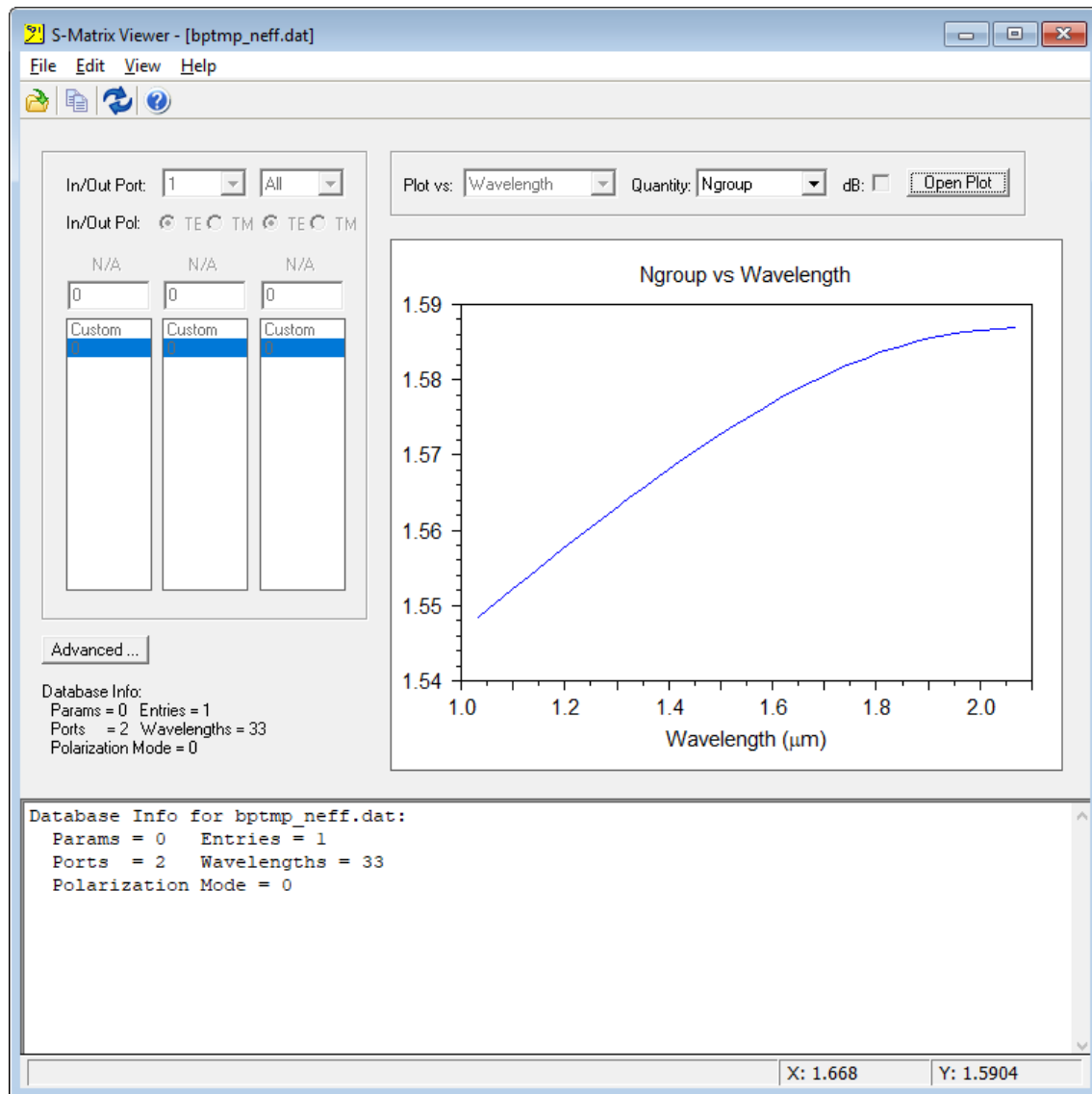


Figure 3-6: Sample dispersion results.

3.C. Display Options

The display options for FemSIM are controlled via the FEM Display Options dialog which can be opened via the **Display...** button in the FemSIM Simulation Parameters dialog.

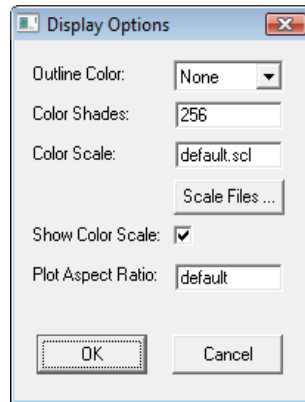


Figure 3-7: The FEM Display Options dialog.

The display options are:

Outline Color

This option indicates whether or not to draw an outline of the structure on top of the field display, and if so, what color scheme to use.

Color Shades

This option indicates the number of color shades to be used to render the displayed field.

Color Scale

This option sets the color scale to be used when rendering the displayed modes in the simulation window. Scale files can either be entered by name, or chosen via the **Scale Files...** button.

Show Color Scale

This option indicates that the color scale should be displayed in the simulation window.

Plot Aspect Ratio

Mode and mesh plots do not normally display in a 1:1 aspect ratio, and this option sets the aspect ratio of the plot to 1 or any other desired value. For stored plots generated by previous simulations, or generated without having set this option, the aspect ratio can be changed in WinPLOT under Options/Axes/Aspect Ratio... or via the `/aspect` option. Please consult the WinPLOT manual for more information.

3.D. Using Monitors to Make Measurements

FemSIM includes the ability to make measurements on computed modes such as overlap integrals, confinement factors, and spot-sizes.

These measurements are done using the same type of monitors that BeamPROP uses. FemSIM uses a subset of all the available monitor options which are described here.

3.D.1. Using Monitors in FemSIM

The use of monitors in FemSIM requires two ingredients: a pathway to determine the position and size of the region where a particular measurement is to be made, and the actual monitor which defines the type of measurement to be made. These monitors can be output from individual simulations and can also be scanned over using MOST.

Defining Pathways

A pathway defines critical geometric information for FemSIM monitors. In the case of overlap monitors, it defines the origin at which the test field and computed mode field are aligned before the overlap is taken. In the case of confinement factors, the pathway defines the area of interest. In the case of spot-size, a pathway is not required.

The definition of pathways is covered in detail in the CAD manual.

Creating Monitors

Monitors can be created by clicking the **Edit Monitors** button in the left toolbar. This opens the Monitors dialog shown below

Pathway Monitors

Monitor Number: 1 New << >> Delete

Monitor Options

Type: Slab Mode Power Pathway: 1 << >>
Tilt: No Background N: default
Component: Major Cover N: default
Mode: 0 Delta N: default
Mode Radial: 1 Width: default
Data File Spec: (Prefix/Ext) ... Custom ... Height: default
Normalization: Input Power Position X: default
Position Y: default
Neff: default
Layers:
Shape: Default

OK Cancel Symbols ...

Figure 3-8: The Monitors dialog where monitors are defined.

Rather than document each feature in this dialog, the common tasks are outlined in the next section(s) with detailed instructions. Note that the **Monitor Type** option is an editable drop-down list which, in addition to being set via the drop-down can also be a Symbol Table expression. See Chapter 3 in the RSoft CAD manual for details about how to use this feature. The built-in values for this option are described in [Section 3.D.6](#).

Accessing Monitor Results

The monitor results are displayed in the simulation window as well as in an ASCII data file `<prefix>.mon`, where `<prefix>` is the **Output Prefix**. This file has one line per mode with different monitors in different columns.

3.D.2. Computing Overlap Integrals

This type of monitor computes an overlap integral of a test field with the computed modes. Two types of test fields are currently supported, and correspond to monitor **Type** choices:

- *Gaussian*

This monitor type computes the overlap integral of a Gaussian test field with the computed modes. The **X Position** and **Y Position** of the Gaussian as well as the **Width** and **Height** can be set in the in right column. The default values for these parameters correspond to the selected pathway.

- *File Power*

This monitor type computes the overlap integral of test data contained in one or more files with the transverse components of the computed modes. The **X Position** and **Y Position** of the test field can be set in the right column. The default values for these parameters correspond to the selected pathway. The data file(s) to be used can be specified via the **Data File Spec** option. These files can be automatically set by a file **Prefix** and **Extension** or manually using the controls provided by clicking **Custom....**

- *Automatic Method:*

The automatic method provides a simple way to define the monitor data files used for individual components by setting a file prefix and extension. The files must have the default output naming style used by RSoft software for field output:

`<prefix>_<comp>.<ext>` where `<comp>` is the field component. For example, if the data files are named `myprefix_ex.m00`, `myprefix_ey.m00`, etc., enter `myprefix` and `m00` for the **Prefix** and **Extension** respectively. These settings can be easily set by choosing one of the field component files via the ‘...’ button. As a convenience, the data files the software will look for based on the prefix and extension specified can be seen by clicking the **Custom** button. Note that this feature should also work with older naming conventions such as `<prefix>.m00`, `<prefix>.n00`, etc, though the new style is preferred.

The software will attempt to locate the files corresponding to the necessary **E** and **H** field components. If any of the necessary component files are missing, the software will attempt to either ignore or calculate the required components. Generally, all necessary components should be made available; in some cases such as low index-contrast simulations, all components may not be necessary.

- *Custom Method:*

To manually define the launch files used for individual components, click the **Custom** button, uncheck **Auto**, and define the necessary components. The major and minor components refer to the X and Y components of the electric **E** and magnetic field **H** respectively.

It is possible to use a variable to define a launch file: set the launch file option to `$<variable>` where `<variable>` is a variable created in the Symbol Table set equal to the name of the desired launch file.

Monitor **Type** choices of *Slab Power*, *Fiber Power*, *Launch Power*, and any *Phase* types are not enabled for FemSIM and will always produce results of 0.

3.D.3. Computing Confinement Factors

This type of monitor computes the confinement factor within the region defined by the pathway's monitor position and dimensions, and corresponds to the Monitor **Type** of *Partial Power*. The results are always normalized to the total power in the mode.

The pathways settings can be overridden by setting the **X Position**, **Y Position**, **Width**, and/or **Height**. For multilayer structures, the **Layers** option specifies a range of layers to be included in the power calculation as either a single layer number or a starting layer number, a hyphen, and an ending layer number (e.g. '2-5').

3.D.4. Computing Mode Spot-Size

These types of monitor computes the mode spot-size in either X or Y, and corresponds to the Monitor **Type** of *1/e Width* and/or *1/e Height*. A pathway is not required for this type of monitor.

3.D.5. Computing Mode Area

The *Effective Area* monitor type is unique to FemSIM and will compute the effective mode area for each mode as:

$$A_{eff} = \frac{\left(\int |E|^2 dA \right)^2}{\int |E|^4 dA}$$

3.D.6. Built-In Values for Editable Monitor Type Option

The value of the built-in **Monitor Type** options are:

Monitor Type List Option	Built-In Value
<i>File Power</i>	MONITOR_FILE_POWER
<i>File Phase</i>	MONITOR_FILE_PHASE
<i>Slab/Fiber Mode Power</i>	MONITOR_WGMODE_POWER
<i>Slab/Fiber Mode Phase</i>	MONITOR_WGMODE_PHASE
<i>Gaussian Power</i>	MONITOR_GAUSS_POWER
<i>Gaussian Phase</i>	MONITOR_GAUSS_PHASE
<i>Launch Power</i>	MONITOR_LAUNCH_POWER
<i>Launch Phase</i>	MONITOR_LAUNCH_PHASE
<i>Partial Power</i>	MONITOR_WG_POWER
<i>Total Power</i>	MONITOR_TOTAL_POWER
<i>Effective Index</i>	MONITOR_FIELD_NEFF
<i>Field 1/e Width</i>	MONITOR_FIELD_WIDTH
<i>Field 1/e Height</i>	MONITOR_FIELD_HEIGHT

3.E. Output Options

The output options for FemSIM are controlled via the Output Options dialog which can be opened via the **Output...** button in the FemSIM Simulation Parameters dialog.

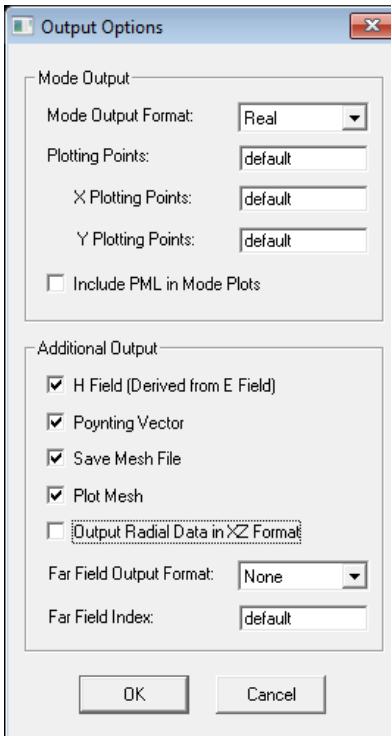


Figure 3-9: The FEM Output Options dialog.

The output options are:

Mode Output Format

Sets the format that FemSIM uses to output modes. Modes (for **E** and **H**) and vector plots of the transverse **E** and **H** fields are saved with the following file names:

Plot File	Data File
<prefix>_ex.p##	<prefix>_ex.m##
<prefix>_ey.p##	<prefix>_ey.m##
<prefix>_ez.p##	<prefix>_ez.m##
<prefix>_hx.p##	<prefix>_hx.m##
<prefix>_hy.p##	<prefix>_hy.m##
<prefix>_hz.p##	<prefix>_hz.m##
<prefix>_sz.p##	<prefix>_sz.m##
<prefix>_et.p##	<prefix>_ex.m##, <prefix>_ey.m##

<prefix>_ht.p## <prefix>_hx.m##, <prefix>_hy.m##

The effective indices for the computed modes are output to the simulation window as well as to the file <prefix>.nef.

An additional plot file <prefix>_et.pcs and/or <prefix>_ht.pcs is also created which produces a vector plot of the mode field with a contour plot of the vector magnitude.

The `/vectc` WinPLOT option can be used to analyze polarization using two field components, say E_x and E_y . See Section 3.E.2 in the WinPLOT manual for details, contact us with questions.

Plotting Points, X Plotting Points, and Y Plotting Points

Sets the number of points used to display and save mode profiles. The **Plotting Points** option controls both the X and Y axes simultaneously; the **X Plotting Points** and **Y Plotting Points** options can be used to control each axis separately.

If all three of these options are set to the keyword `default`, the number of X/Y plotting points used is linked to the **X/Y Grid Size**: a finer grid size will result in a greater number of points in the plots. For coarse grid sizes, FemSIM will use a minimum of 97 points along any axis.

Explicitly setting these options to a specific number will set the number of elements used for the mode profiles (# of points = # of elements + 1); effects of the grid sizes will not be incorporated. Note that when using the mesh symmetry options described in [Section 3.A.2](#), half the specified number of elements are used.

Include PML in Mode Plots

Enables the inclusion of the PML region, if any, into mode plots.

H Field (Derived from E Field)

Enables the output of the **H** field when the **Formulation** is set to *E-Field* in the Advanced Parameters dialog. This option is not available when the **Formulation** is set to *H-Field*.

Poynting Vector

Outputs the Poynting vector for each computed mode. The file names are <prefix>_sz.m## with corresponding plot files <prefix>_sz.p##.

Save Mesh File and Plot Mesh

These options enable the output of the mesh plot.

Output Radial Data in XZ Format

This options enables the output of a XZ cross-section of radial cavity modes. The Y position of the cross-section can be set via the `output_as_3D_ypos` symbol.

Far Field Output Format and Far Field Index

Enables the output of a far-field transformation of mode(s) found. The format can be *None* (default, disables far-field output), *Intensity*, and *Amp/Phase*. The far-field is calculated in a homogenous air region by default. To have the far-field calculated in another medium, set the **Far Field Index** option in the Output Options dialog to the desired refractive index. If this option is set to a value different than the index of the near-field plane, the effect of Fresnel reflection is included.

3.F. Advanced Options

FemSIM has several advanced options which extend the basic simulation capability. These advanced options are set in the Advanced Parameters dialog which can be opened via the **Advanced...** button in the FemSIM Simulation Parameters dialog.

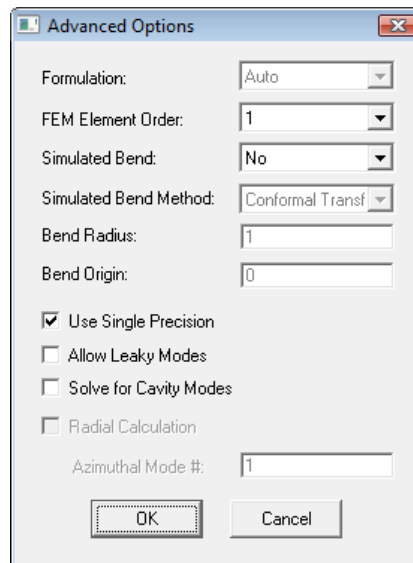


Figure 3-10: The FEM Advanced Options dialog.

3.F.1. Advanced Algorithm Options

The advanced algorithm options are:

Formulation

This sets the field formulation used to solve for modes. The default is based on the electric field (E), but the magnetic field H can be used as well.

FEM Element Order

This field sets the basis function order for an element which affects accuracy. A value of 1 corresponds to first order, 2 corresponds to second order. The default value is 1. Note that there is a tradeoff between the potentially higher accuracy obtained with second order compared to first order and the correspondingly larger time and memory requirements.

Use Single Precision

This option enables the use of single precision numbers during the calculation. Disabling this option uses double precision. This option can also be controlled globally via the RSoft CAD preferences.

Allow Leaky Modes

This option allows leaky modes to be found and is off by default. This option should only be used when finding leaky modes, using it unnecessarily can over-complicate the calculation and make it hard to converge to a mode solution. Also, when using this option, it is necessary to enable the use of PML by changing the **PML Width** to a non-zero value. See [Section 3.A.3](#) for details about using PML to solve for leaky modes.

Solve for Cavity Modes

This option enables the solution of cavity modes propagating in the simulation plane rather than waveguide modes propagating out of the simulation plane. When this option is selected, effective index should be interpreted as the eigenfrequency, f . Note that radially symmetric cavity modes (a VCSEL mode for example) can be output in absolute 3D coordinates by setting `output_as_3D` to 1.

Allow Leaky Modes is automatically enabled for cavity mode calculations.

3.F.2. Advanced Simulated Bend Options

FemSIM supports two simulated bend formulations which allow for the calculation of modes of curved waveguides/fibers:

- *Conformal Transformation:*

This method uses a conformal index mapping technique to solve for bending modes and is described in the Advanced Options chapter in the CAD manual. This method is suitable for geometries where the width of the bending structure is much smaller than the radius.

- *Direct:*

This method solves for bending modes directly in radial coordinates and is described in this section.

Enable the **Simulated Bend** option in the Advanced Options dialog to perform a simulated bend calculation. The **Simulated Bend Method** can then be set to either *Conformal Transformation* or *Direct*.

Note that the local **Simulated Bend** option in the Additional Component Properties dialog for individual components is not used for a FemSIM calculation. Use the setting in the Advanced Options dialog or the Global Index Generation dialog.

Defining the Bending Radius

Care must be taken to ensure that the radius is defined correctly so as to obtain accurate field and effective index results.

By default, the center of the curved geometry is located at $X = -R$ where R is the **Bend Radius**. If the outer edge of the waveguide is to be located at a radius of R , the outer edge must be positioned at $X = 0$ in the CAD. The **Bend Origin** option can be set to the desired offset to alter this behavior. For example, if the center of a waveguide of width W is at $X=0$, then setting **Bend Origin** to $w/2$ will effectively locate the outer edge of the waveguide at radius R .

The effective index output by this method is defined as

$$N_{eff} = \frac{m}{k_0 R_{eff}}$$

where m is a complex ‘angular propagation constant’ defined so that the field varies as $e^{im\theta}$, k_0 is the propagation constant, and R_{eff} is an effective radius which defaults to the **Bend Radius** but can be set by the variable `fem_neff_reff`.

When using this method, there are several ways to set these options to obtain correct results. The preferred method is to set the **Bend Radius** to the actual radius R , and position the waveguide such that the location where the bend radius is R is at $X=0$. For example, if the radius is measured to the edge of the waveguide, position this edge at $X=0$. When used in this configuration, the effective index values output should use the correct radius. Note that it is not necessary to use this option this way as described above, but care should be taken to ensure the correct radius is used to calculate the effective index values as described above.

Output Coordinates

The mode profiles output when using this method will be in radial coordinates: R is along the X axis in the CAD ($R=0$ is at $X=0$), θ is the radial coordinate (defined in the XZ plane in the CAD), and Z is along the Y axis in the CAD.

Additional Tips

Solving for bending modes can sometimes be tricky. Here are some tips for getting better results:

- *Choose good Starting Neff values*

The algorithm is sensitive to the initial guess, or **Starting Neff (Real)** and **Starting Neff (Imag)** values. The equivalent index profile of a bending waveguide is tilted and its modal profile shifts to one side, often with a leaky wave traveling to the same side. According to the weighted index concept, the effective index of the bending mode is higher than its straight counterpart since more of the field is within the ‘higher’ index region. A slightly higher initial guess is therefore more suitable for a bending mode calculation (when compared to the straight mode). How much higher the guess should be depends on many factors such as the modal confinement, the bending radius, etc. If the guess is too low, a lower order mode could be found. If it is too high, a ‘PML mode’ which appears primarily in the PML region, could be found.

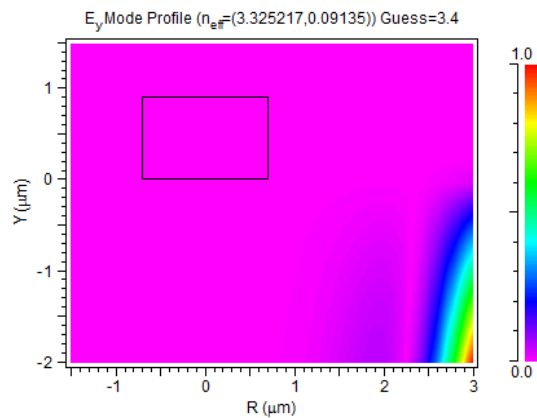


Figure 3-11: A ‘PML Mode’.

- *Use PML*

PML boundary conditions are required to find bending modes. See [Section 3.A.3](#) for more details about PML.

- *Scan from the larger radius to the smaller radius when using the effective index of the straight waveguide as your starting neff value.*

This is advised since the initial guess from the straight waveguide is closer to the effective index of the bending mode at the larger radius. Also, the Increment should be chosen so that it is small enough so that the result from one scan point will be a good initial guess for the next scan point.

- *Do not use MOST Clustering*

MOST can use clustering to parallelize parameter scans. The **Seed Neff Values in MOST Scan** option requires modes to be solved for in a serial fashion and does not work when MOST clustering is used.

3.G. Using Multiple Cores/CPU's (Multi-Threading)

FemSIM is a multi-threaded program that utilizes multiple CPUs and cores within a computer to increase simulation speed. By default, all available cores on a computer will be used. For most uses, this default is sufficient and no additional configuration is necessary.

Multi-threading options are set in the Multi-Threading Settings dialog which can be opened by via the menu option Run/Multi-Threading Settings in the RSoft CAD. The relevant options are:

These are computer-specific settings: changing any of these settings on a particular computer will affect all simulations performed on that computer.

FemSIM # Threads

This is the main multi-threading control and sets the number of threads that FemSIM will use. By default, it is set by the **Default # Threads** option which is automatically set to the number of available cores detected on your computer. Settings this value to 1 effectively disables multi-threading.

Multi-Threading Notes for FemSIM:

- The actual speed increase from using multiple threads depends on many factors including the problem geometry, problem size, and computer used.
- When running multiple simultaneous simulations, it is recommended to ensure that they do not compete for computing power. In such cases you should reduce the **FemSIM # Threads** parameter such that the total number of threads used does not exceed the number of cores available in your computer. Note that it might be faster to run a series of single simulations that utilize all cores rather than multiple simultaneous simulations that use less cores. A simple test run should be sufficient to determine the optimal configuration for a specific problem.
- When running MOST clustered scans (licensed separately), multi-threading is disabled by default. To enable its use, set the number of threads in the MOST dialog in the clustering options section. This option only applies to clustered MOST scans and not to standard MOST scans since only one simulation is performed at once.

3.H. Supported Material Types

These materials can be defined via the Material Editor; see Chapter 8 in the CAD manual for a detailed overview of how to use the Material Editor. This section contains FemSIM-specific notes for these material types.

Linear (Non-Dispersive) Materials

Linear non-dispersive materials are defined by a single real or complex number. See Section 4.B.2 in the CAD manual for details about defining a simple linear material or see Chapter 8 in the CAD manual for defining a linear material via the Material Editor.

Dispersive Linear Materials

Dispersive materials can be defined in a variety of ways including arithmetic expressions, data files, or Sellmeier coefficients. Note that while each FemSIM simulation is at a single wavelength, dispersive properties can be studied by scanning over the simulation wavelength. See [Section 3.B.4](#) for an automated way to do this.

Special Materials

The Material Library in the RSoft CAD contains several built-in materials, including special materials such as Graphene. More details about the Graphene model used, as well as how to set its relevant parameters, can be found in Chapter 8 of the RSoft CAD manual.

3.I. RSoft Utilities

RSoft software includes a number of utilities that can be used for a variety of tasks, including, but not limited to:

These utilities are documented fully in Appendix E of the RSoft CAD manual.

- **Simple Data File Operations** such as cropping/regridding/slicing data files, reordering data columns, and arithmetic operations with data columns/files.
- **Simple Post-Processing** such as far-field calculation, overlap integrals, spot size calculation, dispersion parameter calculation, peak-extraction from spectra, and analysis of FullWAVE state files.
- **Data Conversion** such as bare-matrix to RSoft format, complex matrix conversion, and bitmap image to RSoft data format.

- **Customized Simulation Output:** Specialized simulation outputs that are not automatically created by a simulation can be calculating using these utilities.
- **Field Conversion Interfaces** with Synopsys' CODE V and LightTools products.
- **Field Scattering (BSDF) Interfaces** that allow non-periodic excitation for periodic DiffractMOD and FullWAVE simulations as well as interfaces with Synopsys' LightTools and CODE V products.
- **Interfaces with Other Tools:** Many tools have text inputs and outputs. These utilities can be used to convert data to/from the RSoft format.
- **Custom Post-Processing** by combining several of these utilities to create custom output!

These utilities provide a convenient way to perform and automate common tasks. Many of these utilities can also be spawned directly from MOST, the RSoft scanning tool, making scan post-processing essentially transparent.

4

Basic Tutorials

This chapter contains several basic tutorial examples which illustrate the basic use of the FemSIM simulation engine. These tutorials do not have any corresponding *.ind and/or data files, and are meant to be worked through step-by-step by the user. It is important to perform them in order, even if a particular topic is of most interest to you. It is also recommended that users work through the basic tutorials provided in this chapter before moving to the advanced tutorials in [Chapter 5](#).

Basic Tutorial 1: Simple Waveguide Mode Calculation

This tutorial illustrates basic CAD layout features by creating a new circuit and demonstrates a sample workflow to perform a simulation and obtain simulation results. Our goal will be to layout a simple buried channel waveguide with a cladding index of 1, a core index of 1.5, a width of 2 μm , a height of 1 μm , and then compute its mode at a wavelength of 1.0 μm .

This tutorial will introduce basic concepts about the RSoft CAD and the FemSIM simulation engine that are critical for every user to understand; it is recommended that every new user work through this tutorial. For additional information about the RSoft CAD, see the CAD manual.

CAD Window Basics

We begin by opening up the RSoft CAD window. Windows users can do this via the Start menu and Linux users can issue the command `xbcad` at a command prompt.

The CAD window, shown in Fig. 1, contains the following elements:

- *A menubar near the top of the window*

The menubar allows access to various standard file and editing operations such as opening and saving index files and cutting and pasting waveguide components in a circuit.

- *Two toolbars with icons just below the menubar*

The top two toolbar contains icons for these standard file and editing operations, viewing options, as well as other common functions.

- *A toolbar along the left edge of the window*

The left toolbar contains icons for standard CAD operations such as selection and zooming, as well as icons for choosing the different waveguide components that can be added to the circuit. Additional icons control the simulation and related aspects of the program.

- *A status line at the bottom of the window*

The status line provides information on the current layout mode, along with coordinate display during drawing.

For a complete description of the functionality of the RSoft CAD environment, see the RSoft CAD manual.

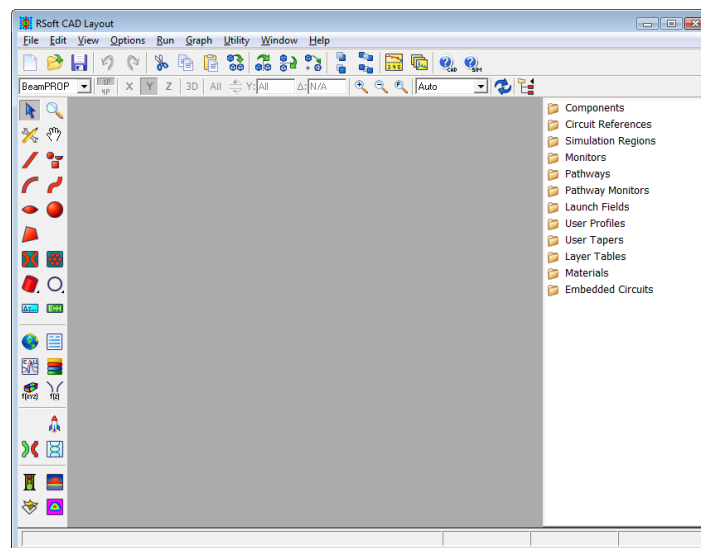


Figure 1: The RSoft CAD program window, showing the menubar at the top, the top and left toolbars, and the status line at the bottom.

Creating a New Circuit

To create a new circuit, click on the **New Circuit** icon in the top toolbar (the leftmost icon on the top toolbar). Alternatively, choose **File/New** from the menu. The startup dialog appears, as shown in Fig. 2. This dialog requests basic information about the circuit to be modeled, including the wavelength of the light, the background refractive index, the default index change between the background material and the structure, and the approximate overall dimensions of the device.

It should be noted that this dialog is also used for RSoft's other packages such as BeamPROP, FullWAVE, BandSOLVE, GratingMOD, and DiffractMOD, and so all fields are not applicable to a FemSIM simulation.

Choosing Initial Settings:

For this simulation, we will use the following settings:

Parameter	Value	Description
Simulation Tool	<i>FemSIM</i>	Sets FemSIM as the simulation engine to be used.
Model Dimension	<i>3D</i>	Creates a 3D structure for use with a 2D mode calculation.
Free Space Wavelength	1.0	Sets the wavelength [μm] for the simulation.
Background Index	1.0	Represents the real refractive index of the background material where no structure is defined. In this case we assume that we have an infinite cladding and so this represents the cladding index.
Index Difference	0.5	Represents the default difference between a component and the background material. In this case this represents the difference between the core index and the cladding index, and so the structure will have an index of 1.5. This is simply the default value to be used for a component; each component can have a unique value if desired.
3D Structure Type	<i>Channel</i>	Sets the profile to be used for 3D components. This choice indicates that, by default, a component should have a rectangular cross-section with a horizontal and vertical size given by the Component Width and Component Height fields respectively.
Component Width	2	Sets the default width of a component [μm]; each

Component Height

1

component can have a unique value if desired.

Sets the default height of a component [μm]; each component can have a unique value if desired.

Once these settings are made, click **OK** to continue. Note that changes can be made to these settings via the Global Settings dialog box which can be opened by clicking the ‘Globe’ icon in the left toolbar.

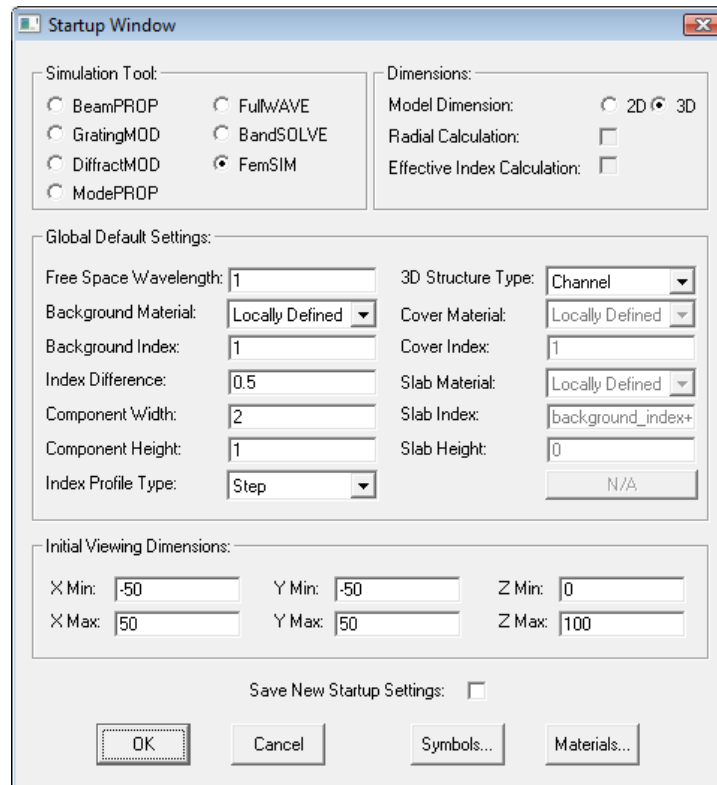


Figure 2: The startup dialog which appears whenever a new circuit is created, and requests basic information about the circuit to be modeled.

The Layout Window:

A layout window is created within the main RSoft CAD program window, as shown in Fig. 3. A coarse coordinate grid is indicated by dots, and the X and Z axes are indicated by light gray lines. When the mouse is moved into the layout window, the cursor becomes a cross-hair, and the coordinate display in the status line indicates the mouse position in real coordinates [μm].

Creating Variables via the Symbol Table

An important concept in the RSoft CAD is the Symbol Table. The Symbol Table, shown in Fig. 4, allows the user to create/modify both built-in and user-defined variables. These variables can

be used to define virtually any parameter within the CAD. Each symbol can be a function of any other symbol, creating a parametric design environment.

The Symbol Table can be opened via the **Edit Symbols** icon in the CAD window. Figure 4 shows the symbol table for this design file which contains several symbols already defined that correspond to the settings made in the **New Circuit** dialog box above.

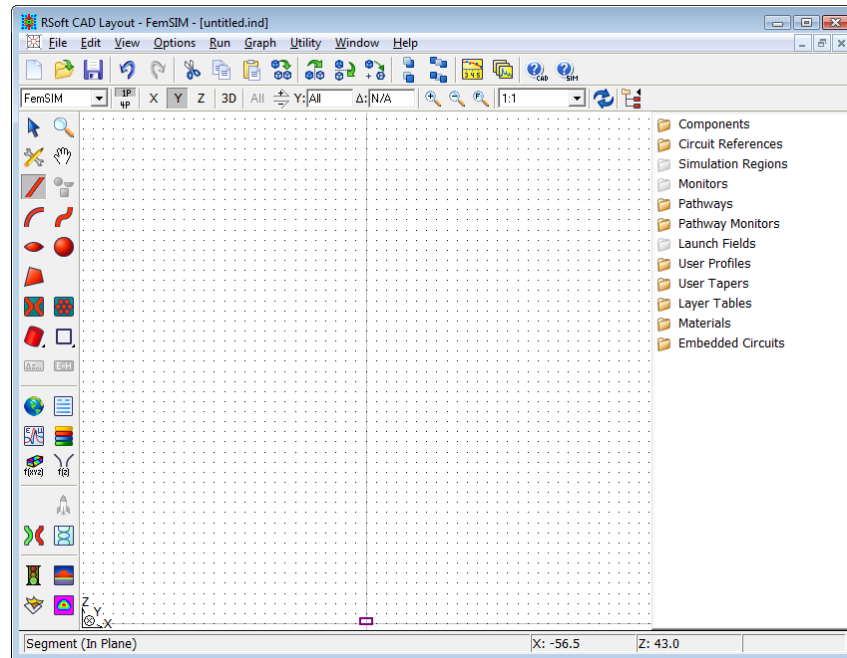


Figure 3: The layout window where waveguide components are added to the circuit.

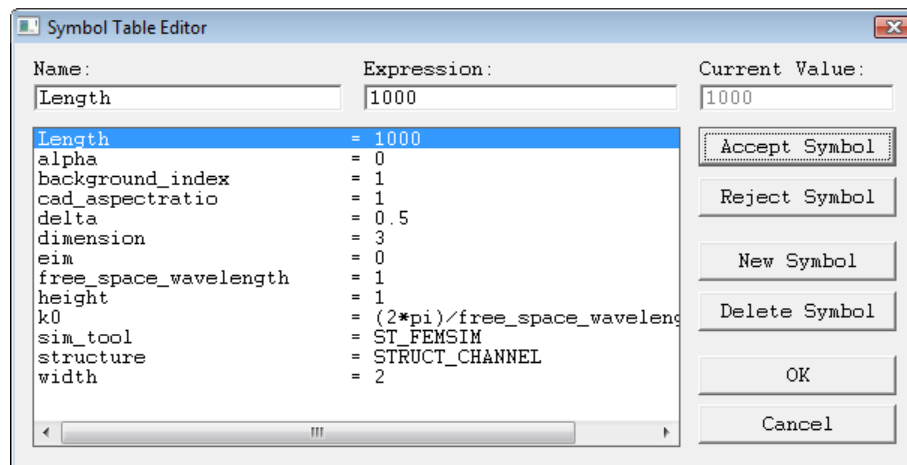


Figure 4: The Symbol Table Editor. The variable `Length` has been defined with a value of 1000.

In this tutorial, we are going to create a variable which will be used to set the length of the waveguide segment. This variable will not affect the simulation, but it will be a useful exercise. Define a symbol `Length`, and set it equal to `1000`. To do this, click **New Symbol**, enter the symbol **Name** (`Length`) and **Expression** (`1000`), and then click **Accept Symbol**. The variable `Length` should now appear in the symbol table.

The names of symbols in the symbol table are case-sensitive. Also, symbols which begin with a capital letter are displayed at the top of the list, and so it is recommended that any user-created variables start with a capital letter so that they will be easier to find in the symbol table.

Adding the Waveguide

We can now add the waveguide segment to the CAD layout window.

Drawing the Waveguide:

The CAD has several convenient ways to add a component. For this tutorial we choose the straight segment component, or **Segment Mode**, which has already been selected on the left toolbar by default. To draw the segment, move the cross-hair cursor to the beginning of the straight waveguide section, which we will conveniently locate at coordinates ($X=0$, $Z=0$). After positioning the cursor, press but do not release the left mouse button. Now move the cursor to the end of the straight waveguide section, located around the coordinates ($X=0$, $Z=1000$). We will fix the length of this waveguide segment using the variable we defined previously.

As you move the cursor, the center line of the waveguide is repeatedly drawn in blue and erased (this process is termed rubberbanding), and the status line coordinate display is continuously updated. When the cursor is positioned at the desired end point, release the mouse button. The waveguide segment is drawn as shown in Fig. 5, with the center line displayed in black, the left and right edges in dark red, and the interior in light red.

If a mistake is made when adding any waveguide component, simply click on the **Undo Last Change** icon in the top toolbar (the reversing arrow), and the waveguide circuit will be restored to its previous state.

Setting the Length of the Waveguide:

For this example, we are going to set the length of the waveguide to the variable `Length` (`1000  $\mu\text{m}$`) that we previously defined. Open the Segment Properties box for this waveguide by right clicking on the segment in the CAD window. The dialog box shown in Fig. 6 will appear. This dialog box enables the user to set the waveguide properties such as refractive index and waveguide geometry for the waveguide we have drawn.

The CAD is an object oriented design environment which so each segment can have a different Segment Properties dialog box, and therefore a different set of parameters. For the purposes of

this example, we will not explore each the use of all the fields in this dialog. A detailed description of this dialog can be found in Chapter 5 of the CAD manual.

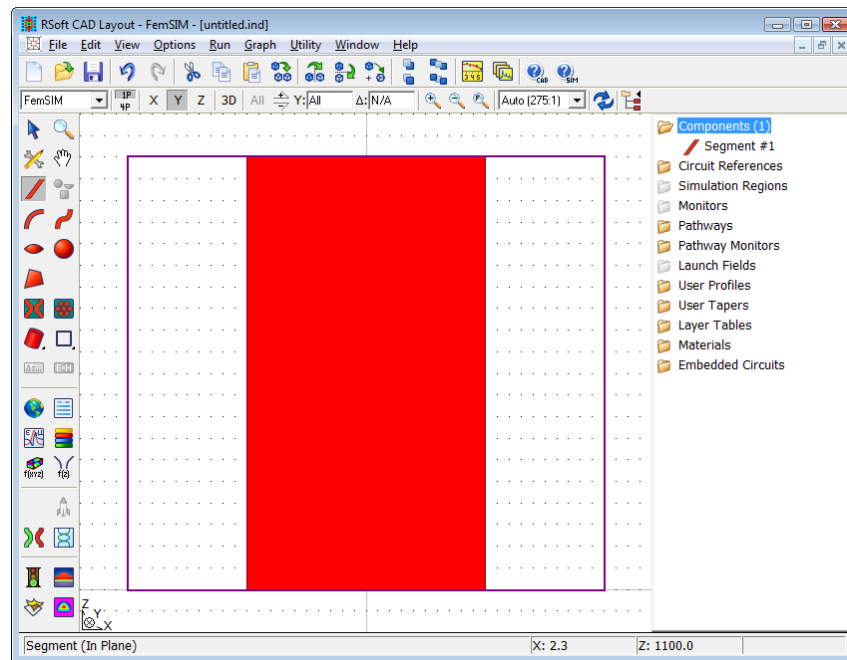


Figure 5: A straight waveguide segment as it appears after drawing, with the center line drawn in black, the left and right edges in dark red, and the interior in light red.

In order to set the length of this segment, make sure the **Reference Type** of the Z coordinate of the Ending Vertex to be *Offset*, and in **Relative To:** the starting vertex of **Component 1**. This tells the software that the Z coordinate of the ending vertex of this segment, or Segment #1, should be computed as an offset from the starting vertex, or vertex 0, of this segment, or component 1. This offset will therefore equal the length of the segment. To set the offset, which is the length in this case, set the **Offset Value** equal to the variable `Length` we previously defined. The dialog box should now appear as in Fig 6.

You may notice that the waveguide you have drawn does not have the same shape as the waveguide shown in Fig. 5. This is most likely due to the aspect ratio and viewing window of the display window. To set the aspect ratio of the CAD window, use the pulldown menu on the view toolbar in the CAD. You may also have to zoom in to see the whole structure. This can be done by clicking the View/Full button in the CAD toolbar.

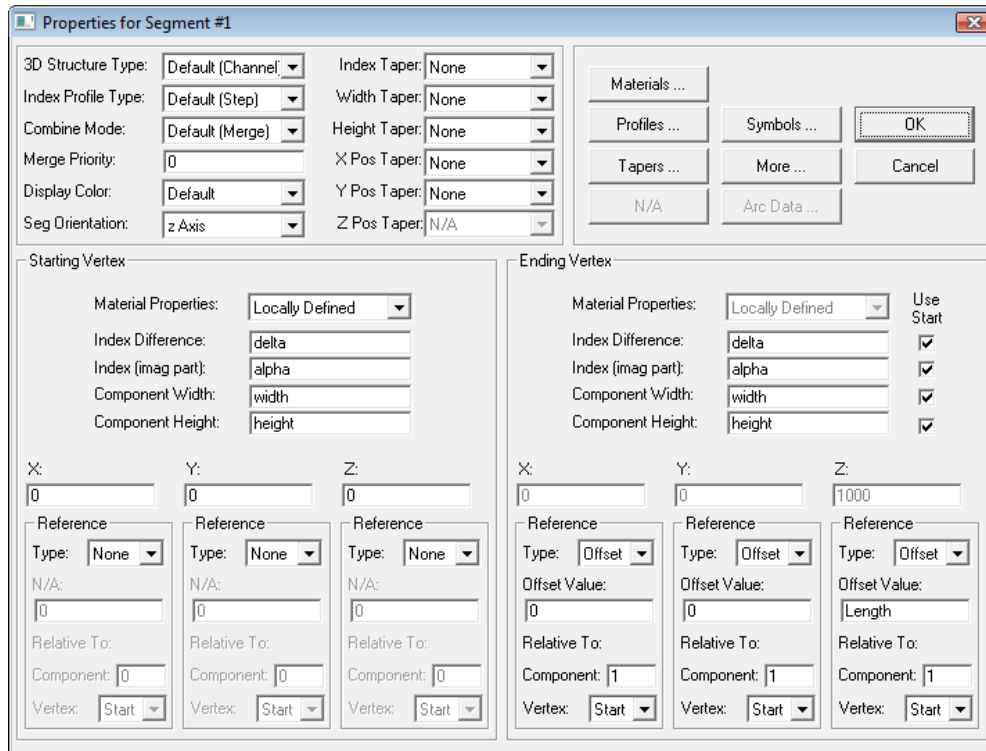


Figure 6: The Segment Properties dialog box for the waveguide segment in this design.

In general, the next step would be to continue to add elements to the CAD layout in order to create the desired device. This process is described further in the CAD manual. For the purposes of this example, there are no more elements to add.

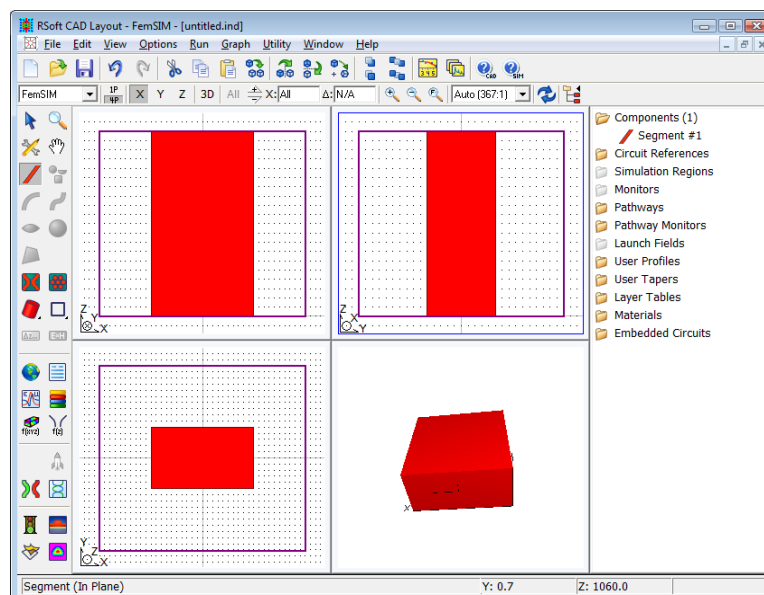


Figure 7: The CAD in multi-pane mode.

Viewing the Structure in 3D

The default view of the structure in the CAD is along the Y axis (the XZ plane). However, the structure can be viewed along the X and Z axes individually, as well as along all three axes. To change this view, click the X, Y, and/or Z buttons in the view toolbar in the CAD. To enter multi-pane mode, click the 4P button; to return to single-pane mode, click the 1P button. More information about these views can be found in the CAD manual.

For FemSIM, the Z axis view is probably the most useful. This view can be seen in the lower left corner of Fig. 7, and can be viewed along in single-pane mode by clicking the Z button in the view toolbar.

Creating the Mesh and Computing the Mode

Once the layout is complete, we can initiate a simulation by clicking the **Perform Simulation** button on the left toolbar to open the FemSIM Simulation Parameters dialog. This dialog allows the basic parameters required for a numerical simulation such as the mesh parameters and controls other important aspects of the simulation as well. Intelligent values for parameters such as grid sizes are automatically computed by the program based on the device parameters and are entered as defaults.

When performing an actual design, it is important to perform a convergence study with the numerical parameters to ensure that accurate results are obtained.

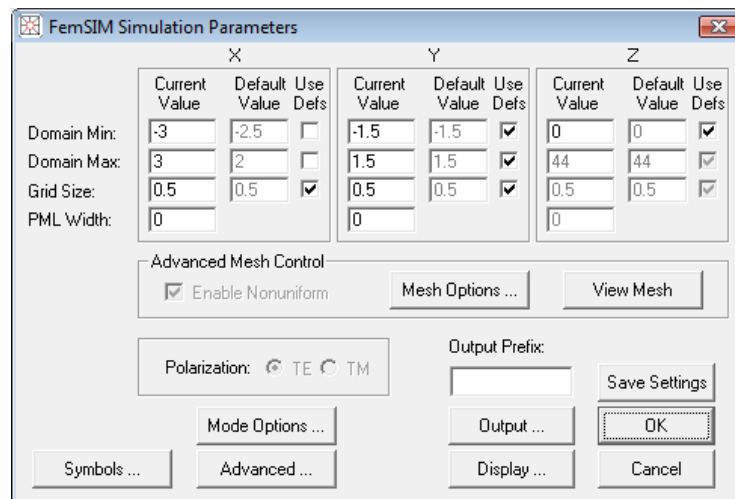


Figure 8: The FemSIM Simulation Parameters dialog where numeric simulation parameters are set.

At this point you can choose to use the default values, or enter a different value for any parameter. For this tutorial, we will change the **Grid Size** values from their default of $0.5 \mu\text{m}$ to a smaller value of $0.2 \mu\text{m}$. However, we will use different domain settings. We will also change

the default settings for the X domain to $(-3, 3)$ and Y domain to $(-1.5, 1.5)$. To do this, first locate the entry field corresponding to the each value to be changed. Next, move the mouse cursor into this field and click on it in order to make it active. Now edit the field by deleting the current value and entering the new value. Change all the parameters so that the FemSIM simulation window appears as shown in Fig. 8.

In order to view the mesh, click the **View Mesh** button. The mesh will be computed and displayed as shown in Fig. 9.

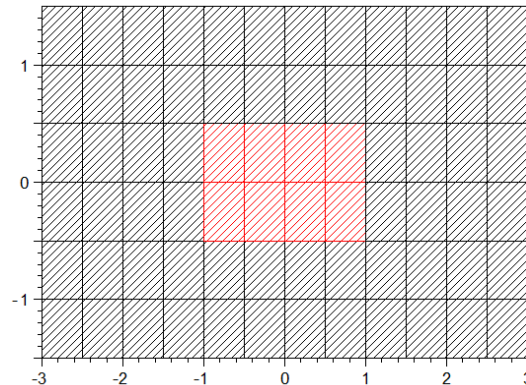


Figure 9: The computed mesh for this simulation.

In this case it is not necessary to subdivide the mesh since this is a rectangular structure, and so the option **Subdivide Mesh** in the FEM Mesh Options dialog is disabled by default. Tutorial 2 will calculate a fiber mode which uses an irregular mesh.

If the mesh accurately reflects the structure, which it does in this case, the mode can be calculated. If not, the mesh parameters should be changed until a satisfactory mesh is produced.

To save simulation results, enter a prefix such as 'test' in the **Output Prefix** field of the simulation parameters dialog and select the desired data files via the **Output...** button. This will save results to disk in files whose names start with the prefix entered. We will open the saved results later in the tutorial.

The default values for the rest of the settings are sufficient for this tutorial. Once the numerical parameters are set, click the **OK** button to begin the simulation.

The Simulation Window

After a simulation is initiated, the computation begins in a new window separate from the main program, as shown in Fig. 10. The bottom portion of the window shows the convergence process as modes are found. Once the first mode is found, one component will be displayed.

Other data can be seen as well:

- *Viewing Other Field Components:*

Once a mode is found, its 3 components can be viewed in the FemSIM Simulation window. To do this, use the buttons in the simulation toolbar. Note that the profiles shown have different maximum amplitudes shown on the color scale, and are normalized to the component with the highest value. For example, the E_x component of the mode found in this example has a maximum value of 1, where the E_y component has a maximum value of 0.045. Thus, this mode is quasi-TE mode. The E_z component can be viewed as well.

- *Viewing a Field Cross-Section:*

Right clicking on a axis will display the cross-section of the mode at that point. Right clicking again will return the view to the contour plot.

- *Zooming in on an area of interest:*

The mouse can be used to zoom in by dragging a box around the area of interest. Right clicking will zoom back out.

The simulation program runs as a background task, so that you can switch away to another program, or back to the main program to edit another structure or begin another simulation. When the simulation is finished, the titlebar of the simulation program indicates “Computation Completed”, and displays the coordinates of the cursor for reference.

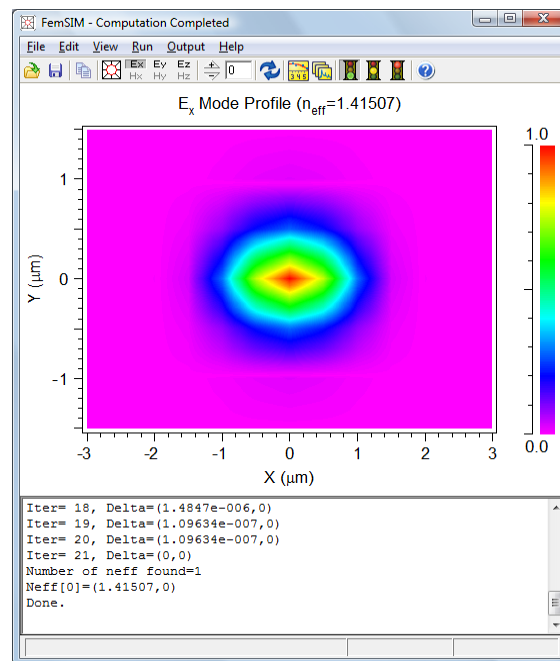


Figure 10: The separate simulation program, after computing the mode. The graphical display shows a color-coded contour plot of the mode field.

Accessing Saved Data

All data saved by FemSIM is in ASCII format, and so it is easy to access. For most output, two files are created: a raw data file, and a plotting command file that instructs the WinPLOT plotting program how to display the data. Generally, plotting command files have an extension that starts with the character 'p'.

Opening Plots:

To view or print the saved simulation results, click on the **View Graphs** icon in the top toolbar of the CAD window (the yellow graph), and select the desired graph from the dialog that is presented. In this case, several plotting files with the **Output Prefix** entered should be present, including `test_ex.p00`. This file contains the E_x component of the computed mode. Choose this file and click **OK** to open it.

A WinPLOT window which will show the time monitor data. The mouse can be used to zoom in the graph to areas of interest. To zoom out again, simply right-click.

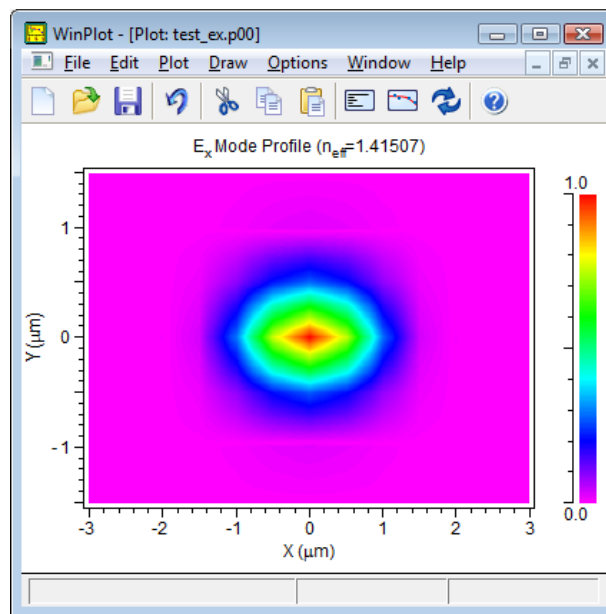


Figure 11: A WinPLOT window which shows the E_x component of the mode profile from this simulation. See the next sections for more details on how to access this data directly..

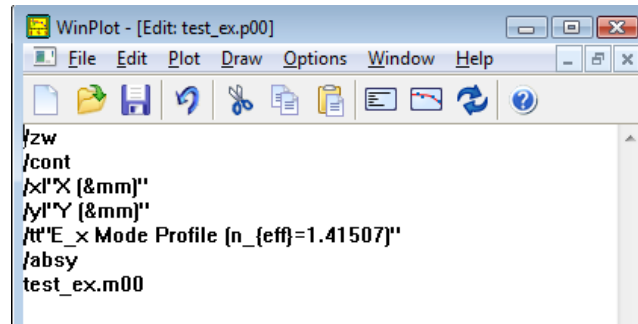


Figure 12: The plotting commands contained in the file `test_ex.p00` used to display the time monitor data stored in the file `test_ex.m00`.

Viewing the Plotting Commands:

The plotting command file `test_ex.p00` that was opened in WinPLOT is actually just a text file that contains plotting commands. To view this file, either open it in a text editor, or click the **View Editor** button in the WinPLOT window. The contents of this file are shown in Fig. 12. These commands indicate the axis labels to use, the legend text, as well as the raw data file `test_ex.m00`. Clicking the **View Plot** button returns the view to the plot.

Descriptions for these and other plotting commands can be found in the WinPLOT manual.

Viewing the Raw Data:

The raw data file `test_ex.m00` can be opened in any text editor, and contains the mode profile data:

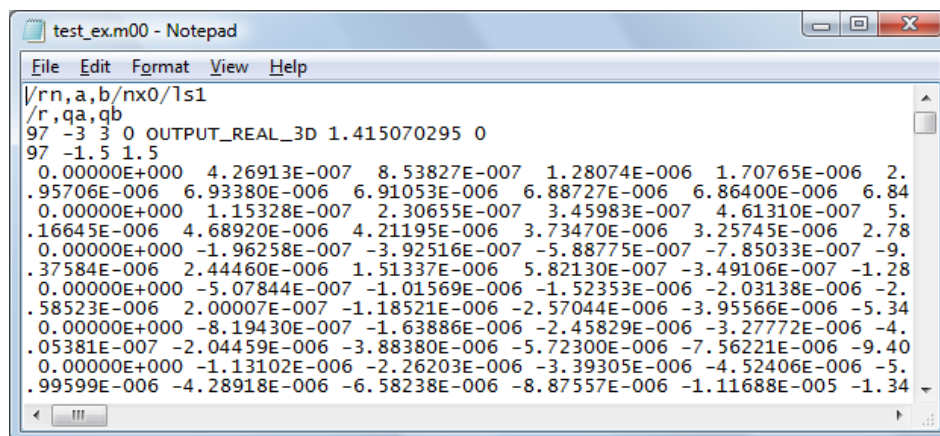


Figure 13: The raw data produced by this simulation.

Viewing other Types of Output:

Most other types of output produced by FemSIM are parallel to this output. Each field component can also be saved, as well as multiple modes.

Areas for Further Exploration:

Several areas to explore include:

- *Solving for Additional Modes*

FemSIM can easily solve for multiple modes supported by a structure. In the FemSIM Simulation Parameters window, click the **Mode Options...** button, and change the **# of Modes to Output** to 2. Once the simulation is complete, you can view the two modes via the buttons in the FemSIM simulation window. View the various components of the two modes, and note that the second mode found is a quasi-TM mode since its primary component is along E_y .

- *Showing the Structure Outline:*

It can be useful to display the structure outline in the simulation window. To do this, open the FemSIM Simulation Parameters dialog and click the **Display...** button. Set the **Outline Color** to *Default* and click **OK** twice to perform the simulation. The structure outline will now be displayed in the simulation window.

- *Changing the Display Aspect Ratio:*

To change the display aspect ratio during a simulation, open the FullWAVE Simulation Parameters dialog and click the **Display...** button. Set the **Plot Aspect Ratio** to the desired value. The value of `default` corresponds to a value which maximizes the size of the simulation display.

- *Changing the Color Scale*

A color scale is used to map the field values displayed during a simulation to particular colors for display. To change the color scale used, open the FemSIM Simulation Parameters dialog and set the **Color Scale** field to the desired file. The **Scale Files...** button can be used to locate the color scale needed. For information regarding the use and creation of color scale files can be found in an Appendix in the CAD manual.

Basic Tutorial 2: Simple Fiber Mode Calculation

This tutorial illustrates basic meshing concepts using a fiber object. Our goal will be to layout a simple fiber with a cladding index of 1, a core index of 1.5, a diameter of 8.2 μm , and then compute its fundamental mode at a wavelength of 1.55 μm .

This tutorial introduces basic meshing concepts. Familiarity with the concepts presented in Tutorial 1 is assumed. If you have not already read through Tutorial 1, please do so now.

Creating the Structure

Open the CAD window and click the **New Circuit** icon on the top toolbar and make the following settings:

Parameter	Value	Description
Simulation Tool	<i>FemSIM</i>	Sets FemSIM as the simulation engine to be used.
Free Space Wavelength	1.55	Sets the global wavelength [μm] for the simulation.
Background Index	1.44	Represents the real refractive index of the background material where no structure has been defined. In this case we will assume that we have an infinite cladding and so this represents the cladding index.
Index Difference	0.01	Represents the default difference between a component and the background material. In this case this represents the difference between the core index and the cladding index, and so the structure will have an index of 1.45. This is simply the default value to be used for a component; each component can have a unique value if desired.
3D Structure Type	<i>Fiber</i>	Sets the profile to be used for 3D components. The choice <i>Fiber</i> indicates that, by default, a component should have a circular cross section with a horizontal and vertical diameter given by the Component Width and Component Height fields respectively. By

Component Width	8.2	setting these fields equal, a circular cross-section is achieved. Represents the default width of a component [μm]; each component can have a unique value if desired.
Component Height	width	Represents the default height of a component [μm]; each component can have a unique value if desired. The variable <code>width</code> is the built-in variable which corresponds to the value of the Component Width .

Once these settings are made, click **OK** to continue. Note that changes can be made to these settings via the Global Settings dialog box which can be opened by clicking the ‘Globe’ icon in the left toolbar.

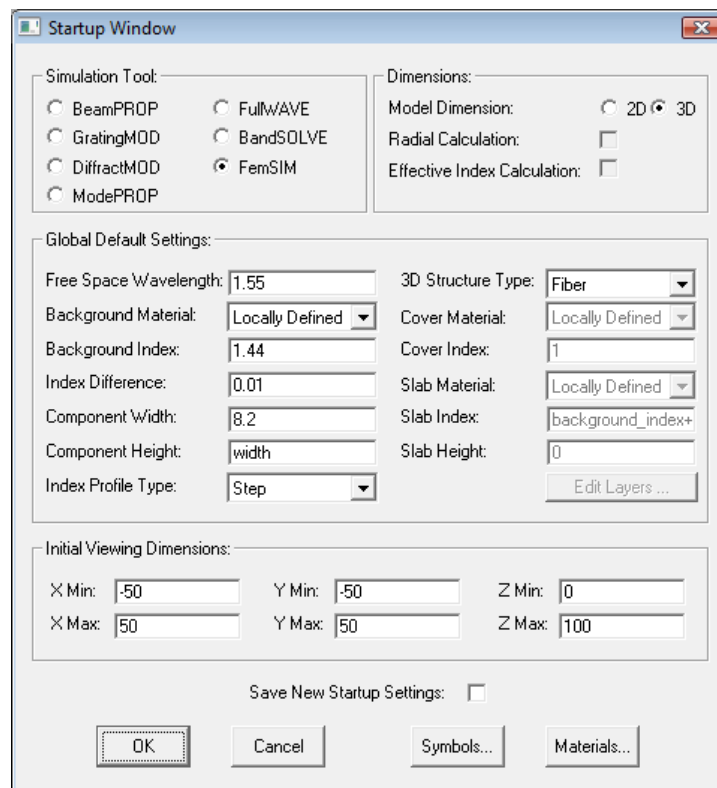


Figure 1: The startup dialog which appears whenever a new circuit is created, and requests basic information about the circuit to be modeled.

Creating Variables via the Symbol Table

For this design we will create a variable which sets the length of our fiber segment. This variable will not affect the simulation, but it will be a useful exercise. Click the **Edit Symbols** button in the left toolbar and create a symbol `Length` with a value of $1000\ \mu\text{m}$. The symbol table should appear as shown in Fig. 2.

Any variable in the symbol table can be set to a function of any other variable. For example the variable `k0` is set to a value of $(2*\pi)*\text{free_space_wavelength}$ corresponding to the magnitude of the wavenumber **k** in free space at the wavelength value chosen. A complete list of built-in functions and constants can be found in an Appendix in the CAD manual, and built-in variables for FemSIM can be found in [Appendix C](#) of this manual.

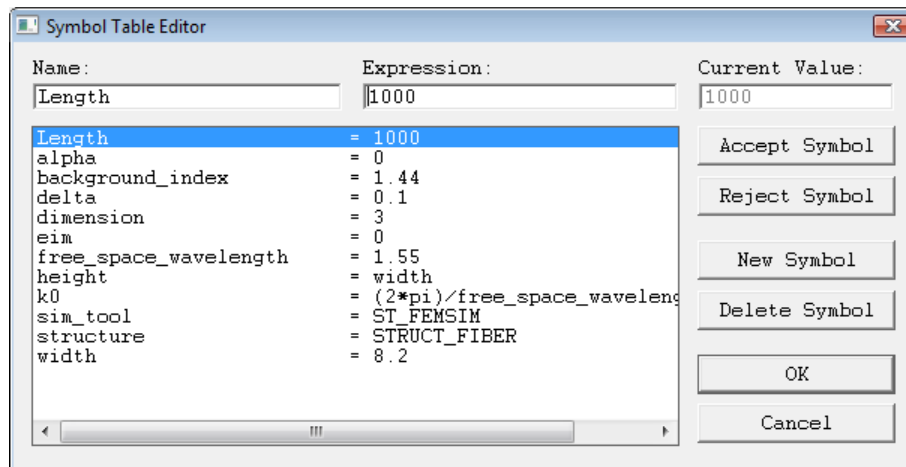


Figure 2: The Symbol Table Editor. The variable `Length` has been defined with a value of 1000.

Adding the Fiber Segment:

To add the fiber segment, make sure you are in **Segment Mode**. Once the fiber has been created, we can set its length to the symbol `Length` ($1000\ \mu\text{m}$) as desired. Right click on the segment to access its Segment Properties dialog, and set the starting vertex coordinates to $(0, 0)$ and the ending vertex to be *Offset* from the starting vertex by a value of $(0, \text{Length})$. When done, the dialog should appear as shown in Fig. 3.

Viewing the Structure along the Z axis

This structure can be viewed along the Z axis by clicking the Z button in the view toolbar.

Creating the Mesh and Computing the Mode

To view the mesh, click the **Perform Simulation** button in the left toolbar. Like Tutorial 1, we will change some of the default parameters. Change the default simulation domain to (-10, 10) for both the X and Y coordinates.

When performing an actual design, it is important to perform a convergence study with the numerical parameters to ensure that accurate results are obtained.

The rest of the parameters can be left to their default values. Click the **View Mesh** button to see the mesh as shown in Fig. 5.

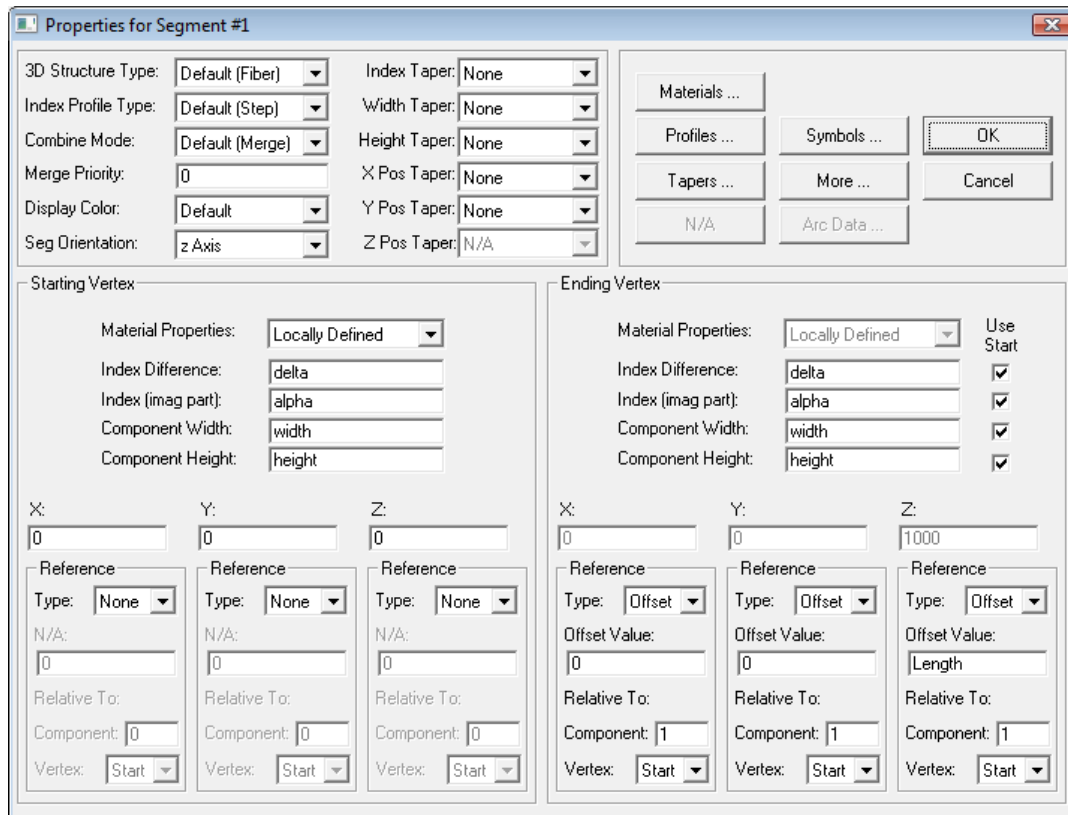


Figure 3: The Segment Properties dialog box for the segment in this design.

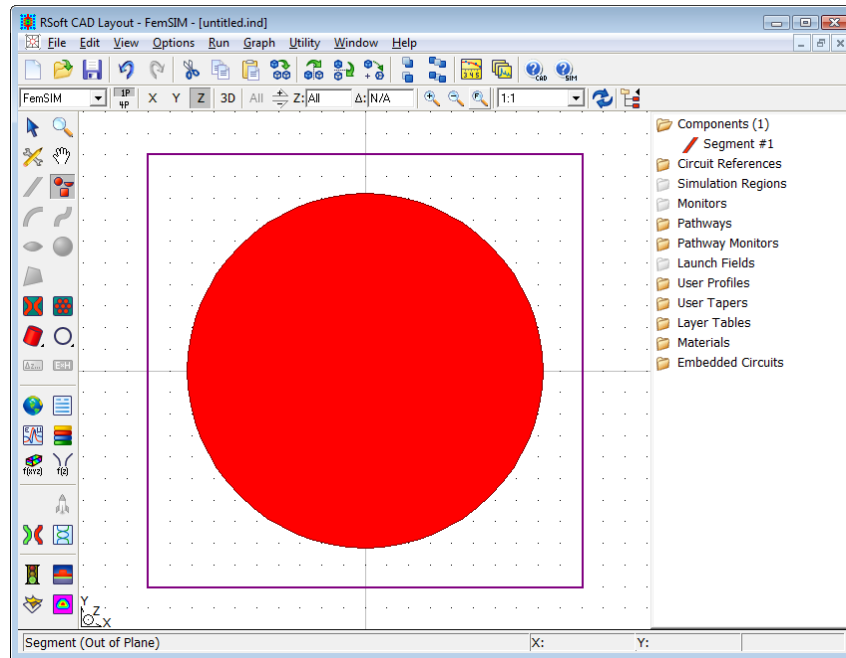


Figure 4: The fiber structure viewed along the Z axis.

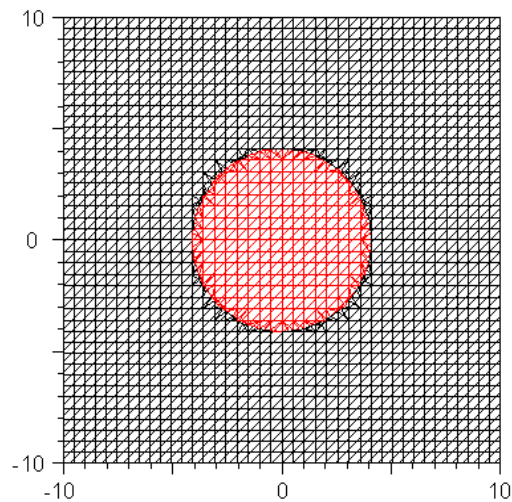


Figure 5: The computed mesh for this simulation.

The mesh options can be changed in the FEM Mesh Options dialog which can be accessed by clicking the Mesh Options... button in the FemSIM Simulation Parameters dialog. Since the **3D Structure Type** is set to *Fiber*, the **Subdivide Mesh** option is enabled, resulting in an irregular mesh that conforms to the curved interface between the core and cladding regions.

Once the numerical parameters are set, and the mesh accurately represents the structure, enter an **Output Prefix** of 'test' and click **OK** to begin the simulation.

The Simulation Results

After a simulation is initiated, the computation begins in a new window separate from the main program, as shown in Fig. 5. The bottom portion of the window shows the convergence process as modes are found. Once the first mode is found, it will be displayed. The field components of each mode can be viewed via the buttons on the simulation toolbar. Right clicking on a axis will display the cross-section of the mode at that point. Right clicking again will return the view to the contour plot. The mouse can also be used to zoom in by dragging a box around the area of interest. Right clicking will zoom back out.

The simulation program runs as a background task, so that you can switch away to another program, or back to the main program to edit another structure or begin another simulation. When the simulation is finished, the titlebar of the simulation program indicates "Computation Completed", and displays the coordinates of the cursor for reference.

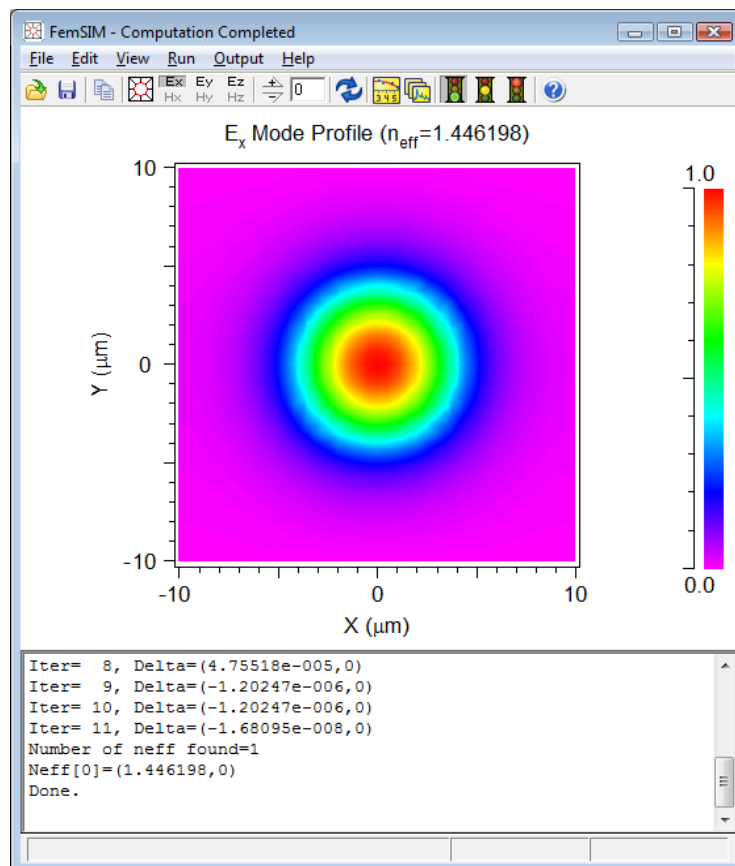


Figure 6: The separate simulation program, after computing the mode. The graphical display shows a color-coded contour plot of the mode field.

Areas for Further Exploration:

Several areas to explore include:

- *Use Symmetric Boundary Conditions*

Since this both the structure and mode profile are symmetric about both the X and Y axes, it is possible to use symmetric boundary conditions to reduce simulation time and memory requirements. Open the Advanced Mesh Controls dialog by clicking the **Mesh Options...** button in the FemSIM Simulation Parameters dialog and set both the **Symmetry at X=0** and **Symmetry at Y=0** fields to *Symmetric*. View the mesh and run the simulation again to see the effect of these options.

- *Use the DataBROWSER to Quickly View All Output Data*

The RSoft DataBROWSER allows the user to quickly sort through all the output files from a simulation. It can be access via the **Launch DataBROWSER** button on the top toolbar in the CAD. Please consult the DataBROWSER manual for more details.

5

Advanced Tutorials

This chapter contains several advanced tutorial examples which illustrate the use of the FemSIM simulation engine. Corresponding to each example there is one or more *.ind and/or data files, which are located in the `<rsoft_dir>\examples\FemSIM\Tutorial` subdirectory.

Unlike the Tutorials in Chapter 4, these tutorials need not be performed in any order. However, it is recommended that you perform the basic tutorials in Chapter 5 before attempting to work through these tutorials as they explain basic concepts in more detail.

While the following examples attempt to cover the key aspects of the software, the breadth of the package allows enormous flexibility in both the number of applications that can be covered as well as the approach used to tackle a given application. If you do not find your application adequately addressed by the following examples, please contact photonics_support@synopsys.com and we will assist you in finding a solution if possible.

Tutorial 1: Leaky Modes in a Multilayer/Rib Waveguide

This tutorial will go through the process of using FemSIM to solve for the leaky modes supported by a multilayer GaAs and AlGaAs waveguide. This tutorial will discuss how to distinguish between highly lossy substrate radiation modes and the leaky guided modes of interest.

This tutorial is based on the following reference:

J. Heaton, M. Bourke, S. Jones, B. Smith, K. Hilton, G. Smith, J. Birbeck, G. Berry, S. Dewar, and D. Wight, “Optimization of Deep-Etched, Single-Mode GaAs/AlGaAs Optical Waveguides Using Controlled Leakage into the Substrate,” *Journal of Lightwave Technology*, **17**, (1999) 267-281.

Structure Overview

This waveguide is a multilayer rib structure that is deep-etched on top of a GaAs substrate. The core and claddings of the device are composed of layers of AlGaAs with different alloy concentrations in order to provide confinement. Due to the large index of the substrate, we will see leakage into the substrate. We will solve for the modes of the waveguide at a wavelength of 1.064 μm .

Creating the Structure in the CAD Environment

This structure will be created in the CAD using the built-in Multilayer structure type. Open the CAD window and click the **New Circuit** icon on the top toolbar and make the following settings:

Parameter	Value
Simulation Tool	<i>FemSIM</i>
Model Dimension	<i>3D</i>
Free Space Wavelength	1.064
3D Structure Type	<i>Multilayer</i>
Component Width	4.4

While the **Component Height** and **Slab Height** appear in this dialog, we will set them directly in the next section via the Symbol Table since they will depend on several symbols we have yet to define.

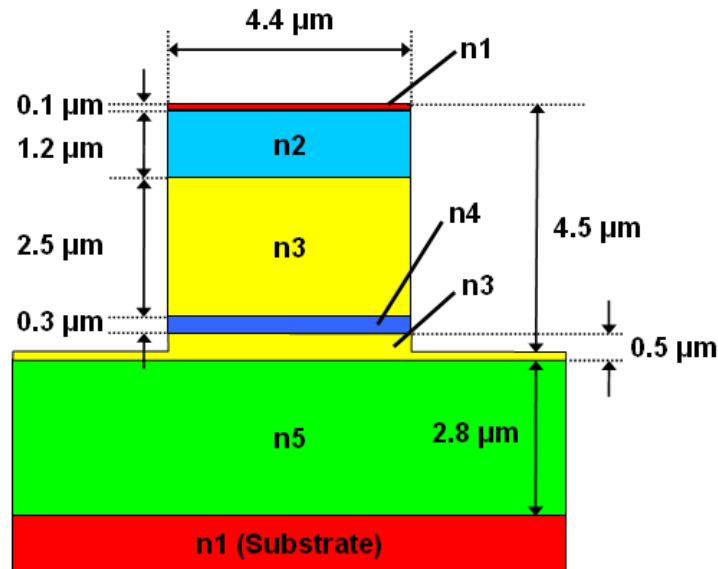


Figure 1: The structure used in this tutorial. The index values at a wavelength of $1.064 \mu\text{m}$, along with AlGaAs alloy compositions as appropriate, are: $n_1 = 3.4804$ (GaAs), $n_2 = 3.3675$ ($x=20\%$), $n_3 = 3.4519$ ($x=5\%$), $n_4 = 3.3955$ ($x=15\%$), $n_5 = 3.4434$ ($x=6.5\%$). The cover material is air ($n=1$).

Defining the Multilayer Structure

Next, press the **Edit Layers** button in the left toolbar to open the Layer Table Editor so we can define the layer structure of the waveguide. In order to do this, we will define several variables to simplify the process. Click the **Symbols...** button and define the following variables:

Variable	Value	Description
Depth	4.5	Sets the etch depth
Length	1000	Sets length of waveguide
H	0.5	Thickness of the spacer layer.
H_0	2.5	Thickness of the core layer
H_1	0.3	Thickness of lower cladding 1
H_2	2.8	Thickness of lower cladding 2
H_C	1.2	Thickness of upper cladding
H_W	0.1	Thickness of cap layer
N	3.4519	Index of spacer layer

N_0	3.4519	Index of core layer
N_1	3.3955	Index of lower cladding 1
N_2	3.4434	Index of lower cladding 2
N_C	3.3675	Index of upper cladding
N_Sub	3.4804	Index of substrate
N_W	3.4804	Index of cap layer
background_index	N_Sub	Sets the substrate index
height	$H_W + H_C + H_0 + H_1 + H_2$	Sets the height of the waveguide
slab_height	$H_W + H_C + H_0 + H_1 + H_2 - \text{Depth}$	Sets the slab height

Once all of the symbols have been entered, press **OK** to return to the Layer Table Editor. Using the New Layer and Accept Layer buttons, create the Layer Table 0 as shown in Fig. 2.

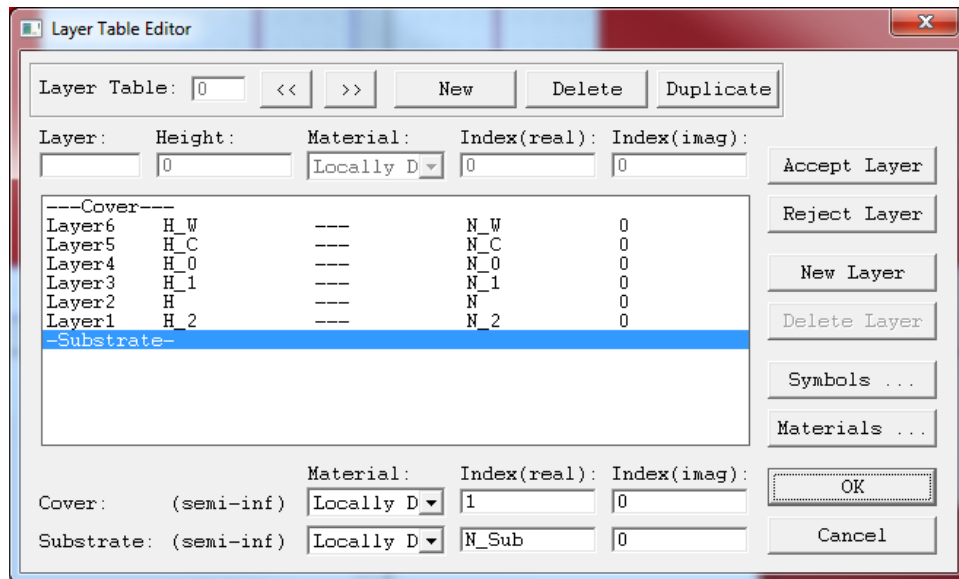


Figure 2: Layer table for GaAs/AlGaAs structure.

Once the layer information has been entered, click **OK** to return to the twice to go to the CAD window.

Creating the Waveguide

Draw in a straight segment starting at $X=0$, $Z=0$ and going to $X=0$, $Z=\text{Length}$. To make these settings, right-click on the segment to access its Segment Properties dialog. While in the

Segment Properties dialog, note that the Layer Table settings for both the starting and ending vertex are already set to 0 by default which is the layer table we have defined.

Checking the Index Profile

In order to ensure that we have created the correct multilayer profile, we can compute the index profile of the structure. Though different then the simulation mesh to be used, it will give us confidence that the structure has been correctly defined. Click the **Display Material Profile** button on the left toolbar and change the X **Grid Size** to 0.1 and the Y **Grid Size** to 0.01 to better resolve the layered structure. Next, since most of the layers have very similar index values, change the range of index values displayed. Press the **Display...** button and set **Index Display Min** to the lowest index in the structure, n_c . Press **OK** in this window and the next to view the index profile as shown in Fig. 3.

Setting the Mesh Parameters

FemSIM solves for the modes of the structure on a mesh. Click the **Perform Simulation** icon in the left toolbar to open the FemSIM Simulation Parameters dialog.

For the most part, the default mesh parameters will be used. In this case we will use a slightly larger X domain to accommodate more modes, as well as a smaller Y grid size to better resolve the thin layers. Change the X **Domain Min** to -4, the X **Domain Max** to 4, and the Y **Grid Size** to 0.05. Also, since we are looking for lossy modes, PML must be used at the boundaries. Change the X **PML Width** to 2.5 and the Y **PML Width** to 0.25. After these changes are made, click the **View Mesh** to view the simulation mesh to be used. Note that this includes both the simulation domain and the PML region.

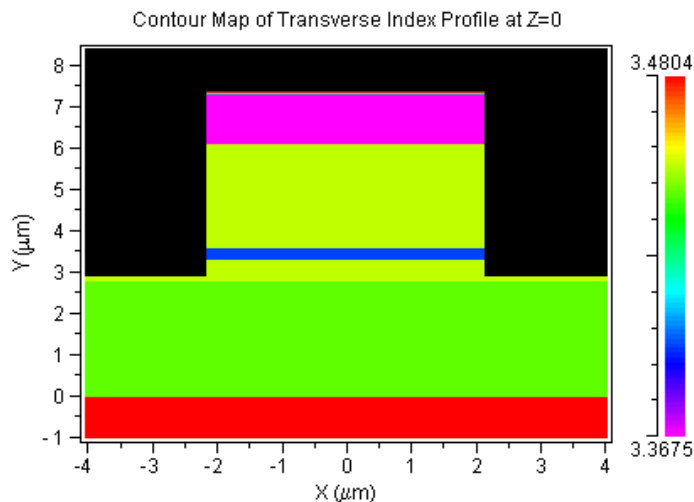


Figure 3: The index profile of the GaAs/AlGaAs multilayer waveguide. The various layers are clearly visible. The black region is the air cover, and lies outside the range of the scale used.

Calculating the Fundamental TE & TM Modes

Once the mesh parameters have been set, the mode options need to be defined. Click the **Mode Options...** button in the FemSIM Simulation Parameters dialog to open the FEM Mode Options dialog. Since we want to find two leaky modes (TE and TM) that are guided in the core layer, set the **# of Modes to Find** to 2 and set the **Starting Neff (Real)** to the index of the core layer (n_0). Click **OK** to and then click **Advanced...** to open the Advanced Parameters dialog. Enable the **Allow Leaky Modes** option, and then click **OK** twice to initiate the mode computation.

At the end of the computation, two modes will be found. You can view these modes and their various components using the Output menu option in the FemSIM Simulation window. Note that one mode is primarily defined along X and is therefore TE, and the other mode is primarily defined along Y and is therefore TM.

Calculating Higher Order Modes

A key point to understand about computing higher order leaky modes for this GaAs based device is that FemSIM will compute modes that are not necessarily guided. This is because, by default, FemSIM finds modes based on their proximity to the **Starting Neff (Real)** value. Leaky structures such as this waveguide will exhibit radiation modes that have a real effective indices near this value, but that also have extremely high losses. A convenient way to ignore these modes is to sort the modes by in order of increasing loss.

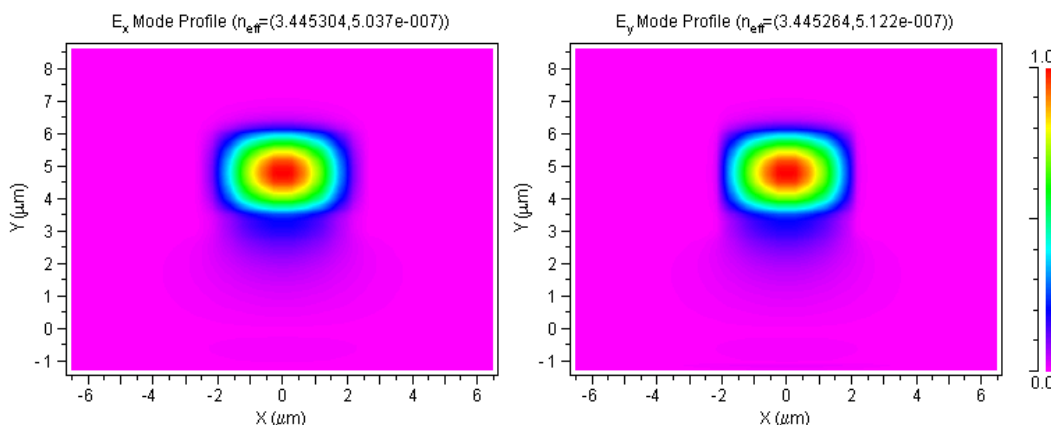


Figure 4: Fundamental modes found by FemSIM: a) E_x component of TE mode b) E_y component of TM mode. These and all other components can be viewed via the Output menu option in the FemSIM simulation window. The raw data for these modes is also saved in the directory from which the simulation is performed.

In order to illustrate this concept, we will attempt to solve for the first four modes of the waveguide. We will do this first without sorting the modes, and then again with sorting the modes.

Solving for Modes Without Sorting By Loss

Click the **Perform Simulation** icon in the left toolbar and click the **Mode Options...** button to open the FEM Mode Options dialog. To solve for five modes, change the **# of Modes to Find** to 4. Press **OK** twice to start the simulation.

After this computation completes, the four modes found can be viewed via buttons in the FemSIM simulation window toolbar. Two of these modes are the same fundamental modes found in the previous simulation, and the remaining two are shown in Fig. 5.

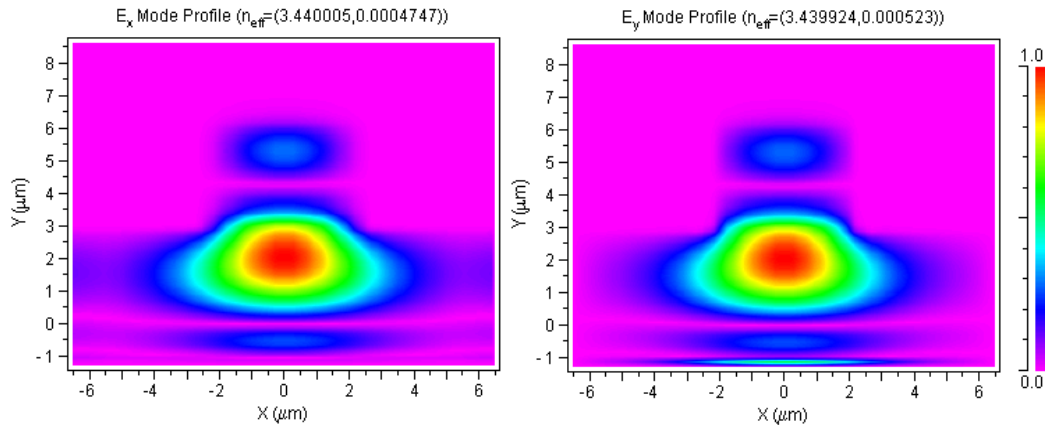


Figure 5: The next two higher order modes found by FemSIM without sorting modes by loss. The first two modes are exactly the same as the previous simulation and are shown in Fig. 4.

An examination of the two modes in Fig. 5 shows us two things:

- The two modes shown in Fig. 5a and 5b are TE and TM modes that appear to have a large amount of field leaking into the substrate.
- The imaginary index of the two modes shown in Fig. 5 is much more than the two modes shown in Fig. 4.

From these observations, we can conclude that while the modes in Fig. 4 are guided leaky modes, the modes in Fig. 5 are highly leaky substrate modes. If we rerun the simulation and sort the modes by their imaginary index, we can easily ignore the highly leaky substrate modes to see if this structure supports higher order guided leaky modes.

Solving for Modes and Sorting By Loss

In order to ignore the highly leaky substrate modes, we will solve for more modes to find additional guided modes, sort the modes we do find by their loss, and then output only the first four modes with lowest loss. In the FEM Mode Options dialog change the **# of Modes to Find** to 10, **Mode Sorting** to *Lowest Loss*, and the **# of Modes to Output** to 6. While we could only output 4 modes, we choose 6 in order to better understand what is happening. Press **OK** twice to rerun the mode computation. At the end of the simulation, the six modes found can be viewed

using the buttons in the FemSIM simulation parameters window toolbar. The first two modes are the same fundamental modes shown in Fig. 4, the next two modes are first order guided leaky modes and are shown in Fig. 6, and the last two modes are the same highly leaky substrate modes shown in Fig. 5.

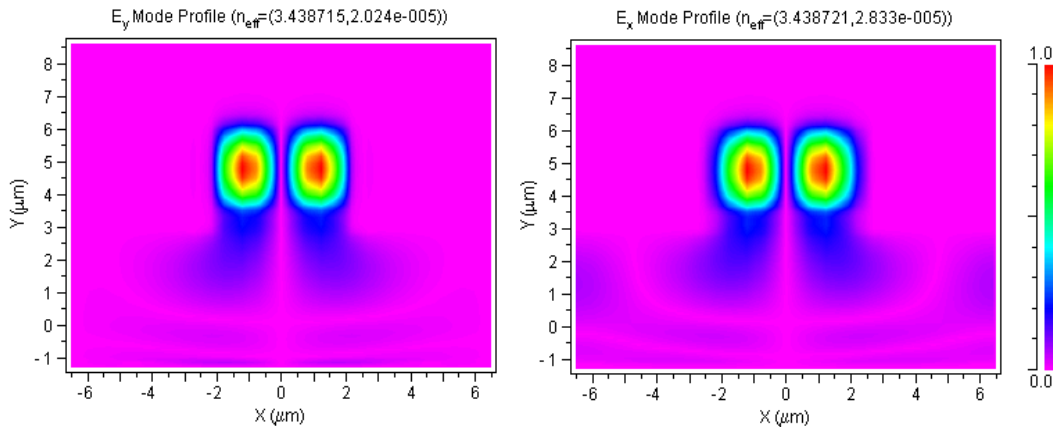


Figure 6: First order guided leaky TM and TE modes found by FemSIM.

By looking at the modes loss carefully, and by solving for additional modes, we were able to find a set of propagating modes that will be guided by this waveguide.

Areas for Further Exploration

Several areas to explore include:

- *Using a Larger domain*

One thing that should be investigated is the effect of larger domain sizes. While this is especially true for the X domain since the fundamental modes are close to the boundaries, it is also true for the Y **Domain Min** value.

- *Perform a Convergence Study with the Mesh Sizes*

A convergence study, as always, should be run on both the X and Y grid sizes to ensure accurate results.

- *Increasing PML Size*

The effect of a larger PML region can be studied. If the PML width is increased, how does the results converge to a particular solution?

- *Solve for Additional Modes*

Increasing the number of modes to be found increased the basis over which guided modes could be found. By increasing the number of modes further, additional higher order guided modes can be found.

Tutorial 2: Modes of a Slot Waveguide Structure

This tutorial will go through the process of setting up a slot waveguide structure. In exploring the computation of the fundamental TE mode of this structure using FemSIM, we will have to explore the convergence of the modes based on the mesh size for tightly confined structures. This tutorial is based on the following reference:

V. Almeida, Q. Xu, C. Barrios, and M. Lipson, “Guiding and confining light in a void nanostructure,” *Optics Letters*, **29** (2004) 1209-1211.

Structure Overview

This device has two silicon channel waveguides that are buried in silica. A large discontinuity exists on the boundaries of the silicon waveguides due to the high index contrast. The waveguides are placed with a small gap between them, which causes the modes and discontinuities to overlap leading to a high confinement.

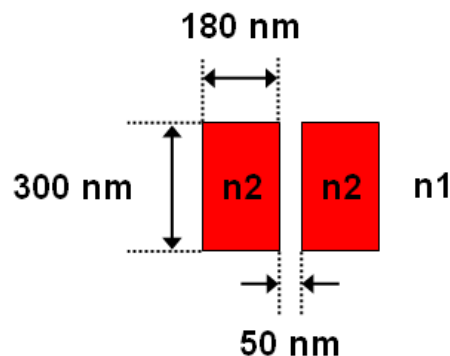


Figure 1: The structure used in this tutorial. It is composed of two identical buried Si channels in a uniform SiO₂ background and separated by a small distance. The index values at a wavelength of 1.55 μm are $n_1 = 2.48$ and $n_2 = 1.44$.

Creating the Structure in the CAD Environment

This structure will be created in the CAD using the built-in channel structure type. Open the CAD window and click the **New Circuit** icon on the top toolbar and make the following settings:

Parameter	Value
Simulation Tool	<i>FemSIM</i>
Model Dimension	<i>3D</i>

3D Structure Type	Channel
Free Space Wavelength	1.55
Component Height	0.3

While the **Component Width**, **Background Index**, and **Index Difference** appear in this dialog, we will set them directly in the next section via several symbols we have yet to define.

Once these settings are made, press **OK** to go to the main CAD window.

Defining Symbols

Next, press the **Edit Symbols** icon and define/set the following variables:

Variable	Value	Description
NC	1.44	Index of the cladding
NH	3.48	Index of the core regions
NGX	2	Number of grid points in the X direction to be used during a simulation within each core waveguide.
NGY	5	Number of grid points in the Y direction to be used during a simulation within each core waveguide.
grid_size	width/NGX	Grid size in the X direction
grid_size_y	height/NGY	Grid size in the Y direction
WH	0.18	Width of Si waveguides
WS	0.05	Width of gap between waveguides
background_index	NC	Index of background material
delta	NH-background_index	Index difference of the core waveguides
width	WH	Width of the silicon waveguides

Once all of the symbols have been entered, press **OK** to go back to the main CAD window.

Creating the Waveguides

Draw a straight segment starting at $(X, Z) = (-(WH+WS)/2, 0)$ and going to $(X, Z) = (-(WH+WS)/2, 1000)$. To make these settings, right-click on the segment to access its Segment Properties dialog. The rest of the parameters of this segment can be left at their default values.

To make the second waveguide, select the first waveguide created and click the **Duplicate Selection** icon on the top toolbar to create a copy. Edit this second segment's properties so that starts at $(X, Z) = ((WH+WS)/2, 0)$ and going to $(X, Z) = ((WH+WS)/2, 1000)$. When complete, the structure will appear as shown in Fig. 2.

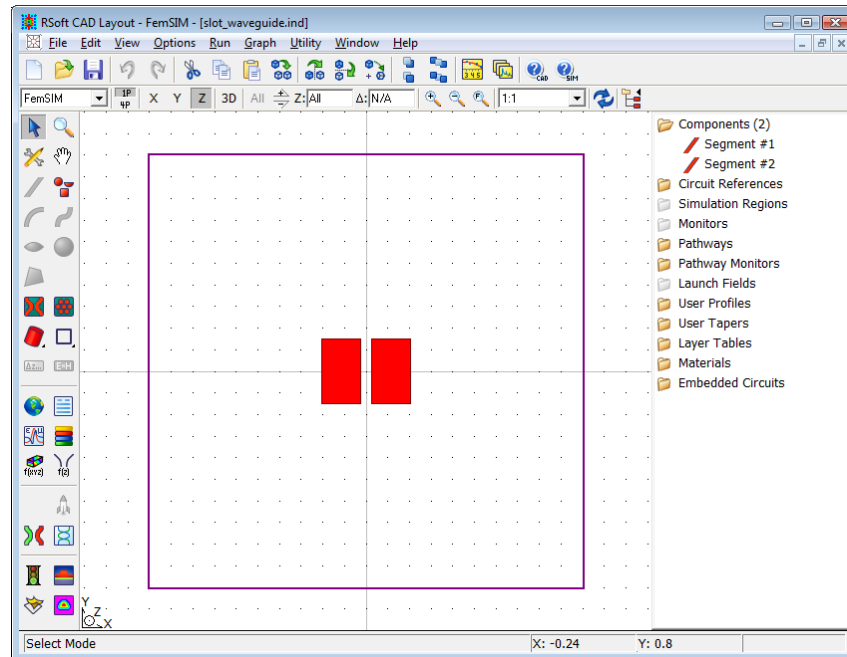


Figure 2: Layout for slotted waveguide as seen along the Z axis in the CAD window.

Checking the Index Profile

Click the **Display Material Profile** icon on the left toolbar and set the X/Y **Domain Min** to -1, the X/Y **Domain Max** to 1, the X **Slice Grid** to `grid_size` and the Y **Slice Grid** to `grid_size_y`. Once these changes are made, press **OK** to view the index profile. The result shows that the grid is relatively coarse right now. We will explore this later in our analysis of the simulation.

Calculating the Fundamental TE Mode with Coarse Grid

FemSIM solves for the modes of the structure on a mesh. Click the Perform Simulation icon on the left toolbar to open the FemSIM Simulation Parameters dialog. For the most part, we will use the default mesh parameters to start out. We will then refine the parameters in order to increase the accuracy of the modes found.

Since this structure is symmetric, we can speed up the computation by using a half-domain. Click the **Mesh Options...** button and set the **Symmetry at X=0** to be *Symmetric*. Press **OK** to return to the FemSIM Simulation Parameters dialog, and click the **View Mesh** button to display the mesh as shown in Fig 4a.

Once the mesh parameters have been set, we can solve for the modes of this device. Since we want to find the fundamental TE mode, no changes need to be made to the mode computation options. Click **OK** to run the computation.

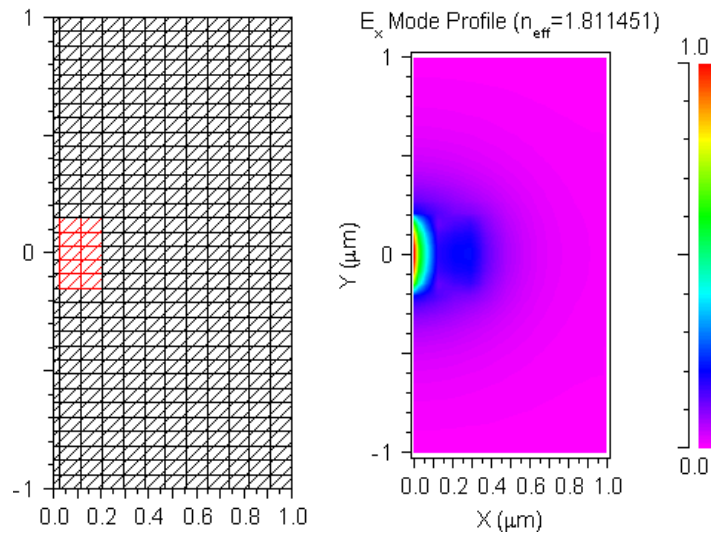


Figure 4: a) Mesh of the slotted waveguide. While the X grid size is larger than the slot, FemSIM automatically places a grid point at the edge of the core waveguide because the **Interface Alignment** mesh option is on by default. b) Mode for slotted waveguide for initial settings.

Performing a convergence study

Figure 5 shows the computed mode for this device. However, since we used a relatively coarse grid size, it will be beneficial to perform a convergence study on the grid size. Our convergence study will consist of continually reduce the grid size along X until the results converge to a particular result. To do this we will increase the value of the symbol `NGX`. Repeat the mode computation for three values of `NGX`: 5, 10, and 25.

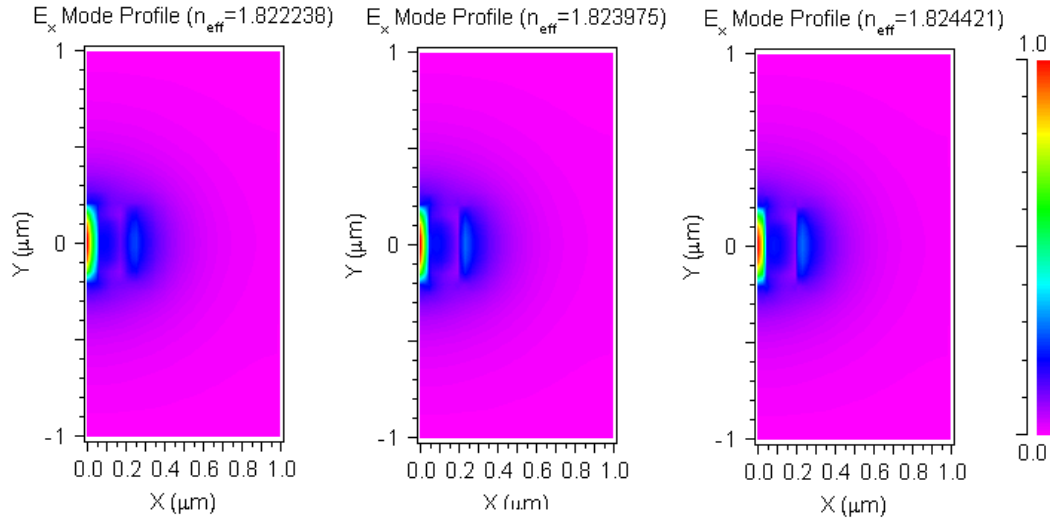


Figure 6: Results from simulations with NGX = a) 5 b) 10, and c) 25.

As you can see, the results are converging as we increase the value of NGX . Also, as we increase NGX , the simulation time and memory also increase. If, in the simulation where NGX is 25, you right-click at $Y=0$, the cross section of the mode at $Y=0$ will be shown.

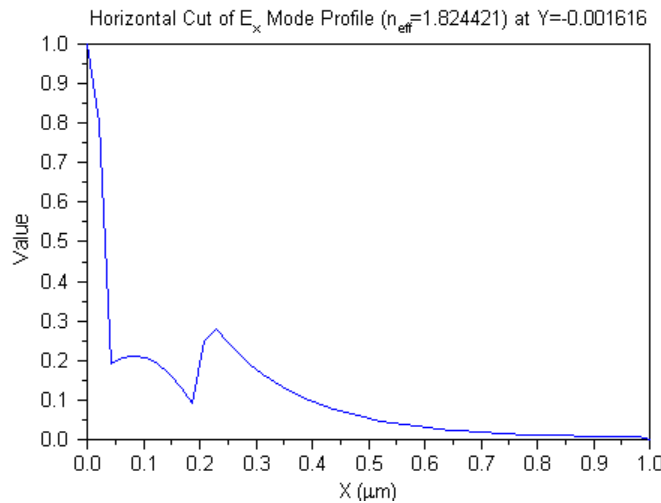


Figure 7: Cross section of fundamental mode at $Y=0$ with NGX set to 25.

The results are clearly beginning to get close to convergence. However, since the gap width is much smaller than the other feature sizes, it will take a lot of mesh points to converge to a solution. Rather than continue to reduce the mesh sizes further, we will utilize the advanced non-uniform grid options.

Convergence Study Using Advanced Grid Features

The simulations that were previously run used a “global” grid size that affected the entire domain. However, as we can clearly see by the field plots shown in Figures 6 and 7, a lot of “interesting” effects occur on the boundaries between the core and the cladding. FemSIM has the ability to use a non-uniform grid at interfaces between material regions which allows for finer resolution in these areas without having to increase the resolution in the entire domain.

To do this, click the **Mesh Options...** button in the FemSIM dialog and set the following parameters:

Options	Value	Description
X Grid Size (Edge)	0.005	Sets the X dimension of a grid element at the edge of a material region.
Y Grid Size (Edge)	0.005	Sets the Y dimension of a grid element at the edge of a material region.
Grid Grading	Yes	Enables the use of a graded grid from the edge of a material region to the center.

After these values are set, set the variable `NGX` to 2 to use the initial “average” grid size value we started with before the previous convergence study. View the simulation mesh and note that the mesh will be non-uniform at material boundaries as expected.

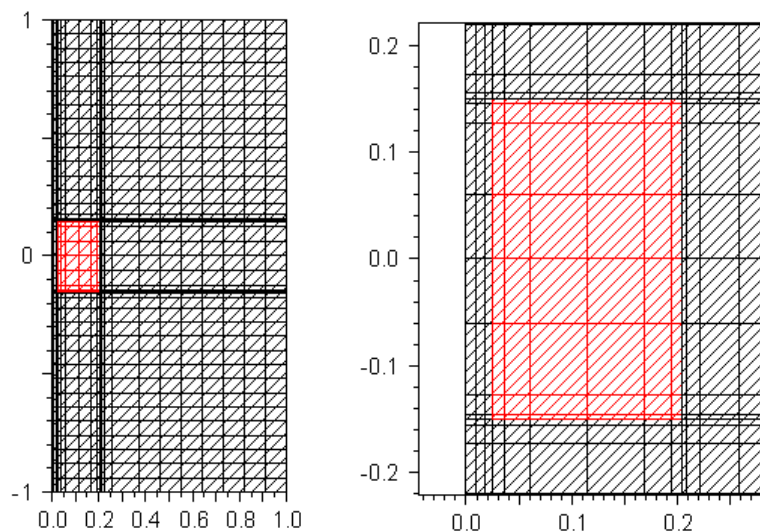


Figure 8: a) Mesh generated using non-uniform bulk factor. b) Close-up of the mesh in the slot region.

Compute the mode again with this mesh and again with X and Y **Grid Size (Edge)** options set to 0.0015.

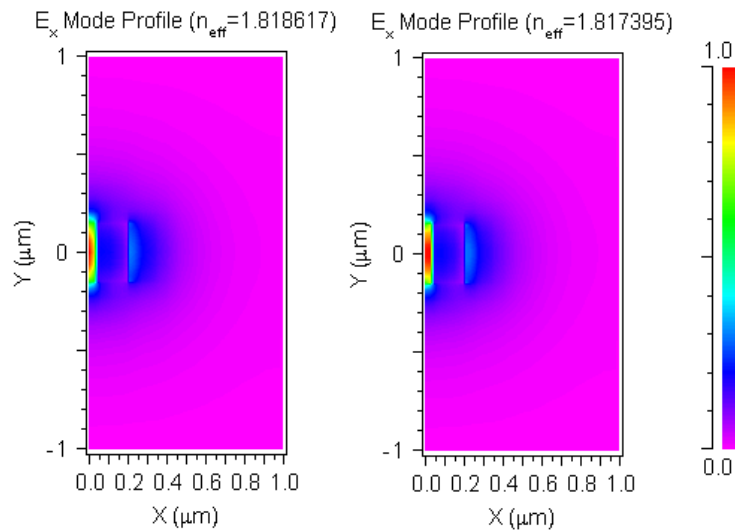


Figure 9: Fundamental modes found using **Grid Size (Edge)** value of a) 0.005, and b) 0.0015.

The computations using the non-uniform grid options not only run relatively quicker than the previous simulations, but they result in a higher resolution field around the two core regions. If, in the second simulation with **Grid Size (Edge)** set to 0.0015, you right-click at $Y=0$, the cross section of the mode at $Y=0$ will be shown.

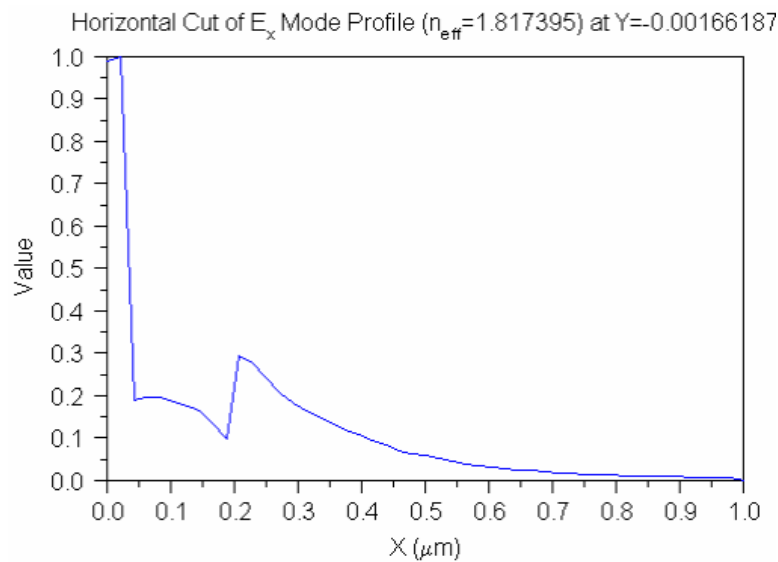


Figure 10: Cross section of fundamental mode at $Y=0$ found with **Grid Size (Edge)** of 0.0015.

The cross-section shown in Fig. 10 matches Figure 2 of the reference cited at the start of this tutorial. By using these advanced parameters, we were able to compute the mode of this device with improved accuracy without changing the simulation speed appreciably.

While this advanced convergence study explored the effect of some of the non-uniform mesh options, the global mesh size is still critical for the mode computation and still needs to be studied via a convergence study.

Areas for Further Exploration

There are several areas that can be explored further in this case. These include:

- *Exploring the mesh size*

The first convergence study performed in this tutorial can be continued without using the advanced features. The accuracy can then be compared to the advanced features that were discussed in this tutorial.

- *Convergence study using non-uniform mesh options*

While the advanced meshing features gave greater accuracy while using a relatively coarser mesh, the result can still be converged further. Additional analysis should be performed to study the effects of reducing the global mesh size as well as the non-uniform mesh.

- *Utilizing Asymmetric Boundary Conditions to find TM modes.*

Since the modes of this structure will be symmetric, asymmetric boundary conditions can be employed in order to compute the TM modes as shown in the reference.

Tutorial 3: VCSEL Cavity Mode

FemSIM has the capability to compute cavity modes for waveguide devices. A common cavity structure found in lasers is vertical cavity surface emitting lasers (VCSELs). In this tutorial, we will go through the process of setting up, designing, and simulating the resonances of a VCSEL device. This tutorial is based on the following reference:

Michael J. Noble, James A. Lott, and John P. Loehr, “Quasi-Exact Optical Analysis of Oxide-Apertured Microcavity VCSEL’s Using Vector Finite Elements,” *IEEE J. of Quant. Elect.*, **34** (1998)

It is assumed that the user is somewhat familiar with the RSoft CAD and the multilayer structure for this tutorial. For more information on the CAD settings, please refer to the RSoft CAD manual. Also, please note that the results of this tutorial might be slightly different then that of the reference due to estimated parameters and different physical constants.

Structure Overview

This VCSEL structure is composed layers of AlO and AlGaAs to make reflective DBR stacks. We will solve for cavity modes at a wavelength of $0.855\text{ }\mu\text{m}$ for several aperture diameters.

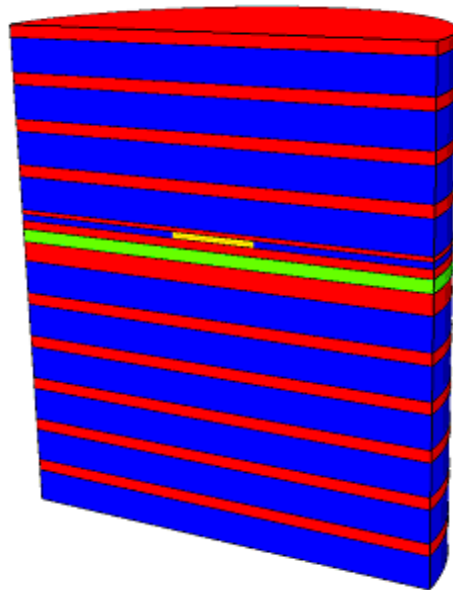


Figure 1: A cross-section of the VCSEL structure used in this tutorial.

Device Setup

When solving for a VCSEL cavity mode, the structure's inherent cylindrical symmetry can be exploited to result in a simpler computation. Rather than simulate the entire structure, only one half of a cross-section is needed. This, as well as the fact that FemSIM will solve for cavity modes defined in the XY plane, means that the built-in multilayer structure type is a natural choice.

Open the CAD window and click the **New Circuit** icon on the top toolbar and make the following settings:

Parameter	Value
Simulation Tool	<i>FemSIM</i>
Model Dimension	<i>3D</i>
Background Index	1.5
Component Width	10
Component Height	10
3D Structure Type	<i>Multilayer</i>
Slab Height	0

Creating Symbols and Layer Tables

In order to use the multilayer structure type, layers need to be defined. Click the **Edit Layers** button to open the Layer Table editor. In order to simplify the creation of these layers, we will use several symbols. Click the **Edit Symbols** button and define/set the following variables:

Variable	Value	Description
L	100	Length of segments
N_AlAs	$\text{sqrt}(9.92)$	Index of AlAs.
N_AlGaAs	$\text{sqrt}(12.489)$	Index of AlGaAs
N_AlO	1.5	Index of AlO
N_GaAs	$\text{sqrt}(13.59)$	Index of GaAs.
Radius	0.4	Radius of aperture
T_AlAs	0.03	Thickness of AlAs
T_AlGaAs	0.055	Thickness of AlGaAs
T_AlGaAs_Bar	$(\text{free_space_wavelength} - (T_Gain * N_GaAs)) / (2 * N_AlGaAs)$	Thickness of AlGaAs barrier
T_AlO	0.155	Thickness of AlO

T_Gain	0.06	Thickness of gain medium
T_Gap	T_AlGaAs_Bar-T_AlAs-T_Space	Thickness of gap layer
T_Space	0.021	Thickness of spacer layer
background_index	N_GaAs	Index of substrate
cover_index	N_GaAs	Index of cover index

Once done creating these symbols, click **OK** to return to the Layer Table editor where the VCSEL layers can now be defined. Set the **Substrate Index** and **Cover Index** to N_GaAs. We will define two layer tables, one for the aperture, and one for the rest of the structure. First we will define the layer table for the aperture. Set layer table 0 to have the following layers:

Layer #	Height	Index
1	T_AlO	N_AlO
2	T_AlGaAs	N_AlGaAs
3	T_AlO	N_AlO
4	T_AlGaAs	N_AlGaAs
5	T_AlO	N_AlO
6	T_AlGaAs	N_AlGaAs
7	T_AlO	N_AlO
8	T_AlGaAs	N_AlGaAs
9	T_AlO	N_AlO
10	T_AlGaAs	N_AlGaAs
11	T_AlO	N_AlO
12	T_AlGaAs_Bar	N_AlGaAs
13	T_Gain	N_GaAs
14	T_Gap	N_AlGaAs
15	T_AlAs	N_AlAs
16	T_Space	N_AlGaAs
17	T_AlO	N_AlO
18	T_AlGaAs	N_AlGaAs
19	T_AlO	N_AlO
20	T_AlGaAs	N_AlGaAs

21	T_AlO	N_AlO
22	T_AlGaAs	N_AlGaAs
23	T_AlO	N_AlO
24	T_AlGaAs	N_AlGaAs

To define the non-aperture layer table, it will be easier to duplicate the layer table we have already created and modify it. Click the **Duplicate** button to create Layer Table 1, and set Layer 15 to have an index of N_{AlO} as shown in Fig. 2. Press **OK** to return to the Global Settings window, and then **OK** again to go to the CAD window.

Creating the Structure

In the CAD window, draw in a straight segment that will represent the entire VCSEL structure (without the aperture). Right click on the segment to access its Segment Properties dialog, and change the starting and ending **Layer Table** to 1 and set the segment to have a length of L . Click the **More...** button, and set the **Display Color** to *Dark Blue*. Click **OK** in this window and the next to return to the CAD.

Draw another straight segment to represent the aperture, and set its starting and ending **Layer Table** to 0 and its starting and ending **Waveguide Width** to $2 \times \text{Radius}$. Set the segment's length to L and click **OK** to return to the CAD.

When both waveguides have been created, the structure should appear in the CAD window as shown in Fig. 2.

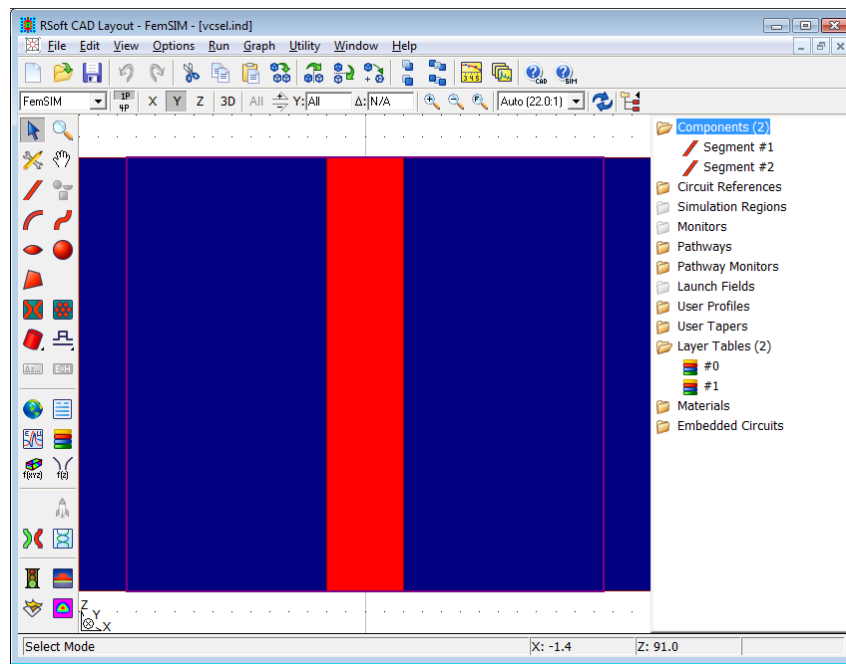


Figure 2: Completed VCSEL setup as seen in the CAD window. The VCSEL structure is defined along the Y axis. Since we will be performing a radial calculation, we will only use the right half of the structure.

Checking the Index Profile

Now we will display the index profile of the device to ensure that the correct structure was created. Click the **Display Material Profile** icon on the left toolbar, and make the following settings:

Parameter	Value	Description
X Domain Min	0	Sets the X domain minimum. This value is chosen so that only the right half of the structure will be used since a radial calculation will be performed.
X Domain Max	2.5	Sets the X domain maximum.
X Grid Size	0.1	Sets the X grid size.
X Slice Grid	0.1	Sets the X grid size for display purposes.
Y Domain Min	-0.5	Sets the Y domain minimum.
Y Domain Max	3	Sets the Y domain maximum.
Y Grid Size	0.01	Sets the Y grid size.
Y Slice Grid	0.01	Sets the Y grid size for display purposes.

Once these changes are made, click **OK** to view the index profile.

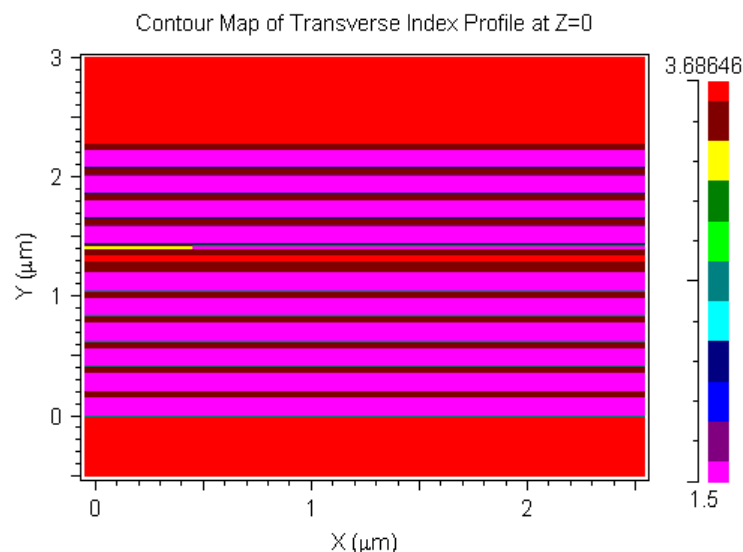


Figure 3: Displayed index profile.

Computing the Supported Cavity Modes

Once the structure has been created, the supported cavity modes of the VCSEL can be computed. Click the **Perform Simulation** icon in the left toolbar and make the following settings:

Parameter	Value	Description
X PML Width	<code>10*grid_size</code>	Sets the PML width at X boundaries.
Y Grid Size	0.1	Sets the Y grid size.
Y PML Width	<code>10*grid_size</code>	Sets the PML width at Y boundaries

Next the mode solving options need to be set. Click the **Mode Options...** button and make the following settings:

Starting Neff (Real)	1/0.855	This is set to the inverse of the wavelength to look for cavity modes around $1/\lambda$. During a cavity mode calculation, this field sets the frequency of the mode in units of μm^{-1} .
# of Modes to Find	5	Indicates that 5 modes should be found.
# of Modes to Output	1	Though 5 modes will be found, only 1 mode will be saved. The mode saved will be the mode with the lowest loss since Mode Sorting is set to <i>Lowest Loss</i> .
Mode Sorting	<i>Lowest Loss</i>	This setting indicates that the modes will be sorted by loss.

Click **OK** and click **Advanced...** to make the following settings:

Parameter	Value	Description
Allow Leaky Modes	Enabled	This enables the use of PML which is needed for cavity mode computations.
Solve for Cavity Modes	Enabled	This enables the computation of cavity modes.
Radial Calculation	Enabled	Enables the radial calculation.
Azimuthal Mode #	1	Sets the radial mode number.

Click **OK** to return to the Simulation Parameters dialog. These parameters indicate that the device has radial symmetry and to assume an azimuthal mode number of 1. Click **OK** to return to the Simulation Parameters dialog.

Checking the Simulation Mesh

FemSIM performs simulations on an irregular mesh, so it is important to look at this mesh to ensure that it correctly represents the structure. To view the mesh, click the **View Mesh** button in the Simulation Parameters window.

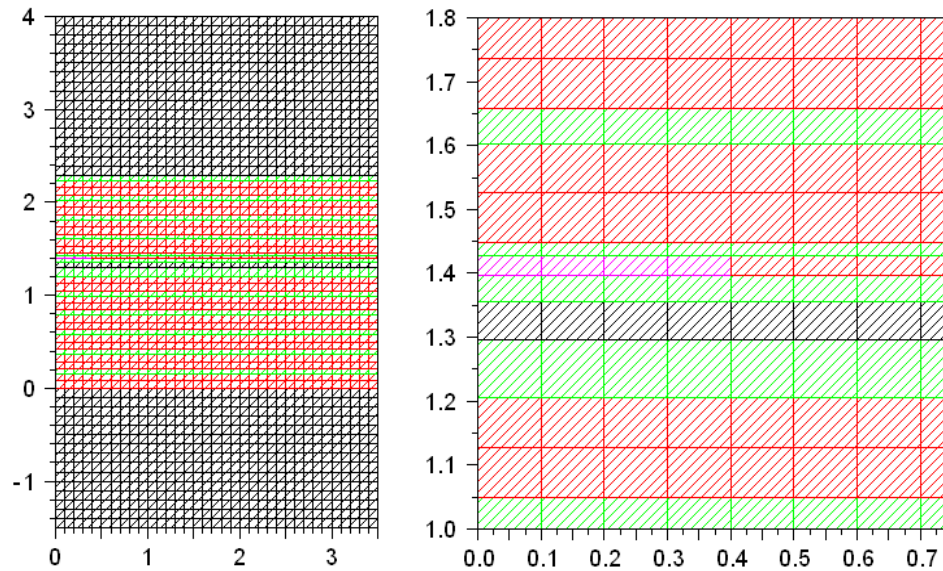


Figure 4: a) The simulation mesh. b) The mesh in the aperture region.

Figure 4 shows that the simulation parameters are quite coarse, especially in the aperture region. This could be fixed by making the global simulation values very fine, but this would make the bulk regions have an unnecessarily small grid slowing down the computation. Instead, we will use advanced grid features.

Click the **Mesh Options...** button in the FemSIM Simulation Parameters dialog and set the X and Y Minimum Divisions to 5 to set the minimum number of elements for a given material region. Click **OK** and then click the **View Mesh** button again and note that the mesh more accurately represents the structure.

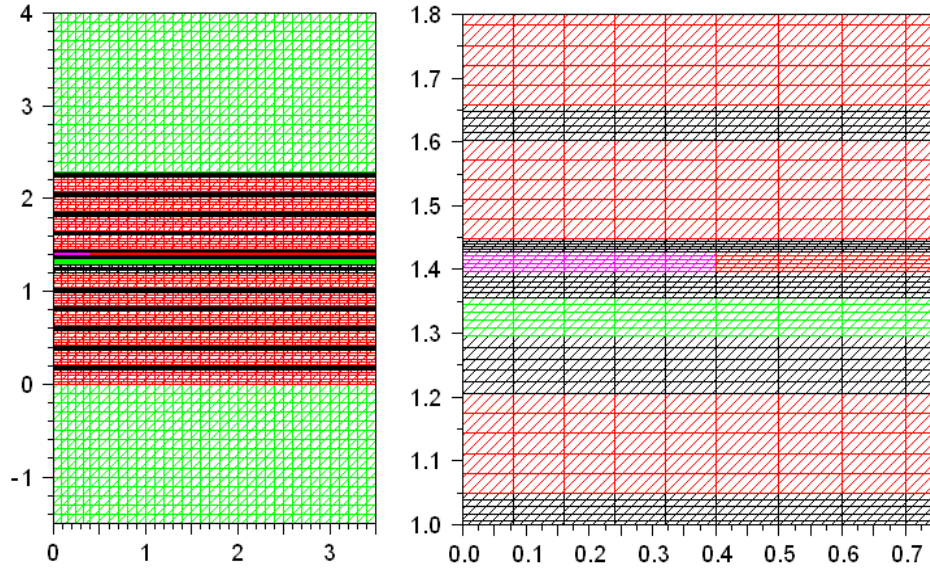


Figure 5: a) The simulation mesh using advanced features. b) The mesh near the aperture.

Performing the Simulation

Now that we have successfully set up the mesh, we can now compute the supported modes of the VCSEL. Press the **OK** button in the Simulation Parameters dialog to start the computation.

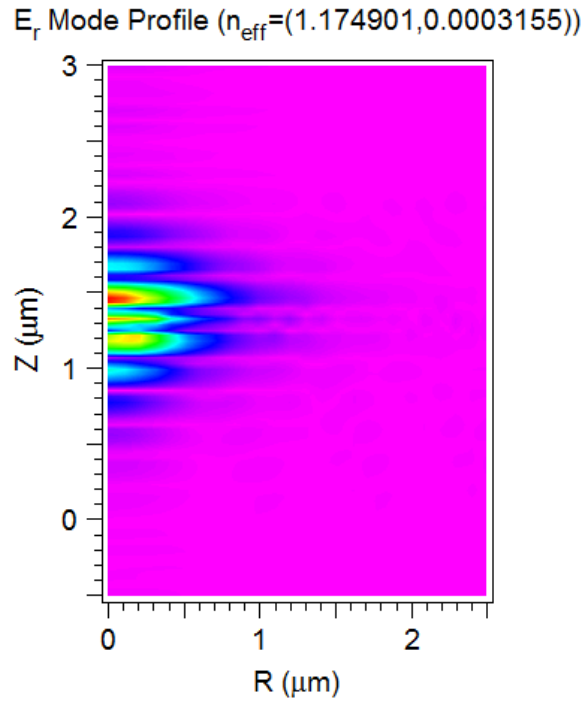


Figure 6: Mode computation with aperture size set to 0.4 μm . To see this E_r component of the mode, click the **Ex** button in the top toolbar.

Note that the real part of the effective index is the frequency in μm^{-1} . This means that this resonance occurs at $0.8511 \mu\text{m}$ as requested.

Scanning Aperture Radius to Study Device Characteristics

In the previous section we found the cavity mode of the VCSEL for a single aperture radius. In this section, we will use the scanning portion of the MOST utility to scan over the aperture size to see how the performance of the VCSEL for different designs. We will study the effect of aperture size on both the resonant wavelengths as well as the threshold material gain. These parameters can be easily calculated from FemSIM's output:

- *Resonant Wavelength*

As we saw before, the resonant wavelength is simply the inverse of the real part of the effective index FemSIM produces.

- *Threshold Material Gain*

According to the reference, this value is the loss in cm^{-1} times 2 (to convert field to intensity) divided by the confinement factor. The loss from FemSIM, in units of μm^{-1} , is $4\pi n_{\text{imag}}/\lambda$ where λ is the resonant wavelength. If we assume a total volume confinement factor of 0.11 for this particular device, then the threshold material gain is equivalent to $80000\pi n_{\text{imag}}/(0.11\lambda)$.

We will configure MOST to produce plots of these two parameters as a function of the aperture radius.

Setting up the Scan

To start the scan process, click the **Launch MOST Optimizer/Scanner** icon on the left toolbar. Under the Independent Variables tab, add the variable `Radius` with the following settings:

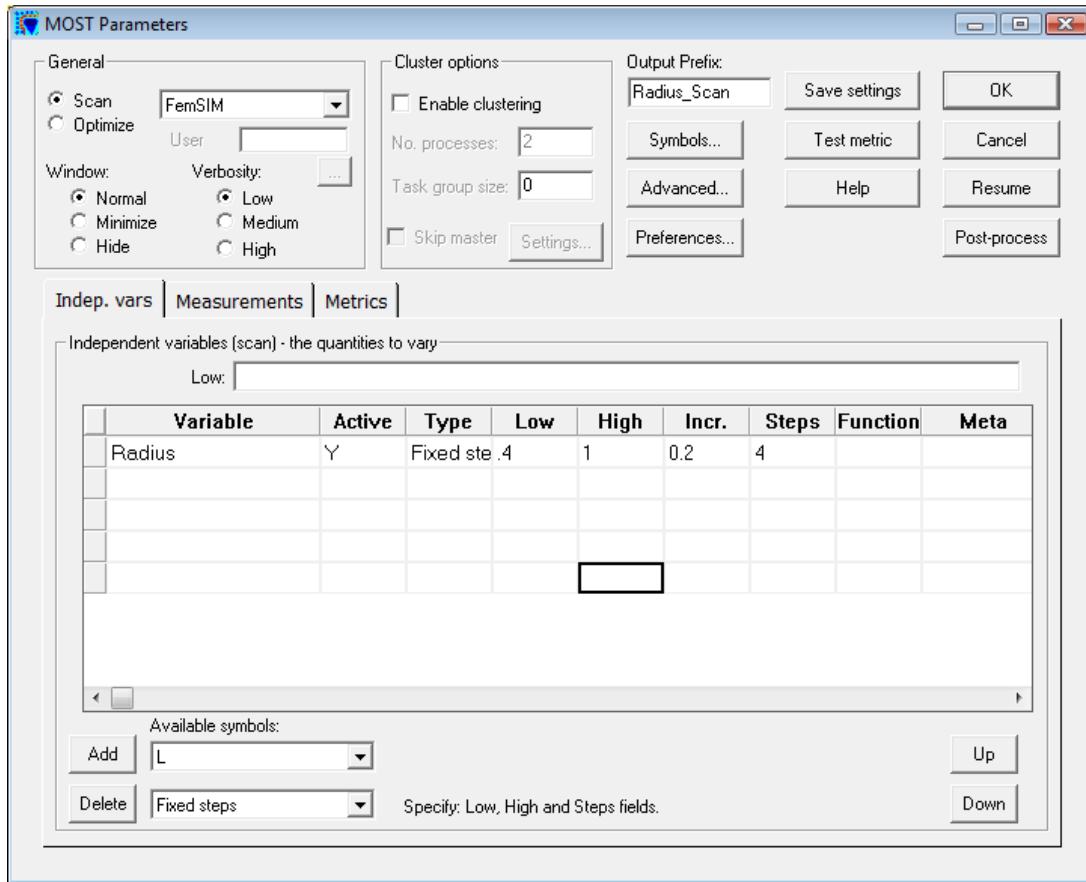


Figure 7: The Independent Variables portion of the MOST dialog.

Parameter	Value	Description
Variable	Radius	Sets the variable to scan over
Type	Fixed Steps	Indicates that the variable will be scanned over a fixed number of steps. This field can be changed by clicking on it.
Low	0.4	Sets the lowest value of the variable to be scanned.
High	1	Sets the highest value of the variable to be scanned.
Steps	4	Sets the number of steps in the scan.

Next, press the **Measurements** tab and enable the *fs_mode_neffr_0* and *fs_mode_neffi_0* measurements as shown in Fig. 8. These measurements correspond to the real and imaginary indices of the first mode found.

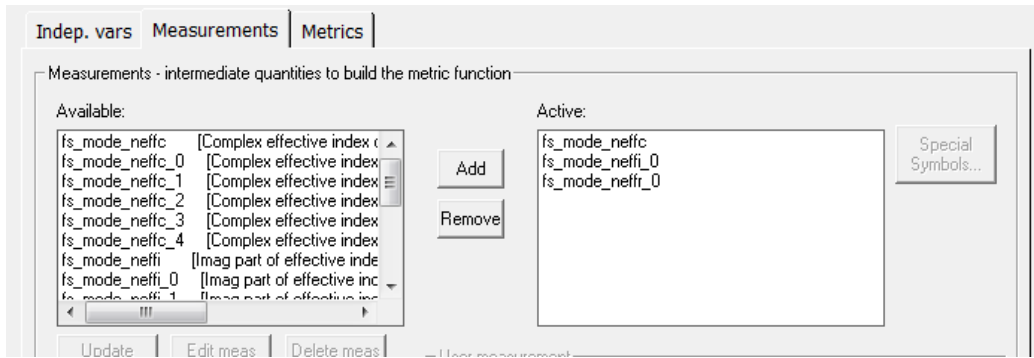


Figure 8: Measurement settings in the MOST dialog.

Now that we have set up the Measurements of the scan, we can now set up the metrics, which are the outputs that we want MOST to calculate: the resonant wavelength and threshold material gain. Click on the **Metrics** tab in MOST. We will create two metrics. First, create a metric called **Resonance** and set it equal to $1/fs_mode_neffr_0$. Next, create **Threshold_M_Gain** and set it equal to $fs_mode_neffi_0 * 4 * \pi * 10000 * fs_mode_neffr_0 * 2 / .11$.

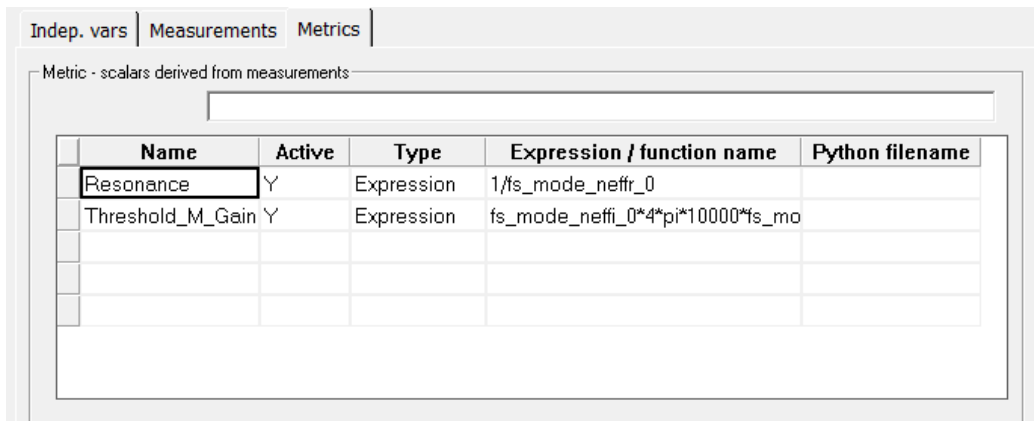


Figure 9: Metric settings in the MOST dialog.

Since we are only interested in the resonant wavelengths during the scan, the speed of the scan can be increased by turning off the mode profile output. This can be done by setting the **# of Modes to Output** in the Mode Options dialog to 0, or by setting the variable `fem_nev_output` to 0.

Once all of these MOST parameters are set, enter an **Output Prefix** of `Radius_Scan` and click **OK** to run the scan. At the end of the scan, open the DataBROWSER by clicking the **Open Databrowser** icon in the MOST dialog. Open the files `Radius_scan_met_Resonance.pcs` and `Radius_scan_met_Threshold_M_Gain.pcs`.

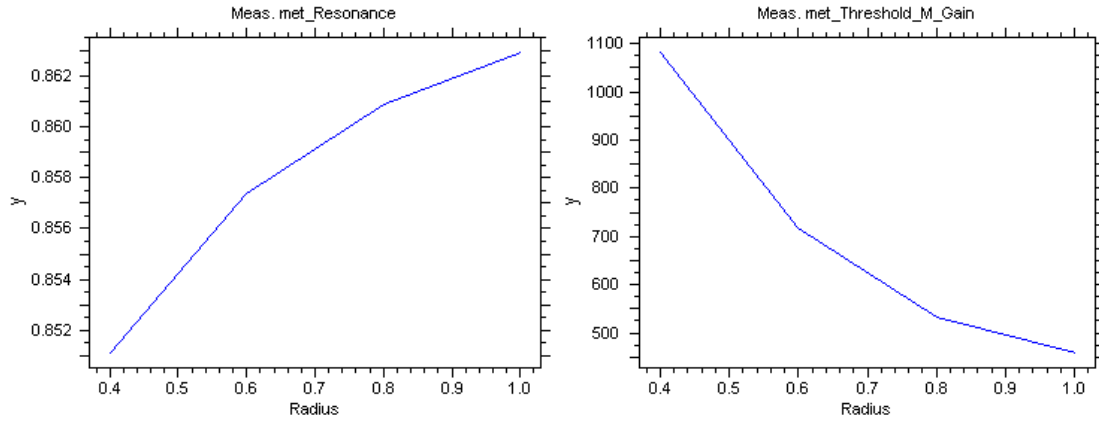


Figure 10: a) The resonance (in μm) versus the aperture radius and b) threshold material gain (in cm^{-1}) versus the aperture radius.

Areas for Further Exploration

- *Perform a Convergence Study*

Use smaller values for the mesh size to see how the accuracy increases. Analyze both the global and advanced options.

- *Try Optimization*

If the copy of the software is licensed for the optimization portion of MOST, use the optimization feature to optimize the device.

Tutorial 4: Bending Modes

The calculation of bending waveguide modes is essential to the design of Planar Lightwave Circuits (PLCs) and other waveguide-based photonic devices. Bending waveguides introduce shifts in the mode shape and bending losses which, as a function of the radius, determine the minimum bending radius as well as the minimum size of the PLC.

The material in this tutorial is based on these references:

- 1 R. Baets and P. E. Lagasse, “Loss calculation and design of arbitrarily curved integrated-optic waveguides”, *J. Opt. Soc. Am.*, **73** (1983)
- 2 W. Lui, C.L. Xu, T. Hirono, K. Yokoyama and W.P. Huang, “Full-Vectorial Wave Propagation in Semiconductor Optical Bending Waveguides and Equivalent Straight Waveguide Approximations”, *IEEE J. Lightwave Tech.*, **16** (1998)

The files used in this tutorial are found in the directory:

```
<rsoft_dir>\examples\FemSIM\tutorial\tut4\
```

Background

FemSIM utilizes two methods to solve for bending modes:

- *Conformal Index-Mapping Method:*

The conformal index-mapping method uses an approximation to translate a bent waveguide onto a straight waveguide as described in Ref 1. This method is valid for bends where the width of the guide is much smaller than the radius of the bend.

- *Direct Method:*

The direct method solves Maxwell’s equations in cylindrical coordinates directly without approximation. This method is valid for all types of bends and is described in Ref. 2.

For both methods, since bending modes are leaky by nature, FemSIM must be used with perfectly matched layer (PML) boundary conditions. Also, it is critical to understand how the bend radius is defined and how the propagation constant is computed; see [Section 3.F.2](#) for details.

Structure Overview

The device studied in this tutorial is based on an example in Ref. 2 and consists of a ridge waveguide as shown in Fig. 1. The ridge waveguide has a width of 1.4 μm and consists of a guiding layer and a cap layer. The guiding layer is composed of InGaAsP ($n = 3.39$) with a height of 0.6 μm . The cap layer is composed of InP ($n = 3.17$) with a height of 0.3 μm . The

semi-infinite substrate is composed of InP. This geometry is achieved by using the *Multilayer Structure Type* in the RSoft CAD. The operating wavelength is $1.55\ \mu\text{m}$.

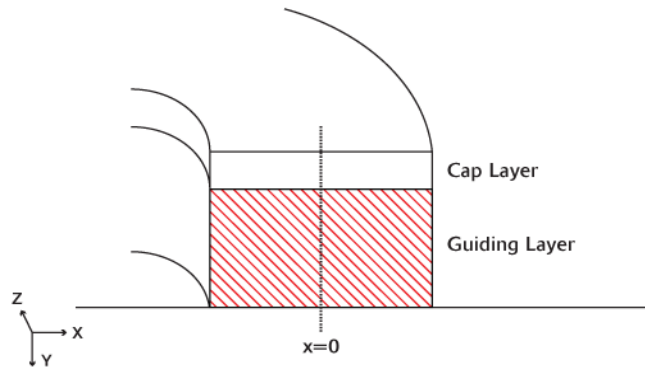
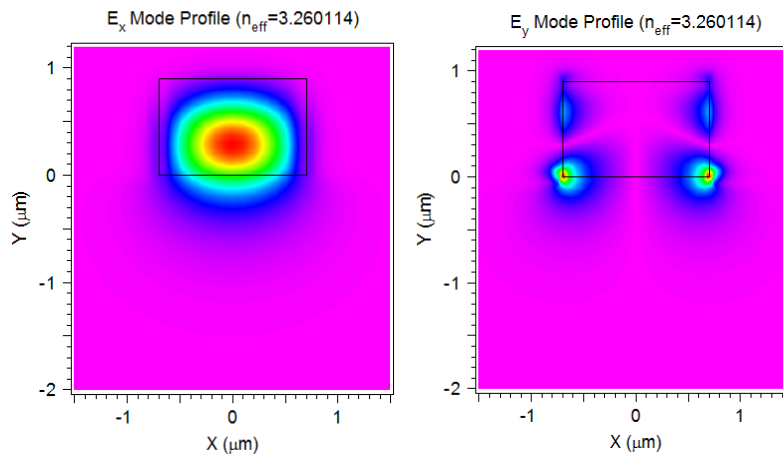


Figure 1: The structure studied in this tutorial.

Calculating the Straight Waveguide Mode

Before studying the modes of the bending waveguide, it will be beneficial to first compute the modes of a straight version of the waveguide. This will give us some general knowledge about the structure as well as give us a reasonable initial guess for the effective index of the mode.

Open the file `straight.ind` and open the FemSIM simulation parameters dialog. This file has been configured to compute the first and second modes of this waveguide. Since FemSIM does not separate modes by polarization, this will correspond to the fundamental TE and TM mode. Click **OK** to start the simulation. The modes, shown in Fig. 2, show that the effective index of both modes is around 3.26.



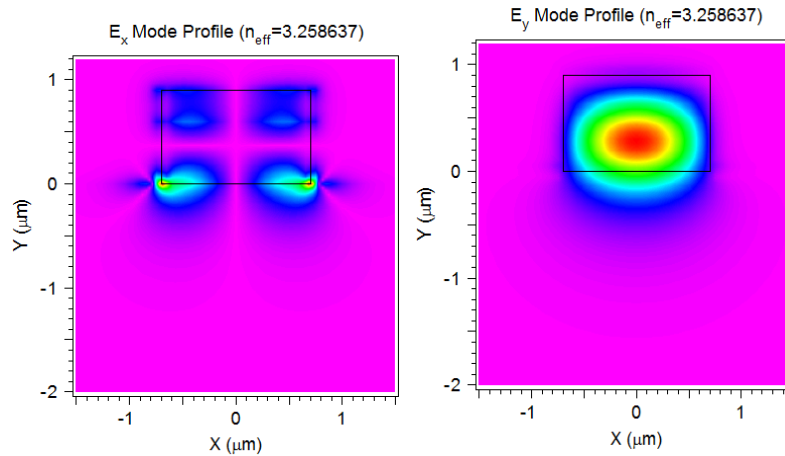


Figure 2: Mode profiles of the TE (top) and TM (bottom) modes of the straight waveguide. The E_x components are shown on the left and E_y components on the right.

Calculating the Bending Mode

With this knowledge about the straight waveguide mode we are ready to proceed to bending mode calculation. Open the file `bendingmode.ind` and open the FemSIM simulation parameters dialog. Note that the X/Z **PML Width** has been set to 0.5. Click the **Advanced...** button and set the following settings:

Option	Value	Description
Allow Leaky Modes	selected	Enables leaky modes to be solved for.
Simulated Bend	Yes	Enables the use of simulated bends.
Simulated Bend Method	<i>Direct</i>	Selects the direct simulated bend method described above which uses an exact solution of Maxwell's equations in cylindrical coordinates.
Bend Radius	R	The bending radius.
Bend Origin	0	The offset relative to X=0 where the bending radius is measured.

Click **OK** and then click **Mode Options...** and note that the **Starting Neff (real)** value is set to 3.26 as found before. Click **OK** twice to run the simulation.

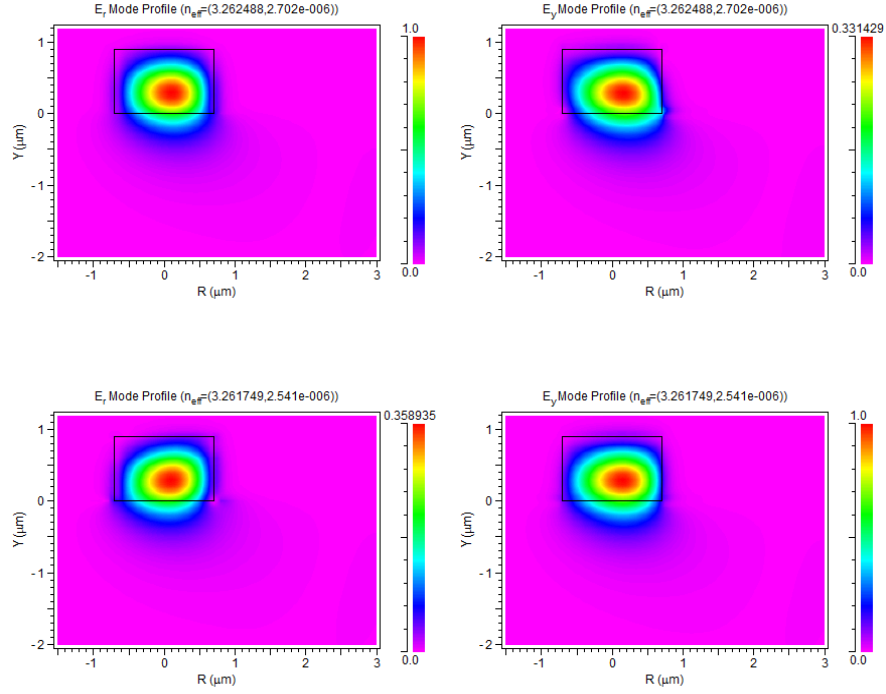


Figure 3: Mode profiles of the TE (top) and TM (bottom) modes of the bending waveguide.

The two bending modes, shown in Fig. 3, illustrate that the polarization of these modes has been tilted due to the bend. This can be seen more clearly from the vector plots shown in Fig. 4.

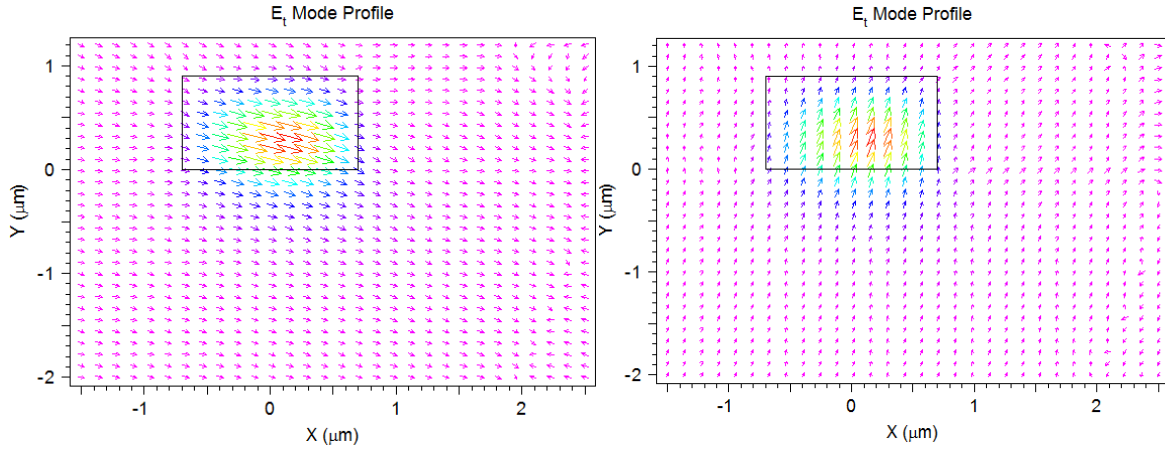


Figure 4: Vector plots of the a) TE, and b) TM bending modes.

Scanning over the Bend Radius

To study the loss due to the bend as a function of the bend radius, we can simply scan over the radius using MOST. For this example, we will also scan over the **Simulated Bend Method** to compare between the two methods.

Open the file `scanR.ind`. This is almost the same file used in the previous section with a few minor differences:

- It contains a MOST scan
- It has the **# Modes to Output** set to 0 to save file space since we are only going to look at the effective indices.
- The **Seed Neff Values in MOST Scan** and **Seed Imag Part of Neff** options in the Mode Options dialog are selected. These options allow the complex effective index found at one scan point to be used as the starting value for the next scan point.

Click the **Launch MOST Optimizer/Scanner** button in the left CAD toolbar to open the MOST dialog. A scan over the variables `R` (bend radius) and `simulated_bend_method` (Bend Method) is already defined. The scan over `R` is from the larger radius to the small radius (see ‘Additional Tips’ in [Section 3.F.2](#)). The scan over `simulated_bend_method` will only use values of 0 and 1 which correspond to the *Conformal* and *Direct* methods respectively.

Variable	Active	Type	Low	High	Incr.	Steps	Function	Meta
R	Y	Fixed step	100	10	-2	46		
simulated_bend_method	Y	Fixed inc	0	1	1	2		

Figure 5: MOST scan settings over Radius and `simulated_bend_method`.

For this scan, we are going to look at the real and imaginary effective index of the first mode. Note that there are only two active measurements: `fs_mode_neffi_0` and `fs_mode_neffr_0`.

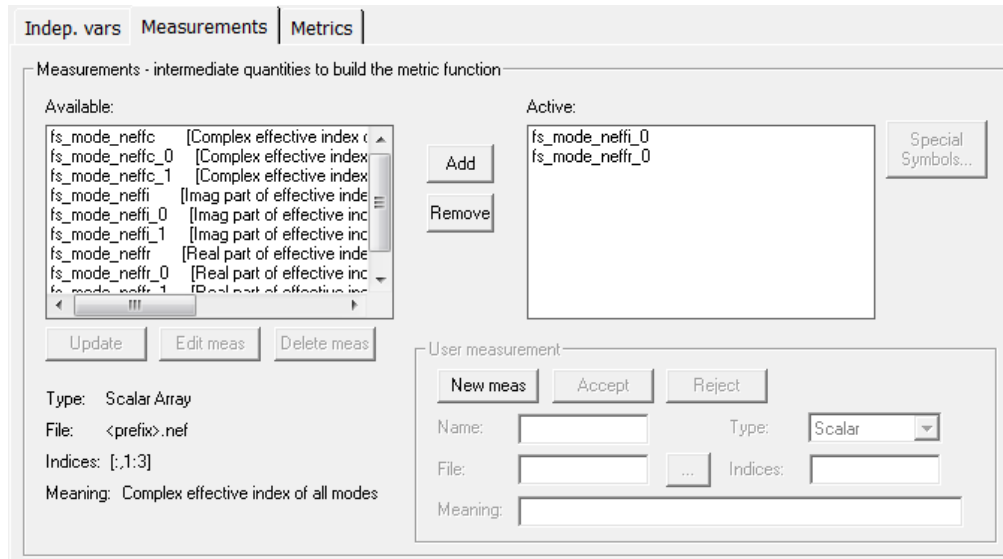


Figure 6: Most measurements for the scan over bending radius.

Enter an **Output Prefix** such as `scan` in the MOST dialog and click **OK** to start the scan. The scan results are shown in Fig. 7. As predicted, the discrepancy gets bigger when the bend radius gets smaller.

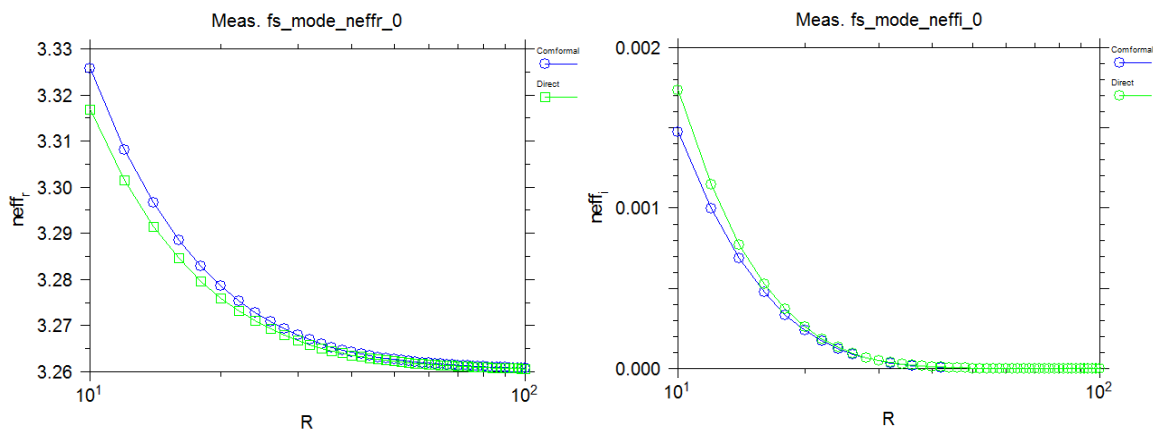


Figure 7: Scan results of the a) real, and b) imaginary effective indices. Note that the plots have been slightly changed as noted below.

The plots shown in Fig. 7 have been modified to better view the data. To make these changes for yourself, all you have to do is edit the WinPLOT commands used to display the data. This can be done by opening the plot in WinPLOT and clicking the View Edit button in the top toolbar to show the commands. Alternatively, you can open the WinPLOT plot file in a text editor. The changes that have been made were to use a logarithmic scale for the X axis (adding the `/lx` command), adding symbols for the data points (adding the `/st1` command), and changing the legend titles. See the WinPLOT manual for a complete list of supported WinPLOT commands.

Appendix A

Tips and Traps

This appendix contains some advice on good habits and sources of confusion for novices and experienced FemSIM users alike.

A.A. Common FemSIM mistakes

- *When solving for leaky modes, make sure appropriate options are enabled.*
When solving for leaky modes, it is necessary to enable the **Allow Leaky Modes** option in the Advanced Parameters dialog, as well as defining a non-zero **PML Width**.
- *Not understanding how FemSIM handles polarization.*
See [Section 3.B.3](#) for a description of how FemSIM handles polarization.

A.B. Some Good FemSIM habits to learn

- *Use the Compute Material Profile option frequently.*
When setting up a structure, it is easy to make mistakes with geometry and refractive index information. Many apparently bizarre results are simply the result of simulating the wrong structure.
- *Use too few grid points at first; use too many at the end.*

General results for your structure can often be obtained with remarkably coarse grids. You can save time when doing initial exploratory studies by not worrying too much about accuracy. At the same time, when a structure includes fine features, the results can be quite inaccurate with an overly coarse grid. You have not completed your design project until you have confirmed the convergence of your results at a higher grid resolution.

- *Learn how to use the command line utilities included with FemSIM*

Since post-processing with these tools can be automated, you can bypass a lot of tedious work by writing a single command. You may even find uses for these utilities in your other work. These utilities are documented in the appendices of the CAD manual.

- *Learn the scripting capabilities of FemSIM*

While FemSIM is designed to be used via a graphical interface (GUI), a long series of simulations can be tedious to perform through the GUI. These types of calculations can easily be automated through the use of a script. FemSIM's scripting capabilities are based on the fact that simulations can be run from the command line. Because of this unique capability, FemSIM can utilize the any native scripting language within the OS you are using. For instance, Windows users can use simple DOS batch files to run FemSIM, while advanced users who may have standard scripting languages installed on their system such as PERL can write powerful scripts in these languages.

- *Read the WinPLOT manual.*

WinPLOT's rather terse command syntax can be daunting at first but allows complex plotting features to be described quickly in just a few lines. Being able to quickly launch plots with old command files from the command line can be a huge time saver.

- *Use color scales to produce attractive plots.*

WinPLOT is capable of producing many types of plots beyond the standard output usually displayed. Color scale files can be found in the installation directory, and are used to change the default scale used when displaying contour plots. These files are easy to create, and provide a quick way to customize the display of data for presentation purposes.

Appendix B

FemSIM File Formats

This appendix documents the file formats used in FemSIM. Except where noted, all files are in ASCII text format. For further information, contact RSoft.

B.A. Standard Data File Format

These are the main file formats used to represent functions of one or two variables, including field profiles, index profiles, and other output types. This format is described in detail in Appendix B of the RSoft CAD manual.

B.B. Mode Files

The mode files produced by FemSIM are in the standard RSoft file format described above. These files are, where ## is the mode number:

Plot File	Data File	Description
<prefix>_ex.p##	<prefix>_ex.m##	Ex field data
<prefix>_ey.p##	<prefix>_ey.m##	Ey field data
<prefix>_ez.p##	<prefix>_ez.m##	Ez field data
<prefix>_hx.p##	<prefix>_hx.m##	Hx field data

<code><prefix>_hy.p##</code>	<code><prefix>_hy.m##</code>	Hy field data
<code><prefix>_hz.p##</code>	<code><prefix>_hz.m##</code>	Hz field data
<code><prefix>_sz.p##</code>	<code><prefix>_sz.m##</code>	Poynting Vector data
<code><prefix>_et.p##</code>	<code><prefix>_ex.m##,</code> <code><prefix>_ey.m##</code>	Vector plot of transverse E field.
<code><prefix>_ht.p##</code>	<code><prefix>_hx.m##,</code> <code><prefix>_hy.m##</code>	Vector plot of transverse E field.

B.C. Neff File Format

The Neff file contains the effective index results from a simulation and is named `<prefix>.nef`. This file has three columns, the first has the mode number and the second and third have the complex-valued effective indices for each mode.

B.D. Monitor File Format

Monitor files contain the result of any defined monitors and is named `<prefix>.mon`. Each row corresponds to each mode, and different columns refer to each defined monitor.

Appendix C

Symbol Table Variables

This appendix contains the symbol table variable names which control the internal operation of FemSIM. Most of these variables correspond to dialog fields which have already been documented.

C.A. FemSIM Simulation Parameters

The following variables correspond to the fields found in the FemSIM Simulation Parameters dialog:

Field Name	Variable Name
X Domain Min	boundary_min
X Domain Max	boundary_max
X Grid Size	grid_size
X PML Width	pml_width_x
Y Domain Min	boundary_min_y
Y Domain Max	boundary_max_y
Y Grid Size	grid_size_y
Y PML Width	pml_width_y
Z Domain Min	domain_min

C.B. Mesh & Boundary Options

The following variables correspond to the fields in the Advanced Mesh Controls dialog box that are not documented in the previous section:

Field Name	Variable Name
X Grid Size (Bulk)	grid_size
X Grid Size (Edge)	grid_edge_x
X Griding Ratio	grid_ratio_x
X Minimum Divisions	grid_mindiv_x
X Grid Grading	grid_bulk_nonuniform_x
X Interface Alignment	grid_align_x
Y Grid Size (Bulk)	grid_size_y
Y Grid Size (Edge)	grid_edge_y
Y Griding Ratio	grid_ratio_y
Y Minimum Divisions	grid_mindiv_y
Y Grid Grading	grid_bulk_nonuniform_y
Y Interface Alignment	grid_align_y
Subdivide Mesh	mesh_sub
Subdivision #	mesh_subn
Subdivision Tolerance	mesh_subtol
Symmetry at X=0	symmetric_x
Symmetry at Y=0	symmetric_y

C.C. Mode Options

The following variables correspond to the fields in the FEM Mode Options dialog box:

Field Name	Variable Name
Starting Neff (Real)	fem_neff

Starting Neff (Imag)	fem_neff_imag
# of Modes to Find	fem_nev
# of Modes to Output	fem_nev_output
Mode Sorting	fem_sortev
Allow Leaky Modes	fem_leaky
Solve for Cavity Modes	fem_cavity
FEM Element Order	fem_order
Neff Tolerance	neff_tol
Max Iterations	fem_max_iterations

The following options correspond to other mode options:

Variable Name	Description
radial	Enables a radial calculation
fem_azimuthal_mode	Sets the azimuthal mode number for a radial calculation
fem_hfield	Enables formulation in terms of H rather than E

C.D. Display Options

These variables correspond to the fields in the FEM Display Options dialog box:

Field Name	Variable Name
Outline Color	color_outline
Color Shades	color_shades
Color Scale	color_scale
Show Color Scale	color_scale_display
Plot Aspect Ratio	plot_aspectratio

Appendix D

Release Notes

This appendix summarizes the changes from previous versions of FemSIM.

Versions 2015.06 and Later

The release notes for version 2015.06 and later versions can be found here:

```
<rsoft_dir>\readme\releasenotes
```

Version 2014.09

- With this release, the RSoft Component Suite will no longer officially support 32-bit operating systems, Windows XP, Windows Vista, or Red Hat Enterprise Linux (RHEL) 4. If you have any questions, please contact rsoft_support@synopsys.com.
- A new Configure Licensing utility (Windows only) provides an easy way to reconfigure licensing options. This includes installing a new license key file and restarting SCL. This option can be opened from the Start Menu and requires administrative rights.
- Several improvements to SCL licensing implementation.
- Updated Linux installation routine.

Version 2013.12

- All products in RSoft's Component Design Suite have been increased to version 2013.12 which is the Synopsys standard.
- Updated Licensing to use Synopsys Common Licensing (SCL).
- Added option to globally use Single Precision for calculations.

Changes from Version 3.2 to Version 9.0

- All products in RSoft's Component Design Suite have been increased to version 9.0 with this release.
- New pathway monitor data file specification to now accept launch files based on a file prefix and extension instead of providing individual files for each necessary field component. This feature works well with the previous feature.
- Improved mesh generation, especially for simulated bends.
- Added new simulated bend option which solves for bending modes in radial coordinates.

Changes from Version 3.1 to Version 3.2

- Added 1D mode solving capability.

Changes from Version 3.0 to Version 3.1

- Introduction of a new material editor where complex materials can be defined and shared between different design files and users. See CAD manual for more details.
- New Vector plots for transverse mode components.
- New Monitor feature for computation of common post-processing parameters such as overlap integrals, confinement factors, and spot-size computations.

Changes from Version 2.0 to Version 3.0

- Improved CAD interface. See CAD manual for details.
- Added a new Advanced Parameters dialog.
- Moved several options previously in the Mode Options dialog to the Advanced Parameters dialog.
- Added a new Output Options dialog.

Changes from Version 1.0 to Version 2.0

- Renamed the .dat output file to .nef for consistency with other tools.
- Added a mode seeding option for use in MOST scans.
- Added major speed improvement and lowered memory usage.

Version 1.0

This is the first public release of the program.

Index

#

Modes to Output, 95
of Modes to Find, 22, 69, 85
of Modes to Output, 22, 55, 69, 85, 88
Threads, 39

1

1/e Height, 31
1/e Width, 31

3

3D, 44, 65, 72, 81
3D Structure Type, 18, 44, 56, 59, 65, 72, 81

A

Accept Symbol, 45
Advanced..., 35, 69, 85, 94
Allow Leaky Modes, 20, 35, 69, 85, 94, 98
Amp/Phase, 32
Antisymmetric, 18
Auto, 30
Azimuthal Mode #, 24, 85

B

Background Index, 44, 56, 72, 81
Bend Origin, 36, 94

bend radius, 36, 94

C

Cancel, 6
channel, 44
color scale, 27, 55
Color Shades, 27
Component, 47
Component Height, 44, 56, 65, 72, 81
Component Width, 44, 56, 65, 72, 81
Conformal, 95
Conformal Transformation, 36
Cover Index, 81
Custom, 30
Custom..., 30

D

Data File Spec, 30
Default, 18, 55
Default # Threads, 39
Dimensions, 4
Direct, 36, 94, 95
Dispersion, 25
Display Color, 81
Display Material Profile, 6, 65, 72, 81
Display..., 27, 55, 65
Domain Max, 17, 68, 72, 81
Domain Min, 17, 68, 71, 72, 81
Duplicate, 81
Duplicate Selection, 72

E

Edit Layers, 65, 81
Edit Monitors, 29
Edit Symbols, 45, 58, 72, 81
Effective Area, 31
E-Field, 32
Ex, 85
Expression, 45
Extension, 30

F

Far Field Index, 32
Far Field Output Format, 32
FEM Element Order, 35
FemSIM, 6, 44, 56, 65, 72, 81
fiber, 18, 56, 59
Fiber Power, 30
Files of type, 6
Formulation, 32, 35
Free Space Wavelength, 25, 44, 56, 65, 72

G

Grid Grading, 77
grid size, 17, 18, 50, 65, 68, 81, 85
Grid Size (Edge), 77

H

H Field (Derived from E Field), 32
Height, 30, 31
H-Field, 32
High, 88

I

Include PML in Mode Plots, 32
Index Difference, 44, 56, 72
Index Display Min, 65
Intensity, 32
Interface Alignment, 18, 74

L

Launch DataBROWSER, 62
Launch MOST Optimizer/Scanner, 88, 95
Launch Power, 30
Layer Table, 81
Layers, 31
Low, 88
Lowest Loss, 69, 85

M

Max Iterations, 22
Measurements, 88
Mesh Options..., 18, 62, 74, 77, 85
Metrics, 88
Mode Options..., 55, 69, 85, 94
Mode Output Format, 32
Mode Sorting, 22, 24, 69, 85
Model Dimension, 17, 44, 65, 72, 81
Monitor Type, 29, 32
More..., 81
multilayer, 65, 81, 92

N

Name, 45
Neff Tolerance, 22
New Circuit, 44, 45, 56, 65, 72, 81
New Symbol, 45
No, 18
None, 18, 32

O

Offset, 47, 58
Offset Value, 47

OK, 6, 44, 50, 53, 55, 56, 59, 65, 69,
72, 74, 81, 85, 88
Open Databrowser, 88
Opts..., 25
Outline Color, 27, 55
Output Prefix, 6, 9, 29, 50, 53, 59, 88
Output Radial Data in XZ Format, 32
Output..., 6, 32, 50

P

Partial Power, 31
Perform Simulation, 6, 50, 59, 68,
69, 85
Phase, 30
Plot Aspect Ratio, 27, 55
Plot Mesh, 32
Plotting Points, 32
PML Width, 20, 35, 68, 85, 94, 98
polarization, 24
Poynting Vector, 32
Prefix, 30

R

Radial Calculation, 24, 85
Reference Type, 47
Relative To:, 47

S

Save Mesh File, 32
Save Settings, 6
Scale Files..., 27, 55
Seed Imag Part of Neff, 22, 95
Seed Neff Values in MOST Scan, 22,
36, 95
Segment Mode, 47, 58
Show Color Scale, 27
simulated bend, 36, 94
Simulated Bend Method, 36, 94, 95
Simulation Tool, 6, 44, 56, 65, 72, 81
Slab Height, 65, 81
Slab Power, 30
Slice Grid, 72, 81
Solve for Cavity Modes, 35, 85
Spawn Result Viewer, 25
Starting Neff, 22
Starting Neff (imag), 22, 36
Starting Neff (Real), 22, 36, 69, 85,
94
Steps, 88
structure type, 92
Subdivide Mesh, 18, 50, 59
Subdivision #, 18
Subdivision Tolerance, 18
Substrate Index, 81
Symbols..., 65
Symmetric, 18, 62, 74

Symmetry at X=0, 18, 62, 74
Symmetry at Y=0, 18, 62

T

Type, 30, 31, 88

U

Undo Last Change, 47
Use Single Precision, 35

V

Variable, 88
View Editor, 53
View Graphs, 53
View Mesh, 6, 22, 50, 59, 68, 74, 85
View Plot, 53

W

Waveguide Width, 81
Wavelength Range, 25
Width, 30, 31

X

X Plotting Points, 32
X Position, 30, 31

Y

Y Plotting Points, 32
Y Position, 30, 31
Yes, 18, 77, 94